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Light-Emitting Device And Display Apparatus

Abstract

A light-emitting device is formed over a first insulating layer and includes a first electrode, a second electrode, and an organic compound layer. The first electrode is formed in contact with the first insulating layer. The organic compound layer is positioned between the first and second electrodes. The second electrode and the organic compound layer are separated from at least one of other light-emitting devices adjacent to the light-emitting device. When seen from a direction substantially perpendicular to a surface of the first insulating layer where the first electrode is formed, outlines of the second electrode and the organic compound layer are substantially aligned with each other. The organic compound layer includes a light-emitting layer and an electron-injection layer. The electron-injection layer includes a mixed layer including a metal, a first organic compound, and a second organic compound. The first organic compound has a phenanthroline ring with an electron-donating group. The second organic compound includes a π-electron deficient heteroaromatic ring.

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Background/Summary

BACKGROUND OF THE INVENTION
1. Field of the Invention
[0001] One embodiment of the present invention relates to a light-emitting device and a display apparatus.
[0002] Note that one embodiment of the present invention is not limited to the above technical field. Examples of the technical field of one embodiment of the present invention include a semiconductor device, a display apparatus, a light-emitting apparatus, a power storage device, a memory device, an electronic device, a lighting device, an input device (e.g., a touch sensor), an input/output device (e.g., a touch panel), driving methods thereof, and manufacturing methods thereof.
2. Description of the Related Art
[0003] Display apparatuses are being developed into a variety of applications these days. For example, a television device for home use (also referred to as TV or television receiver), digital signage, and a public information display (PID) are being developed as large-sized display apparatuses, and a smartphone and a tablet terminal each provided with a touch panel are being developed as small-sized display apparatuses.
[0004] At the same time, an increase in the resolution of display apparatuses is also required. For example, devices for virtual reality (VR), augmented reality (AR), substitutional reality (SR), or mixed reality (MR) are given as devices requiring high-resolution display apparatuses and are being developed actively.
[0005] Development is actively conducted on light-emitting devices (also referred to as light-emitting elements) as display elements used in display apparatuses. Light-emitting devices utilizing electroluminescence (hereinafter referred to as EL; such devices are also referred to as EL devices or EL elements), particularly organic EL devices that mainly use organic compounds, are suitable for display apparatuses because of having features such as ease of reduction in thickness and weight, high-speed response to input signals, and driving with a constant DC voltage power source.
[0006] In order to obtain a higher-resolution display apparatus using an organic EL device, patterning an organic layer by a photolithography method using a photoresist or the like, instead of an evaporation method using a metal mask, has been studied. By using the photolithography method, a high-resolution display apparatus in which the distance between organic compound layers is several micrometers can be obtained (see Patent Document 1, for example).
REFERENCE
Patent Document
[0007] [Patent Document 1]Japanese Translation of PCT International Application No. 2018-521459
SUMMARY OF THE INVENTION
[0008] It has been known that a cathode and an organic compound layer of an organic EL device exposed to atmospheric components such as water and oxygen have affected initial characteristics or reliability, and thus it has been common knowledge that the cathode and the organic compound layer are treated in an inert gas atmosphere or a near-vacuum atmosphere. In particular, an electron-

injection layer, which often includes an alkali metal, an alkaline earth metal, or a compound thereof highly reactive with water or oxygen, rapidly deteriorates and loses the function as the electron-injection layer when the surface of the organic compound layer including the electron-injection layer is exposed to the air.

[0009] However, processing steps by the aforementioned photolithography method inevitably expose the organic EL device to the air.

[0010] An object of one embodiment of the present invention is to provide a novel light-emitting device. Another object of one embodiment of the present invention is to provide a highly efficient light-emitting device. Another object of one embodiment of the present invention is to provide a highly reliable light-emitting device. Another object of one embodiment of the present invention is to provide a highly efficient and highly reliable light-emitting device.

[0011] Another object of one embodiment of the present invention is to provide a novel light-emitting device manufactured through a photolithography process. Another object of one embodiment of the present invention is to provide a highly efficient light-emitting device manufactured through a photolithography process. Another object of one embodiment of the present invention is to provide a highly reliable light-emitting device manufactured through a photolithography process. Another object of one embodiment of the present invention is to provide a high-emission-efficiency and highly reliable light-emitting device manufactured through a photolithography process.

[0012] Another object of one embodiment of the present invention is to provide a novel light-emitting device that can be used in a high-resolution display apparatus. Another object of one embodiment of the present invention is to provide a highly efficient light-emitting device that can be used in a high-resolution display apparatus. Another object of one embodiment of the present invention is to provide a highly reliable light-emitting device that can be used in a high-resolution display apparatus. An object of another embodiment of the present invention is to provide a high-emission-efficiency and highly reliable light-emitting device that can be used in a high-resolution display apparatus.

[0013] An object of another embodiment of the present invention is to provide a highly reliable display apparatus. An object of another embodiment of the present invention is to provide a high-resolution display apparatus. An object of another embodiment of the present invention is to provide a highly reliable and high-resolution display apparatus.

[0014] Note that the description of these objects does not preclude the existence of other objects. One embodiment of the present invention does not necessarily achieve all of these objects. Other objects can be derived from the description of the specification, the drawings, and the claims.

[0015] One embodiment of the present invention is a light-emitting device including a first electrode, a second electrode, and an organic compound layer. The organic compound layer includes a light-emitting layer and an electron-injection layer. The electron-injection layer includes a metal, a first organic compound, and a second organic compound. The first organic compound has a phenanthroline ring with an electron-donating group. The second organic compound includes a T-electron deficient heteroaromatic ring. The first organic compound and the metal form a donor level by interaction and function as an electron donor with respect to the second organic compound.

[0016] One embodiment of the present invention is a light-emitting device including a first electrode, a second electrode, and an organic compound layer. The organic compound layer includes a light-emitting layer and an electron-injection layer. The electron-injection layer includes a metal, a first organic compound, and a second organic compound. The first organic compound has a phenanthroline ring with an electron-donating group. The second organic compound includes a n-electron deficient heteroaromatic ring. The LUMO level of the second organic compound is lower than that of the first organic compound.

[0017] One embodiment of the present invention is a light-emitting device being formed over a first insulating layer and including a first electrode, a second electrode, and an organic compound

layer. The first electrode is formed in contact with the first insulating layer. The organic compound layer is positioned between the first electrode and the second electrode. The second electrode and the organic compound layer are separated from at least one of other light-emitting devices adjacent to the light-emitting device. When seen from a direction substantially perpendicular to a surface of the first insulating layer where the first electrode is formed, an outline of the second electrode and an outline of the organic compound layer are substantially aligned with each other. The organic compound layer includes a light-emitting layer and an electron-injection layer. The electron-injection layer includes a mixed layer including a metal, a first organic compound, and a second organic compound. The first organic compound has a phenanthroline ring with an electron-donating group. The second organic compound includes a T-electron deficient heteroaromatic ring.
[0018] Another embodiment of the present invention is a light-emitting device being formed over a first insulating layer and including a first electrode, a second electrode, and an organic compound layer. The first electrode is formed in contact with the first insulating layer. The organic compound layer is positioned between the first electrode and the second electrode. The second electrode and the organic compound layer are separated from at least one of other light-emitting devices adjacent to the light-emitting device. When seen from a direction substantially perpendicular to a surface of the first insulating layer where the first electrode is formed, an outline of the second electrode and an outline of the organic compound layer are substantially aligned with each other. The organic compound layer includes a light-emitting layer and an electron-injection layer. The electron-injection layer includes a mixed layer including a metal, a first organic compound, and a second organic compound. The first organic compound includes a phenanthroline ring. The minimum value of an electrostatic potential of the first organic compound is smaller than or equal to −0.085 En when a threshold value of electron density distribution in atomic unit is 0.0004 e/a.sub.0.sup.3. The second organic compound includes a π-electron deficient heteroaromatic ring.
[0019] Another embodiment of the present invention is the light-emitting device with any of the above structures, in which the phenanthroline ring of the first organic compound includes an electron-donating group.
[0020] Another embodiment of the present invention is the light-emitting device with any of the above structures, in which the phenanthroline ring is a 1,10-phenanthroline ring and the electron-donating group is bonded to at least one of a 4-position and a 7-position of the 1,10-phenanthroline ring.
[0021] Another embodiment of the present invention is a light-emitting device with any of the above structures, in which the electron-donating group is at least one of an alkyl group, an alkoxy group, an aryloxy group, an alkylamino group, an arylamino group, and a heterocyclic amino group.
[0022] Another embodiment of the present invention is the light-emitting device with any of the above structures, in which the acid dissociation constant pK.sub.a of the first organic compound is greater than or equal to 8.
[0023] Another embodiment of the present invention is the light-emitting device with any of the above structures, in which a spin density of the electron-injection layer measured by an electron spin resonance method is higher than or equal to 5×10.sup.16 spins/cm.sup.3.
[0024] Another embodiment of the present invention is the light-emitting device with any of the above structures, in which the spin density of a mixed film including the metal and the first organic compound measured by an electron spin resonance method is lower than or equal to 2×10.sup.16 spins/cm.sup.3, and the spin density of a mixed film including the metal and the second organic compound measured by an electron spin resonance method is lower than or equal to 2×10.sup.16 spins/cm.sup.3.
[0025] Another embodiment of the present invention is the light-emitting device with any of the above structures, in which the second organic compound includes a phenanthroline ring.
[0026] Another embodiment of the present invention is the light-emitting device with any of the above structures, in which the second organic compound includes a phenanthroline ring.

above structures, in which the glass transition temperature of the second organic compound is higher than or equal to 100° C.

[0027] Another embodiment of the present invention is the light-emitting device with any of the above structures, in which a LUMO level of the second organic compound is lower than that of the first organic compound.

[0028] Another embodiment of the present invention is the light-emitting device with any of the above structures, in which the second organic compound has an electron-transport property.

[0029] Another embodiment of the present invention is the light-emitting device with any of the above structures, in which the acid dissociation constant pK.sub.a of the second organic compound is greater than or equal to 4 and less than 8.

[0030] Another embodiment of the present invention is the light-emitting device with any of the above structures, in which the LUMO level of the second organic compound is higher than or equal to −3.0 eV and lower than or equal to −2.0 eV.

[0031] Another embodiment of the present invention is the light-emitting device with any of the above structures, in which the metal belongs to Group 1, Group 3, Group 11, or Group 13 in a periodic table.

[0032] Another embodiment of the present invention is the light-emitting device with any of the above structures, in which the metal is a typical metal.

[0033] Another embodiment of the present invention is the light-emitting device with any of the above structures, in which the metal is a transition metal.

[0034] Another embodiment of the present invention is the light-emitting device with any of the above structures, in which the organic compound layer includes a p-type layer between the electron-injection layer and the second electrode. The p-type layer includes a third organic compound and a fourth organic compound or a metal oxide. The third organic compound has a hole-transport property. The fourth organic compound includes at least one of a halogen group and a cyano group.

[0035] Another embodiment of the present invention is a light-emitting apparatus including a plurality of light-emitting devices, each of which is any of the light-emitting devices described above. Each of the plurality of light-emitting devices includes, between the first electrode and the second electrode, the organic compound layer including the light-emitting layer and the electron-injection layer. The second electrode and the organic compound layer included in each of the plurality of light-emitting devices are independent between the plurality of light-emitting devices.

[0036] Another embodiment of the present invention is a display apparatus including the light-emitting device with any of the above structures and a transistor or a substrate.

[0037] Another embodiment of the present invention is a display module including any of the above light-emitting devices and at least one of a connector and an integrated circuit.

[0038] Another embodiment of the present invention is an electronic device including any of the above light-emitting devices and at least one of a housing, a battery, a camera, a speaker, and a microphone.

[0039] With one embodiment of the present invention, a novel light-emitting device can be provided. With another embodiment of the present invention, a highly efficient light-emitting device can be provided. With another embodiment of the present invention, a highly reliable light-emitting device can be provided. With another embodiment of the present invention, a highly efficient and highly reliable light-emitting device can be provided.

[0040] With another embodiment of the present invention, a novel light-emitting device manufactured through a photolithography process can be provided. With another embodiment of the present invention, a highly efficient light-emitting device manufactured through a photolithography process can be provided. With another embodiment of the present invention, a highly reliable light-emitting device manufactured through a photolithography process can be provided. With another embodiment of the present invention, a high-emission-efficiency and highly

reliable light-emitting device manufactured through a photolithography process can be provided.
[0041] With another embodiment of the present invention, a novel light-emitting device that can be used in a high-resolution display apparatus can be provided. With another embodiment of the present invention, a highly efficient light-emitting device that can be used in a high-resolution display apparatus can be provided. With another embodiment of the present invention, a highly reliable light-emitting device that can be used in a high-resolution display apparatus can be provided. With another embodiment of the present invention, a high-emission-efficiency and highly reliable light-emitting device that can be used in a high-resolution display apparatus can be provided.
[0042] With another embodiment of the present invention, a highly reliable display apparatus can be provided. With another embodiment of the present invention, a high-resolution display apparatus can be provided. With another embodiment of the present invention, a highly reliable and high-resolution display apparatus can be provided.
[0043] With another embodiment of the present invention, a novel organic compound, a novel light-emitting device, a novel display apparatus, a novel display module, and a novel electronic device can be provided.
[0044] Note that the description of these effects does not preclude the existence of other effects. One embodiment of the present invention does not necessarily have all of these effects. Other effects can be derived from the description of the specification, the drawings, and the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0045] In the accompanying drawings:
[0046] FIGS. **1**A to **1**C illustrate light-emitting devices;
[0047] FIGS. **2**A to **2**C show results of analyzing spin density distribution in composite materials in a ground state;
[0048] FIGS. **3**A and **3**B show results of analyzing electrostatic potential maps of organic compounds in a ground state;
[0049] FIGS. **4**A to **4**C show results of analyzing electrostatic potential maps of composite materials in a ground state;
[0050] FIGS. **5**A and **5**B illustrate light-emitting devices;
[0051] FIGS. **6**A and **6**B are a top view and a cross-sectional view of a display apparatus;
[0052] FIGS. **7**A to **7**G illustrate pixel layout examples;
[0053] FIGS. **8**A to **8**E are cross-sectional views illustrating an example of a method for manufacturing a display apparatus;
[0054] FIGS. **9**A and **9**B are cross-sectional views illustrating the example of the method for manufacturing the display apparatus;
[0055] FIGS. **10**A to **10**D are cross-sectional views illustrating the example of the method for manufacturing the display apparatus;
[0056] FIGS. **11**A to **11**C are cross-sectional views illustrating the example of the method for manufacturing the display apparatus;
[0057] FIGS. **12**A to **12**C are cross-sectional views illustrating the example of the method for manufacturing the display apparatus;
[0058] FIGS. **13**A and **13**B are cross-sectional views illustrating the example of the method for manufacturing the display apparatus;
[0059] FIGS. **14**A and **14**B are perspective views illustrating a structure example of a display module;
[0060] FIGS. **15**A and **15**B are cross-sectional views illustrating structure examples of a display

apparatus;
[0061] FIG. **16** is a perspective view illustrating a structure example of a display apparatus;
[0062] FIG. **17** is a cross-sectional view illustrating a structure example of a display apparatus;
[0063] FIG. **18** is a cross-sectional view illustrating a structure example of a display apparatus;
[0064] FIGS. **19**A to **19**C are a cross-sectional view and top views illustrating a structure example of a display apparatus;
[0065] FIG. **20** is a cross-sectional view illustrating a structure example of a display apparatus;
[0066] FIGS. **21**A to **21**C are a cross-sectional view and top views illustrating a structure example of a display apparatus;
[0067] FIGS. **22**A to **22**D illustrate examples of electronic devices;
[0068] FIGS. **23**A to **23**F illustrate examples of electronic devices;
[0069] FIGS. **24**A to **24**G illustrate examples of electronic devices;
[0070] FIG. **25** shows an ESR spectrum of a sample 1;
[0071] FIG. **26** shows an ESR spectrum of a sample 2;
[0072] FIG. **27** shows an ESR spectrum of a comparative sample 3;
[0073] FIG. **28** shows an ESR spectrum of a comparative sample 4;
[0074] FIG. **29** shows an ESR spectrum of a comparative sample 5;
[0075] FIG. **30** shows an ESR spectrum of a comparative sample 6;
[0076] FIG. **31** shows an ESR spectrum of a comparative sample 7;
[0077] FIG. **32** shows an ESR spectrum of a comparative sample 8;
[0078] FIG. **33** shows an ESR spectrum of a comparative sample 9;
[0079] FIG. **34** shows an ESR spectrum of a thin film obtained by co-evaporation of PCBBiF and OCHD-003;
[0080] FIG. **35** shows the luminance-current density characteristics of a light-emitting device **1** and a reference light-emitting device **2**;
[0081] FIG. **36** shows the luminance-voltage characteristics of the light-emitting device **1** and the reference light-emitting device **2**;
[0082] FIG. **37** shows the current efficiency-current density characteristics of the light-emitting device **1** and the reference light-emitting device **2**;
[0083] FIG. **38** shows the current density-voltage characteristics of the light-emitting device **1** and the reference light-emitting device **2**, and
[0084] FIG. **39** shows the electroluminescence spectra of the light-emitting device **1** and the reference light-emitting device **2**.

DETAILED DESCRIPTION OF THE INVENTION

[0085] Embodiments will be described in detail with reference to the drawings. Note that the present invention is not limited to the following description, and it will be readily appreciated by those skilled in the art that modes and details of the present invention can be modified in various ways without departing from the spirit and scope of the present invention. Therefore, the present invention should not be construed as being limited to the description in the following embodiments.
[0086] In this specification and the like, a device manufactured using a metal mask or a fine metal mask (FMM) is sometimes referred to as a device having a metal mask (MM) structure. In this specification and the like, a device manufactured without using a metal mask or an FMM is sometimes referred to as a device having a metal maskless (MML) structure.

Embodiment 1

[0087] As a method for forming an organic semiconductor film in a predetermined shape, a vacuum evaporation method with a metal mask (mask vapor deposition) is widely used. However, in these days of higher density and higher resolution, mask vapor deposition has come close to the limit of increasing the resolution for various reasons such as the alignment accuracy and the distance between the mask and the substrate. An organic semiconductor device having a finer pattern is expected to be achieved by shape processing of an organic semiconductor film by a photolithography method. Moreover, since a photolithography method achieves large-area processing more easily than mask vapor deposition, processing of an organic semiconductor film by the photolithography method is being researched.

[0088] It has been known that an organic compound layer and a cathode of an organic EL device exposed to atmospheric components such as water and oxygen have affected initial characteristics or reliability, and thus it has been common knowledge that the organic compound layer and the cathode are treated in an inert gas atmosphere or a near-vacuum atmosphere.

[0089] In particular, an electron-injection layer, which sometimes includes an alkali metal, an alkaline earth metal, or a compound thereof (hereinafter also referred to as a Li compound or the like) highly reactive with water or oxygen, easily deteriorates and has a greatly decreased electron-injection property when exposed to the air. In addition, also in the case where another metal with a small work function is used for the cathode, the electron-injection layer might have a decreased electron-injection property and significantly increase the driving voltage when exposed to water, oxygen, or the like.

[0090] Processing steps by the aforementioned photolithography method inevitably expose a light-emitting device in the middle of manufacturing to the air. Furthermore, the photolithography process uses a variety of chemical solutions and includes a cleaning step, which is a severe condition that further promotes deterioration.

[0091] Thus, the processing by the photolithography method performed for the cathode and the organic compound layer might significantly decrease the electron-injection properties of the cathode and the electron-injection layer. As a result, an organic EL device manufactured through processing by a photolithography method has greatly increased driving voltage and is hard to obtain favorable characteristics.

[0092] In a method for avoiding such characteristics degradation, the processing steps by the photolithography method are performed before formation of the electron-injection layer and the cathode. Meanwhile, when a film to be the electron-injection layer and a film to be the cathode are formed and then the films are processed by the photolithography method, an increase in the number of steps in the processing by a photolithography method can be minimized. In addition, the opportunities of exposing the organic compound layer to the chemical solution and the air can be significantly reduced, which enables performance comparable to that of a light-emitting device manufactured not through exposure to the air.

[0093] In view of the above, one embodiment of the present invention provides a light-emitting device including a first electrode, an organic compound layer, and a second electrode from the substrate side, in which the organic compound layer and the second electrode are formed by processing an organic compound film to be the organic compound layer and a conductive film to be the second electrode by a photolithography method.

[0094] FIG. **1**A is a schematic view of a light-emitting device of one embodiment of the present invention. The light-emitting device includes a first electrode **101** over an insulating layer **1000**, and an organic compound layer (also referred to as EL layer) **103** between the first electrode **101** and a second electrode **102**. The organic compound layer **103** includes at least a light-emitting layer **113** and an electron-injection layer **115**. The light-emitting layer **113** contains a light-emitting substance and emits light when voltage is applied between the first electrode **101** and the second electrode **102**. Note that, in the light-emitting device illustrated in FIG. **1**A, the first electrode **101** serves as an anode and the second electrode **102** serves as a cathode.

[0095] The organic compound layer **103** preferably includes, besides the light-emitting layer **113** and the electron-injection layer **115**, functional layers such as a hole-injection layer **111**, a hole-transport layer **112**, and an electron-transport layer **114**, as illustrated in FIG. **1**A. The organic compound layer **103** may include functional layers other than the above functional layers, such as a hole-blocking layer, an exciton-blocking layer, and an intermediate layer. Alternatively, any of the above-described layers may be omitted.

[0096] In the light-emitting device of one embodiment of the present invention, the organic compound layer **103** and the second electrode **102** are formed in the following manner: an organic compound film to be the organic compound layer **103** and a conductive film to be the second electrode **102** are formed, and then these films are processed by a photolithography method. Thus, in a cross-sectional view, an end portion of the second electrode **102** and an end portion of the organic compound layer **103** are aligned with each other in a direction substantially perpendicular to a surface of the insulating layer **1000**, as illustrated in FIG. **1**A.
[0097] As described above, if high reactivity of an alkali metal compound or the like used for an electron-injection layer is one factor of deterioration due to high reaction in an air exposure step, an etching step, a cleaning step, or the like, the use of a substance with low reactivity instead of the alkali metal compound or the like for the electron-injection layer probably inhibits an increase in driving voltage even when the light-emitting device is manufactured through processing by a photolithography method.
[0098] Thus, in the light-emitting device of one embodiment of the present invention, the electron-injection layer **115** is formed using a composite material formed by a combination of a metal, a first organic compound having an electron-donating property and unshared electron pairs, and a second organic compound having an electron-transport property. The metal and the first organic compound form a donor level (a singly occupied molecular orbital (SOMO) level or a highest occupied molecular orbital (HOMO) level) by interaction and function as an electron donor with respect to the second organic compound. Accordingly, the electron-injection layer **115** can be formed to have a favorable electron-injection property and resistance to oxygen and water in the air and water and a chemical solution used during the processing step by a photolithography method; thus, the light-emitting device can have a reduced driving voltage and high emission efficiency.
[0099] Since a metal with a low work function typified by an alkali metal and an alkaline earth metal and a compound of the metal with a low work function are highly reactive with oxygen and water, using the metal or the compound for a light-emitting device manufactured through processing by a lithography method may cause a reduction in emission efficiency, an increase in driving voltage, a reduction in driving lifetime, generation of a non-emission region at an end portion of a light-emitting portion, or the like, leading to degradation in the characteristics or a reduction in the reliability of the light-emitting device. However, in one embodiment of the present invention, even when an alkali metal, an alkaline earth metal, or a compound thereof is used, interaction between the alkali metal, the alkaline earth metal, or the compound, the first organic compound having an electron-donating property and unshared electron pairs, and the second organic compound having an electron-transport property stabilizes the composite material of the metal, the first organic compound, and the second organic compound, so that the electron-injection layer **115** having resistance to oxygen and water in the air and water and a chemical solution used during the process by a lithography method can be formed. When an alkali metal, an alkaline earth metal, or a compound thereof is used as the metal in the light-emitting device of one embodiment of the present invention, the donor level (SOMO level or HOMO level) that is formed by interaction between the alkali metal, the alkaline earth metal, or the compound and the first organic compound having an electron-donating property and unshared electron pairs can be a high energy level, facilitating electron donation to the second organic compound having an electron-transport property. This is preferable because a barrier against electron injection from the second electrode **102** to the organic compound layer **103** can be reduced and electrons from the second electrode **102** can be smoothly injected and transported to the light-emitting layer **113** side.
[0100] As the metal in the light-emitting device of one embodiment of the present invention, it is also possible to use any one of metal elements belonging to Group 3 to Group 11 and metal elements belonging to Group 12 to Group 14. These metals have low reactivity with oxygen and water in the air and water and a chemical solution used in a lithography process. Thus, using any of these metals in the light-emitting device is advantageous in that the metals hardly cause

deterioration due to water and oxygen, which would be a matter of concern in the case of using a metal with a low work function. On the other hand, metal elements belonging to Group 3 to Group 11 and metal elements belonging to Group 12 to Group 14, which are stable and have a low electron-injection property, cause the light-emitting device to have reduced emission efficiency, an increased driving voltage, and a reduced driving lifetime, for example. However, in one embodiment of the present invention, even when any one of metal elements belonging to Group 3 to Group 11 and metal elements belonging to Group 12 to Group 14 is used, a donor level (SOMO level or HOMO level) is formed by interaction between the metal and the first organic compound having an electron-donating property and unshared electron pairs, and electrons can be easily donated to the second organic compound having an electron-transport property. Thus, a barrier against electron injection from the second electrode to the organic compound layer **103** can be reduced and electrons from the second electrode **102** can be smoothly injected and transported to the light-emitting layer **113** side. The above structure is preferably employed, in which case the electron-injection layer **115** having resistance to oxygen and water in the air and water and a chemical solution used in the process by a lithography method can be formed. Thus, one embodiment of the present invention can provide a light-emitting device having high moisture resistance, high water resistance, high oxygen resistance, high chemical resistance, a low driving voltage, and high emission efficiency.

[0101] In the interaction between the metal and the first organic compound having an electron-donating property and unshared electron pairs, the sum of the number of electrons of the compound and the number of electrons of the metal is preferably an odd number, in which case the stabilization energy is lower and a donor level (SOMO level or HOMO level) can be a high energy level. Accordingly, in the case where the number of electrons of the compound is an even number, the metal preferably belongs to an odd-numbered group in the periodic table.

<Estimation of Spin Density and Electrostatic Potential in Interaction Between Metal and Organic Compound by Quantum Chemical Calculation>

[0102] The spin density and the electrostatic potential (ESP) at the time of interaction between a metal, the first organic compound having an electron-donating property and unshared electron pairs, and the second organic compound having an electron-transport property were analyzed by quantum chemical calculation. Note that 4,7-di-1-pyrrolidinyl-1,10-phenanthroline (abbreviation: Pyrrd-Phen), 2,9-di(naphthalen-2-yl)-4,7-diphenyl-1,10-phenanthroline (abbreviation: NBPhen), and silver (Ag) were used as the first organic compound, the second organic compound, and the metal, respectively, in the calculation.

[0103] As the quantum chemistry computational program, Gaussian 09 was used. The calculation was performed using SGI 8600 (produced by Hewlett Packard Enterprise (HPE)). The most stable structures of the first organic compound alone in a ground state, the second organic compound alone in a ground state, a composite material of the first organic compound and the metal in a ground state, a composite material of the second organic compound and the metal in a ground state, and a composite material of the first organic compound, the second organic compound, and the metal in a ground state were calculated by the density functional theory (DFT). As basis functions, 6-311G(d,p) and LanL2DZ were used, and as a functional, B3LYP was used. In the DFT, the total energy is represented as the sum of potential energy, electrostatic energy between electrons, electronic kinetic energy, and exchange-correlation energy including all the complicated interactions between electrons. Also in the DFT, exchange-correlation interaction is approximated by a functional (a function of another function) of one electron potential represented in terms of electron density to enable highly accurate calculations.

[0104] FIGS. **2**A, **2**B, and **2**C respectively show the analysis results of spin density distribution in the composite material of the first organic compound (Pyrrd-Phen) and the metal (Ag) in a ground state, the composite material of the second organic compound (NBPhen) and the metal (Ag) in a ground state, and the composite material of the first organic compound (Pyrrd-Phen), the second

organic compound (NBPhen), and the metal (Ag) in a ground state. In the diagrams, spheres represent atoms included in the compounds, and clouds around some of the atoms represent spin density distribution at the time when the threshold value of electron density distribution in atomic units is 0.003 e/a.sub.0.sup.3 (where e represents elementary charge (1 e=1.60218×10.sup.−19 C) and a.sub.0 represents a Bohr radius (1 a.sub.0=5.29177×10.sup.−11 m)). The clouds represent localization of the doublet ground state of the compounds. Note that no spin density distribution is observed in the first organic compound (Pyrrd-Phen) in a ground state and the second organic compound (NBPhen) in a ground state because the ground states of the first organic compound and the second organic compound are singlet ground states.
[0105] In the composite material of the first organic compound (Pyrrd-Phen) and the metal (Ag) in the doublet ground state, the first organic compound (Pyrrd-Phen) interacts with the metal (Ag), and the metal (Ag) is coordinated to the two nitrogen atoms having unshared electron pairs (the nitrogen atoms (N) at the 1- and 10-positions) in the 1,10-phenanthroline ring of the first organic compound (Pyrrd-Phen), which leads to stabilization and the formation of the composite material. As shown in FIG. **2**A, some spins attributed to an unpaired electron of the metal (Ag) are accordingly distributed over part of the 1,10-phenanthroline ring of the first organic compound (Pyrrd-Phen), particularly the two nitrogen atoms having unshared electron pairs (the nitrogen atoms (N) at the 1- and 10-positions). However, the interaction is weak, and thus, most spins are distributed over the metal (Ag).
[0106] In the composite material of the second organic compound (NBPhen) and the metal (Ag) in the doublet ground state, the second organic compound (NBPhen) interacts with the metal (Ag), and the metal (Ag) is coordinated to the two nitrogen atoms having unshared electron pairs (the nitrogen atoms (N) at the 1- and 10-positions) in the 1,10-phenanthroline ring of the second organic compound (NBPhen), which leads to stabilization and the formation of the composite material.
[0107] As shown in FIG. **2**B, some spins attributed to an unpaired electron of the metal (Ag) are accordingly distributed over part of the 1,10-phenanthroline ring of the second organic compound (NBPhen), particularly the two nitrogen atoms having unshared electron pairs (the nitrogen atoms (N) at the 1- and 10-positions). However, the interaction is weak, and thus, most spins are distributed over the metal (Ag).
[0108] Meanwhile, in the composite material of the first organic compound (Pyrrd-Phen), the second organic compound (NBPhen), and the metal (Ag) in the doublet ground state, the first organic compound (Pyrrd-Phen), the second organic compound (NBPhen), and the metal (Ag) interact with one another, and the metal (Ag) is coordinated to the two nitrogen atoms having unshared electron pairs (the nitrogen atoms (N) at the 1- and 10-positions) in the 1,10-phenanthroline ring of the first organic compound (Pyrrd-Phen) and the two nitrogen atoms having unshared electron pairs (the nitrogen atoms (N) at the 1- and 10-positions) in the 1,10-phenanthroline ring of the second organic compound (NBPhen), which leads to stabilization and the formation of the composite material. As shown in FIG. **2**C, spins attributed to an unpaired electron of the metal (Ag) are accordingly localized in the second organic compound (NBPhen). Furthermore, no spin density distribution is observed in the metal (Ag). It is thus found that the second organic compound (NBPhen) is in a radical anion state owing to the interaction between the first organic compound (Pyrrd-Phen), the second organic compound (NBPhen), and the metal (Ag).
[0109] Next, FIG. **3**A, FIG. **3**B, FIG. **4**A, FIG. **4**B, and FIG. **4**C respectively show the analysis results of the electrostatic potential maps of the first organic compound (Pyrrd-Phen) in a ground state, the second organic compound (NBPhen) in a ground state, the composite material of the first organic compound (Pyrrd-Phen) and the metal (Ag) in a ground state, the composite material of the second organic compound (NBPhen) and the metal (Ag) in a ground state, and the composite material of the first organic compound (Pyrrd-Phen), the second organic compound (NBPhen), and the metal (Ag) in a ground state. In the diagrams, spheres represent atoms included in the compounds, and clouds around some of the atoms represent ESPs in electron density distribution at

the time when the threshold value of electron density distribution in atomic units is 0.003 e/a.sub.0.sup.3. ESP is the energy of interaction between positive point charge with unit quantity of electricity and electron distribution of a molecule, and an electrostatic potential map denotes ESP on an electron density isosurface in colors. In an electrostatic potential map, a region with a negative ESP is denoted in red, a region with a positive ESP is denoted in blue, an atom in the region with a negative ESP has negative charge, and an atom in the region with a positive ESP has positive charge. To show a region with a negative ESP and a region with a positive ESP in FIGS. **3**A and **3**B and FIGS. **4**A to **4**C, which are obtained by gray-scale conversion of electrostatic potential maps produced through analysis, a deep red portion (i.e., the region with a negative ESP) is surrounded by a thick dotted line, and a deep blue portion (i.e., the region with a positive ESP) is surrounded by a thin dashed-dotted line.
[0110] As shown in FIG. **3**A, the ESP around the two nitrogen atoms having unshared electron pairs (the nitrogen atoms (N) at the 1- and 10-positions) in the 1,10-phenanthroline ring of the first organic compound (Pyrrd-Phen) in the singlet ground state is negative. The two nitrogen atoms each had a negative Mulliken partial charge of −0.29 e in atomic units. Accordingly, it is found that the two nitrogen atoms have negative partial charge.
[0111] As shown in FIG. **3**B, the ESP around the two nitrogen atoms having unshared electron pairs (the nitrogen atoms (N) at the 1- and 10-positions) in the 1,10-phenanthroline ring of the second organic compound (NBPhen) in the singlet ground state is negative. The two nitrogen atoms each had a negative Mulliken partial charge of −0.34 e in atomic units. Accordingly, it is found that the two nitrogen atoms have negative partial charge.
[0112] In the composite material of the first organic compound (Pyrrd-Phen) and the metal (Ag) in the doublet ground state, the first organic compound (Pyrrd-Phen) interacts with the metal (Ag), and the metal (Ag) is coordinated to the two nitrogen atoms having unshared electron pairs (the nitrogen atoms (N) at the 1- and 10-positions) in the 1,10-phenanthroline ring of the first organic compound (Pyrrd-Phen), which leads to stabilization and the formation of the composite material. As shown in FIG. **4**A, the ESP around the two nitrogen atoms having unshared electron pairs (the nitrogen atoms (N) at the 1- and 10-positions) in the 1,10-phenanthroline ring of the first organic compound (Pyrrd-Phen) and the metal (Ag) is accordingly negative. The two nitrogen atoms each had a negative Mulliken partial charge of −0.37 e in atomic units and the metal (Ag) had a negative Mulliken partial charge of −0.18 e in atomic units. Accordingly, it is found that the two nitrogen atoms and the Ag atom have negative partial charge.
[0113] In the composite material of the second organic compound (NBPhen) and the metal (Ag) in the doublet ground state, the second organic compound (NBPhen) interacts with the metal (Ag), and the metal (Ag) is coordinated to the two nitrogen atoms having unshared electron pairs (the nitrogen atoms (N) at the 1- and 10-positions) in the 1,10-phenanthroline ring of the second organic compound (NBPhen), which leads to stabilization and the formation of the composite material. As shown in FIG. **4**B, the ESP around the two nitrogen atoms having unshared electron pairs (the nitrogen atoms (N) at the 1- and 10-positions) in the 1,10-phenanthroline ring of the second organic compound (NBPhen) and the metal (Ag) is accordingly negative. The two nitrogen atoms had negative Mulliken partial charges of −0.45 e and −0.39 e in atomic units and the metal (Ag) had a negative Mulliken partial charge of −0.06 e in atomic units. Accordingly, it is found that the two nitrogen atoms and the Ag atom have negative partial charge.
[0114] Meanwhile, in the composite material of the first organic compound (Pyrrd-Phen), the second organic compound (NBPhen), and the metal (Ag), which is one embodiment of the present invention, in the doublet ground state, the first organic compound (Pyrrd-Phen), the second organic compound (NBPhen), and the metal (Ag) interact with one another, and the metal (Ag) is coordinated to the two nitrogen atoms having unshared electron pairs (the nitrogen atoms (N) at the 1- and 10-positions) in the 1,10-phenanthroline ring of the first organic compound (Pyrrd-Phen) and the two nitrogen atoms having unshared electron pairs (the nitrogen atoms (N) at the 1- and 10-

positions) in the 1,10-phenanthroline ring of the second organic compound (NBPhen), which leads to stabilization and the formation of the composite material. Accordingly, as shown in FIG. **4**C, a positive ESP is mainly distributed over the metal (Ag) and the first organic compound (Pyrrd-Phen), and a negative ESP is mainly distributed over the second organic compound (NBPhen). It is also found that the ESP around the two nitrogen atoms having unshared electron pairs (the nitrogen atoms (N) at the 1- and 10-positions) in the 1,10-phenanthroline ring of the second organic compound (NBPhen) is negative, whereas the ESP around the metal (Ag) is positive. The two nitrogen atoms each had a negative Mulliken partial charge of −0.52 e in atomic units, whereas the metal (Ag) had a positive Mulliken partial charge of 0.39 e in atomic units. These results show that charge of the Ag atom is distributed over the two nitrogen atoms.

[0115] From the above, it is found that the combination of the metal, the first organic compound having an electron-donating property and unshared electron pairs, and the second organic compound having an electron-transport property is such that the first organic compound and the metal form a donor level by interaction and function as an electron donor with respect to the second organic compound. In one embodiment of the present invention, the electron-injection layer **115** formed using a composite material of such a combination can have a high electron-injection property and resistance to oxygen and water in the air and water and a chemical solution used in a lithography process; thus, the light-emitting device can have a reduced driving voltage and high emission efficiency.

<Estimation of SOMO or HOMO Level in Interaction Between Metal and Organic Compound by Quantum Chemical Calculation>

[0116] Next, stabilization energy at the time of interaction between a metal, the first organic compound having an electron-donating property and unshared electron pairs, and the second organic compound having an electron-transport property and a donor level (SOMO level or HOMO level) formed at the time of the interaction were estimated by quantum chemical calculation.

[0117] As the quantum chemistry computational program, Gaussian 09 was used. The calculation was performed using SGI 8600 produced by HPE. First, the most stable structures of the first organic compound in a ground state, the second organic compound in a ground state, the metal in a ground state, the composite material of the first organic compound and the metal in a ground state, the composite material of the second organic compound and the metal in a ground state, and the composite material of the first organic compound, the second organic compound, and the metal in a ground state were calculated by the density functional theory (DFT). As basis functions, 6-311G(d,p) and LanL2DZ were used, and as a functional, B3LYP was used. Next, the stabilization energy was calculated by subtracting the sum of the total energy of the organic compound(s) alone and the total energy of the metal alone from the total energy of the composite material of the organic compound(s) and the metal. That is, (stabilization energy)=(the total energy of the composite material of the organic compound(s) and the metal)−(the total energy of the organic compound(s) alone)−(the total energy of the metal alone).

[0118] The stabilization energy of the composite material of the first organic compound and the metal, the composite material of the second organic compound and the metal, and the composite material of the first organic compound, the second organic compound, and the metal and the donor level (HOMO level or SOMO level) of the first organic compound, the second organic compound, the composite material of the first organic compound and the metal, the composite material of the second organic compound and the metal, and the composite material of the first organic compound, the second organic compound, and the metal were calculated. The following tables show the calculation results. Note that the donor levels (HOMO levels or SOMO levels) in the tables are calculated values and may be different from measured values.

[0119] First, the results of the calculation performed using 4,7-di-1-pyrrolidinyl-1,10-phenanthroline (abbreviation: Pyrrd-Phen) as the first organic compound, 2,2′-(1,3-phenylene)bis(9-phenyl-1,10-phenanthroline) (abbreviation: mPPhen2P) as the second organic compound, and zinc (Zn) as the metal are shown in the table below.

TABLE-US-00001 TABLE 1 Stabilization HOMO energy level (eV) (eV) First organic compound (Pyrrd-Phen) — −5.65 Second organic compound (mPPhen2P) — −5.88 Composite material of first organic compound and −0.0030 −4.48 metal (Pyrrd-Phen + Zn) Composite material of second organic compound −0.0012 −5.85 and metal (mPPhen2P + Zn) Composite material of first organic compound, −0.92 −2.43 second organic compound, and metal (Pyrrd-Phen + mPPhen2P + Zn)

[0120] The above table shows that the stabilization energy of the composite material of the metal (Zn) and the first organic compound (Pyrrd-Phen) and that of the composite material of the metal (Zn) and the second organic compound (mPPhen2P) each have a negative value. It is also shown that a mixture of the first or second organic compound and the metal is more energetically stable than the first or second organic compound alone owing to the interaction between the organic compound and the metal, and the difference in the stabilization energy between the mixture and the organic compound alone is small. There is a small difference between the HOMO level formed at the time of the interaction and the HOMO level of the first organic compound (Pyrrd-Phen) or the second organic compound (mPPhen2P), indicating weak interaction between the metal and each of the first organic compound and the second organic compound.

[0121] Meanwhile, the stabilization energy of the composite material of the metal (Zn), the first organic compound (Pyrrd-Phen), and the second organic compound (mPPhen2P), which is preferably used for the electron-injection layer **115** of the light-emitting device of one embodiment of the present invention, is lower than the stabilization energy of the composite material of the metal (Zn) and the first organic compound (Pyrrd-Phen) and the stabilization energy of the composite material of the metal (Zn) and the second organic compound (mPPhen2P), indicating higher energetic stability of the composite material of the metal (Zn), the first organic compound (Pyrrd-Phen), and the second organic compound (mPPhen2P). Thus, the stabilization energy of the composite material of the metal, the first organic compound, and the second organic compound is preferably lower than or equal to −0.50 eV, further preferably lower than or equal to −1.0 eV, still further preferably lower than or equal to −2.0 eV, yet still further preferably lower than or equal to −3.0 eV, yet still further preferably lower than or equal to −4.0 eV. The HOMO level formed here is higher than the HOMO level of each of the first organic compound (Pyrrd-Phen) and the second organic compound (mPPhen2P). The HOMO level is preferably high to achieve a high electron-injection property.

[0122] Next, the results of calculation performed using Pyrrd-Phen as the first organic compound, mPPhen2P as the second organic compound, and calcium (Ca) or magnesium (Mg) as the metal are shown in the table below.

TABLE-US-00002 TABLE 2 Stabilization HOMO Composite material of first organic compound, energy level second organic compound, and metal (eV) (eV) Pyrrd-Phen + mPPhen2P + Ca −3.3 −2.40 Pyrrd-Phen + mPPhen2P + Mg −2.4 −2.43

[0123] The metal is preferably an alkaline earth metal (Ca or Mg), in which case the stabilization energy of the composite material of the metal, the first organic compound, and the second organic compound is lower than or equal to −2.0 eV as shown in the above table and the energetic stability is higher. The HOMO level formed here is higher than the HOMO level of each of the first organic compound and the second organic compound. The HOMO level is preferably high to achieve a high electron-injection property.

[0124] Next, the table below shows the results of calculation performed using Pyrrd-Phen as the first organic compound, mPPhen2P as the second organic compound, and a metal belonging to an odd-numbered group (Group 1, 3, 5, 7, 9, 11, or 13), specifically lithium (Li), aluminum (Al), silver (Ag), copper (Cu), or indium (In).

TABLE-US-00003 TABLE 3 Stabilization SOMO Composite material of first organic compound, energy level second organic compound, and metal (eV) (eV) Pyrrd-Phen + mPPhen2P + Li −3.7 −2.32 Pyrrd-Phen + mPPhen2P + Al −4.1 −2.82 Pyrrd-Phen + mPPhen2P + Ag −1.2 −2.35 Pyrrd-

Phen + mPPhen2P + Cu −4.0 −2.39 Pyrrd-Phen + mPPhen2P + In −1.1 −2.83

[0125] The metal is preferably a metal belonging to an odd-numbered group, in which case the stabilization energy of the composite material of the metal, the first organic compound, and the second organic compound is lower than or equal to −1.0 eV, lower than or equal to −2.0 eV, lower than or equal to −3.0 eV, or lower than or equal to −4.0 eV as shown in the above table and the energetic stability is higher. The SOMO level formed here is higher than the HOMO level of each of the first organic compound and the second organic compound. The SOMO level is preferably high to achieve a high electron-injection property.

[0126] In a general manufacturing process of a light-emitting device, an organic compound film to be an organic compound layer of the light-emitting device is formed by a vacuum evaporation method in many cases. Thus, it is preferable to use a material that can be easily deposited by vacuum evaporation, i.e., a material with a low melting point. The metal elements belonging to Group 11 and Group 13 have low melting points and thus can be suitably used for vacuum evaporation. The metal elements belonging to Group 11 and Group 13 are preferable because they are stable with respect to oxygen and water in the air. A vacuum evaporation method is preferably used, in which case a metal atom and an organic compound can be easily mixed.

[0127] Furthermore, Ag or In can be used also as a cathode material. Using the same material for the electron-injection layer **115** and the second electrode **102** is preferable, in which case the manufacture of the light-emitting device can become easier. Moreover, the manufacturing cost of the light-emitting device can be reduced.

[0128] Next, details of the structure of the electron-injection layer **115** which can be used for the light-emitting device are described.

<<Structure of Electron-Injection Layer 115>>

[0129] As described above, a composite material of a metal, the first organic compound, and the second organic compound is preferably used for the electron-injection layer **115**. The metal and the first organic compound interact with each other and form a donor level (SOMO level or HOMO level), and the combination of the metal and the first organic compound functions as an electron donor with respect to the second organic compound having an electron-transport property. With the use of a composite material of such a combination for the electron-injection layer **115**, the light emitting device can have a low driving voltage and high emission efficiency.

[0130] More specifically, a mixed layer including a metal, the first organic compound, and the second organic compound is preferably used as the electron-injection layer **115**. The formation of the mixed layer of the metal, the first organic compound, and the second organic compound facilitates interaction between these substances, so that the first organic compound and the metal function as an electron donor with respect to the second organic compound; thus, a barrier against electron injection from the second electrode **102** to the organic layer **103** can be further reduced. This facilitates electron injection into the organic compound layer **103** and accordingly enables the light-emitting device to have a further reduced driving voltage and further increased emission efficiency. The electron-injection layer **115** can be less likely to be crystallized when being the mixed layer of the metal, the first organic compound, and the second organic compound than when having a stacked-layer structure of the metal, the first organic compound, and the second organic compound. Accordingly, the electron-injection layer **115** is less likely to be crystallized even when affected by oxygen or water in the air and a chemical solution or water during processing by a lithography method for forming the organic compound layer **103** and the second electrode **102**. An increase in driving voltage or a reduction in current efficiency of the light-emitting device due to crystallization of the electron-injection layer **115** can be prevented. Thus, in the light-emitting device where the organic compound layer **103** and the second electrode **102** are formed by a lithography method, the composite material of the metal, the first organic compound, and the second organic compound can be used for the electron-injection layer **115** more suitably as a mixed layer than as a stacked-layer structure.

<Metal>
[0131] As the metal, a typical metal or a transition metal can be used.
[0132] As the typical metal, an alkali metal (Group 1 element) such as Li, Na, K, or Cs, an alkaline earth metal (Group 2 element) such as Mg, Ca, or Ba, a Group 12 element such as Zn, an earth metal (Group 13 element) such as Al or In, a Group 14 element such as Sn, or a compound of a Group 1, 2, 13, or 14 element can be used.
[0133] When an alkali metal, an alkaline earth metal, or a compound thereof is used as the metal, the donor level formed by interaction between the alkali metal, the alkaline earth metal, or the compound and the first organic compound can be a high energy level, facilitating electron donation to the second organic compound. This is preferable because electrons from the second electrode **102** can be smoothly injected and transported to the light-emitting layer **113** side, and accordingly the light-emitting device can have a low driving voltage and high emission efficiency.
[0134] As the transition metal, any of Group 3 elements, including Y and lanthanoids such as Eu and Yb, Group 7 elements such as Mn, Group 8 elements such as Fe, Group 9 elements such as Co, Group 10 elements such as Ni and Pt, Group 11 elements such as Cu, Ag, and Au, and a compound of a Group 3, 7, 8, 9, 10, or 11 element can be used. The transition metal is preferable because it has low reactivity with components of the air such as water and oxygen.
[0135] Among the above-described examples, it is further preferable to use a metal belonging to an odd-numbered group (Group 1, Group 3, Group 5, Group 7, Group 9, Group 11, or Group 13). Among transition metals belonging to the odd-numbered groups, a metal having one electron (unpaired electron) in an orbital of the outermost shell is particularly preferable because a combination of this metal and the first organic compound easily forms a SOMO level.
[0136] A metal that has a low melting point and can be deposited by a vacuum evaporation method is preferably used because a mixed layer of this metal and an organic compound is easy to form. Specifically, for example, the metals belonging to Group 11 and Group 13 have low melting points and thus can be suitably used for vacuum evaporation. The metal elements belonging to Group 11 and Group 13 are preferable because they are stable with respect to oxygen and water in the air.
<First Organic Compound>
[0137] As the first organic compound, an organic compound having an electron-donating property and unshared electron pairs can be used. More specifically, as the first organic compound, an organic compound having a phenanthroline ring can be used. Among organic compounds having a phenanthroline ring, an organic compound having a 1,10-phenanthroline ring, the two nitrogen atoms of which can be coordinated to a metal, is particularly preferably used to facilitate interaction with the metal.
[0138] As the first organic compound, an organic compound having a phenanthroline ring with an electron-donating group is further preferably used. Specifically, introducing an electron-donating group to a 1,10-phenanthroline ring can increase the electron density of the phenanthroline ring and the efficiency of the interaction with the metal. Furthermore, an electron-donating group is preferably bonded to at least one of the 4- and 7-positions of the 1,10-phenanthroline ring. Introducing electron-donating groups to the 4- and 7-positions can increase the electron density of the nitrogen atoms at the 1- and 10-positions, which are the para-positions with respect to the 4- and 7-positions. In addition, steric congestion around the nitrogen atoms at the 1- and 10-positions can be inhibited, and the electron density around the nitrogen atoms can be increased. This structure facilitates the interaction with the metal and is thus preferable. In this specification and the like, a phenanthroline ring with an electron-donating group can be referred to as a phenanthroline ring into which an electron-donating group is introduced or a phenanthroline ring to which an electron-donating group is bonded.
[0139] The minimum value of ESP of the first organic compound is preferably small (i.e., the minimum value is preferably a negative value the absolute value of which is large), in which case the efficiency of the interaction with the metal is high. In an organic compound having a

phenanthroline ring, the electrostatic potential around the nitrogen atoms of the phenanthroline ring, which is likely to be negative, can be further lowered (i.e., the absolute value of the negative value can be increased) by introduction of an electron-donating group to the phenanthroline ring. Note that the electrostatic potential is the energy of interaction between positive point charge with unit quantity of electricity and electron distribution of a molecule. An electrostatic potential value also depends on the threshold value of electron density distribution. To increase the efficiency of the interaction with the metal, the minimum value of the electrostatic potential of the first organic compound is preferably smaller (negatively larger) than the minimum value of the electrostatic potential of a phenanthroline ring having no substituent. Specifically, when the threshold value of electron density distribution in atomic units is 0.0004 e/a.sub.0.sup.3, the minimum value of the electrostatic potential is preferably smaller than or equal to −0.085 E.sub.h (E.sub.h is the Hartree energy (1 E.sub.h=27.211 eV)), further preferably smaller than or equal to −0.090 E.sub.h. When the threshold value of electron density distribution is 0.003 e/a.sub.0.sup.3, the minimum value of the electrostatic potential is preferably smaller than or equal to −0.12 E.sub.h, further preferably smaller than or equal to −0.13 E.sub.h.
[0140] The first organic compound is preferably strongly basic, in which case the first organic compound interacts with holes to significantly reduce the hole-transport property in the electron-injection layer **115** and prevent hole transport from the electron-injection layer **115** to the second electrode **102**, enabling high efficiency of the light-emitting device. Specifically, the acid dissociation constant pK.sub.a of the first organic compound is preferably greater than or equal to 8, further preferably greater than or equal to 10, still further preferably greater than or equal to 12.
[0141] Specific examples of the electron-donating group include an alkyl group, an alkoxy group, an aryloxy group, an alkylamino group, an arylamino group, and a heterocyclic amino group. Note that examples of the electron-donating group that is preferably introduced to the phenanthroline ring are not limited to the above examples. The electron-donating group may be any group that can increase the electron density of the phenanthroline ring by being introduced to the phenanthroline ring. The electron-donating group may be introduced to the phenanthroline ring via an arylene group such as a phenylene group, and the arylene group is preferably a p-phenylene group.
[0142] The alkyl group refers to a monovalent group obtained by eliminating one hydrogen atom from an alkane (C.sub.nH.sub.2n+2). Specific examples of the alkyl group include a methyl group, an ethyl group, a propyl group, an isopropyl group, a butyl group, a sec-butyl group, an isobutyl group, a tert-butyl group, a pentyl group, an isopentyl group, a sec-pentyl group, a tert-pentyl group, a neopentyl group, a hexyl group, an isohexyl group, a sec-hexyl group, a tert-hexyl group, a neohexyl group, a 3-methylpentyl group, a 2-methylpentyl group, a 2-ethylbutyl group, a 1,2-dimethylbutyl group, and a 2,3-dimethylbutyl group.
[0143] The alkoxy group refers to a monovalent group with a structure where an alkyl group is bonded to an oxygen atom. Specific examples of the alkoxy group include a methoxy group, an ethoxy group, an n-propoxy group, an isopropoxy group, an n-butoxy group, a sec-butoxy group, an isobutoxy group, a tert-butoxy group, an n-pentyloxy group, an isopentyloxy group, a sec-pentyloxy group, a tert-pentyloxy group, a neopentyloxy group, an n-hexyloxy group, an isohexyloxy group, a sec-hexyloxy group, a tert-hexyloxy group, and a neohexyloxy group.
[0144] The aryloxy group refers to a monovalent group with a structure where an aryl group is bonded to an oxygen atom. The aryl group refers to a monovalent group obtained by eliminating one hydrogen atom from one of carbon atoms forming the ring(s) of a monocyclic or polycyclic aromatic compound. Specific examples of the aryloxy group include a phenoxy group, an o-tolyloxy group, an m-tolyloxy group, a p-tolyloxy group, a mesityloxy group, an o-biphenyloxy group, an m-biphenyloxy group, a p-biphenyoxyl group, a 1-naphthyloxy group, a 2-naphthyloxy group, and a 2-fluorenyloxy group. Note that the aryloxy group may further have a substituent, and specific examples of the substituent include an alkyl group, an alkoxy group, and a phenyl group.
[0145] The alkylamino group refers to a monovalent group obtained by eliminating one hydrogen

atom from the nitrogen atom of a primary amine in which one alkyl group is bonded to the nitrogen atom, or from the nitrogen atom of a secondary amine in which two alkyl groups are bonded to the nitrogen atom. Specific examples of the alkylamino group include a dimethylamino group and a diethylamino group.

[0146] The arylamino group refers to a monovalent group obtained by eliminating one hydrogen atom from the nitrogen atom of a primary amine in which one aryl group is bonded to the nitrogen atom, or from the nitrogen atom of a secondary amine in which two aryl groups are bonded to the nitrogen atom. Specific examples of the arylamino group include a diphenylamino group, a bis(α-naphthyl)amino group, and a bis(m-tolyl) amino group. Note that the arylamino group may further have a substituent, and specific examples of the substituent include an alkyl group, an alkoxy group, and a phenyl group.

[0147] Note that an amino group having a structure where both an alkyl group and an aryl group are bonded to the nitrogen atom can be regarded as an alkylamino group or an arylamino group. Specific examples of such an amino group include an N-methyl-N-phenylamino group.

[0148] A heterocyclic amino group refers to a monovalent group obtained by eliminating one hydrogen atom from one nitrogen atom among the atoms forming a ring(s) of a heterocyclic amine. Here, the heterocyclic amine refers to a monocyclic or polycyclic heterocyclic compound in which at least one of the atoms forming the ring(s) is a nitrogen atom bonded to a hydrogen atom. Specific examples of the heterocyclic amino group include groups represented by Structural Formulae (R-1) to (R-27) below. Note that the heterocyclic amino group may further have a substituent, and specific examples of the substituent include an alkyl group, an alkoxy group, and a phenyl group.

##STR00001## ##STR00002## ##STR00003##

[0149] In some cases, the property of donating electrons to the phenanthroline ring is lower in a heterocyclic amino group which has aromaticity and in which an unshared electron pair of the nitrogen atom contributes to the aromaticity than in a heterocyclic amino group which has aromaticity and in which an unshared electron pair of the nitrogen atom does not contribute to the aromaticity. Therefore, among the above heterocyclic amino groups, a heterocyclic amino group which has aromaticity and in which an unshared electron pair of the nitrogen atom does not contribute to the aromaticity is further preferable. Specifically, the group represented by Structural Formula (R-1), (R-2), (R-3), (R-4), (R-5), (R-8), (R-9), (R-10), (R-12), (R-14), (R-15), (R-16), (R-17), (R-18), or (R-22) is further preferably used as the electron-donating group. Among these groups, the group represented by Structural Formula (R-3), (R-4), (R-8), or (R-22) is preferably used because the group has a high electron-donating property and can further increase the electron density of the phenanthroline ring.

[0150] Specific examples of the electron-donating group include groups represented by Structural Formulae (R-28) and (R-29) below.

##STR00004##

[0151] Note that an organic compound having a phenanthroline ring that can be used as the first organic compound may have both the above-described electron-donating group and another substituent. Note that introduction of an electron-withdrawing group (e.g., a cyano group or a fluoro group) to the phenanthroline ring is not preferable because the introduction reduces the electron density of the phenanthroline ring and inhibits the interaction with the metal in some cases. Specific examples of the substituent that can be introduced to the phenanthroline ring together with the above electron-donating group include an aryl group. Specific examples of the aryl group include a phenyl group, an o-tolyl group, an m-tolyl group, a p-tolyl group, a mesityl group, an o-biphenyl group, an m-biphenyl group, a p-biphenyl group, a 1-naphthyl group, a 2-naphthyl group, and a 2-fluorenyl group. Note that the aryl group may further have a substituent, and specific examples of the substituent include an alkyl group, an alkoxy group, and a phenyl group.

[0152] The first organic compound may have a structure where a plurality of phenanthroline rings are bonded to each other via a single bond or a divalent group. Specific examples of the divalent group include an alkylene group and an arylene group.

[0153] The alkylene group refers to a divalent group obtained by eliminating two hydrogen atoms from an alkane. Specific examples of an alkylene group include a divalent group having a structure obtained by eliminating one hydrogen atom from any of the above specific examples of an alkyl group.

[0154] The arylene group refers to a divalent group obtained by eliminating two hydrogen atoms from an aromatic hydrocarbon. Specific examples of an arylene group include a divalent group having a structure obtained by eliminating one hydrogen atom from any of the above specific examples of an aryl group. Note that the arylene group may further have a substituent, and specific examples of the substituent include an alkyl group, an alkoxy group, and a phenyl group.

[0155] Specific examples of an organic compound having a phenanthroline ring that can be used as the first organic compound are represented by Structural Formulae (100) to (109). Note that the organic compound that can be used as the first organic compound is not limited to those examples.

##STR00005## ##STR00006##

<<Estimation of Characteristics by Quantum Chemical Calculation>>

[0156] The minimum values of electrostatic potentials (ESP) of the organic compounds represented by Structural Formulae (100) to (109) were estimated by quantum chemical calculation. For comparison, the minimum values of ESP of BPhen, mPPhen2P, NBPhen, and Phen were also estimated. The structural formulae of BPhen, mPPhen2P, NBPhen, and Phen are shown below.

##STR00007##

[0157] As the quantum chemical computational program, Gaussian 09 was used. The calculation was performed using SGI 8600 produced by HPE. The most stable structures of the organic compounds in a ground state were calculated by the density functional theory (DFT). As a basis function, 6-311G(d,p) was used, and as a functional, B3LYP was used.

[0158] The table below lists the estimation results of the minimum values of ESP of the organic compounds, which were obtained through analysis of the electrostatic potentials of the organic compounds in a ground state. Note that the electrostatic potential is the energy of interaction between positive point charge with unit quantity of electricity and electron distribution of a molecule. An electrostatic potential value also depends on the threshold value of electron density distribution. The table below shows electrostatic potentials in electron density distribution at the time when the threshold value of electron density distribution in atomic units is 0.0004 e/a.sub.0.sup.3 or 0.003 e/a.sub.0.sup.3. Note that the threshold value of electron density distribution is referred to as “threshold value of density” in the table below.

TABLE-US-00004 TABLE 4 Minimum value of ESP Minimum value of ESP (E.sub.h) (Threshold value of (E.sub.h) (Threshold value of density = 0.0004 e/a.sub.0.sup.3) density = 0.003 e/a.sub.0.sup.3) Pyrrd-Phen (100) −0.091 −0.12 PrdP2Phen (101) −0.094 −0.13 DMeAPhen (102) −0.089 −0.12 p-MeO-Phen (103) −0.089 −0.12 4,7hpp2Phen (104) −0.096 −0.13 Hid2Phen (105) −0.094 −0.13 CzPhen (106) −0.072 −0.10 mhppPhen2P (107) −0.057 −0.096 9Ph2hppPhen (108) −0.057 −0.096 2,9hpp2Phen (109) −0.061 −0.097 BPhen −0.083 −0.11 mPPhen2P −0.057 −0.094 NBPhen −0.053 −0.093 Phen −0.081 −0.11

[0159] From the above table, it is found that the minimum values of ESP of the organic compounds represented by Structural Formulae (100) to (105) are each smaller than or equal to −0.085 En when the threshold value of electron density distribution is 0.0004 e/a.sub.0.sup.3 and that using any of these organic compounds as the first organic compound is the most preferable. On the other hand, the minimum values of ESP of the organic compounds represented by Structural Formulae (106) to (109) are each larger than −0.085 E.sub.h.

[0160] It is shown that the organic compounds represented by Structural Formulae (100) to (105) have the most preferable values because of having an electron-donating group at each of the 4- and 7-positions of the 1,10-phenanthroline ring.

[0161] The organic compound represented by Structural Formula (106) has N-carbazolyl groups as electron-donating groups at the 4- and 7-positions of the 1,10-phenanthroline ring. In the N-carbazolyl group, in which an unshared electron pair of the nitrogen atom contributes to aromaticity, the property of donating electrons to the phenanthroline ring is lower than that in a group in which an unshared electron pair of a nitrogen atom does not contribute to aromaticity, inhibiting a reduction in the minimum value of ESP of the organic compound represented by Structural Formula (106).
[0162] The organic compounds represented by Structural Formulae (107) to (109) each have electron-donating groups at the 2- and 9-positions of the 1,10-phenanthroline ring. In the case where the electron-donating groups are introduced to the 2- and 9-positions of the 1,10-phenanthroline ring, the property of donating electrons to the nitrogen atoms at the 1- and 10-positions of the phenanthroline ring is low as compared with the case where the electron-donating groups are introduced to the 4- and 7-positions. It is thus further preferable that substitution sites of electron-donating groups be the 4- and 7-positions of a 1,10-phenanthroline ring.
[0163] Note that the LUMO level of the second organic compound is further preferably lower than that of the first organic compound. In that case, electrons can be easily donated from the donor level formed by the first organic compound and the metal to the second organic compound. The LUMO level of the second organic compound is preferably lower than that of the first organic compound so that the second organic compound can have an electron-transport property.
[0164] For example, the LUMO level of the first organic compound is preferably higher than or equal to −3.0 eV and lower than or equal to −2.0 eV, further preferably higher than or equal to −2.7 eV and lower than or equal to −2.0 eV. The LUMO level of the second organic compound is preferably higher than or equal to −3.0 eV and lower than or equal to −2.0 eV, further preferably higher than or equal to −3.0 eV and lower than or equal to −2.5 eV. In that case, electrons can be easily donated from the donor level formed by the first organic compound and the metal to the second organic compound. This can increase the electron-transport property of the second organic compound.
[0165] Note that the HOMO level and the LUMO level of an organic compound are generally estimated by cyclic voltammetry (CV), photoelectron spectroscopy, optical absorption spectroscopy, inverse photoemission spectroscopy, or the like. When values of different compounds are compared with each other, it is preferable that values estimated by the same measurement be used.
[0166] In the case where the acid dissociation constant pK.sub.a of an organic compound is unknown, the acid dissociation constants pK.sub.a of skeletons in the organic compound are calculated and the largest acid dissociation constant pK.sub.a can be regarded as the acid dissociation constant pK.sub.a of the organic compound.
[0167] The acid dissociation constant may be obtained by calculation. For example, the acid dissociation constant pK.sub.a can be obtained by the following calculation method.
[0168] The initial structure of a molecule serving as a calculation model is the most stable structure (the singlet ground state) obtained by first-principles calculation.
[0169] For the first-principles calculation, Jaguar, which is the quantum chemical computational software (Schrödinger, Inc.), is used, and the most stable structure in the singlet ground state is calculated by the density functional theory (DFT). As a basis function, 6-31G** is used, and as a functional, B3LYP-D3 is used. The structure subjected to quantum chemical calculation is sampled by conformational analysis in mixed torsional/low-mode sampling with Maestro GUI produced by SchrÖdinger, Inc.
[0170] In the calculation of pK.sub.a, one or more atoms in each molecule are designated as basic sites, MacroModel is used to search for the stable structure of the protonated molecule in water, conformational search is performed with OPLS2005 force field, and a conformational isomer having the lowest energy is used. Jaguar's pK.sub.a calculation module is used. After structure

optimization is performed by B3LYP/6-31G*, single point calculation is performed by cc-pVTZ(+) and the pK.sub.a value is calculated using empirical correction for functional group(s). In the case where one or more atoms are designated as basic sites in a molecule, the largest of obtained values is used as a pK.sub.a value. The obtained pK.sub.a values are shown below.

[0171] The acid dissociation constant pK.sub.a of 2,9hpp2Phen (Structural Formula (109)) is 13.35, that of 4,7hpp2Phen (Structural Formula (104)) is 13.42, that of Pyrrd-Phen (Structural Formula (100)) is 11.23, that of mPPhen2P is 5.16, that of NBPhen is 5.59, and that of BPhen is 5.62.

<Second Organic Compound>

[0172] As the second organic compound, an organic compound having an electron-transport property can be used. The organic compound having an electron-transport property is preferably a substance having an electron mobility higher than or equal to 1×10.sup.−7 cm.sup.2/Vs, further preferably higher than or equal to 1×10.sup.−6 cm.sup.2/Vs, when the square root of electric field strength [V/cm] is 600. Note that any other substance can also be used as long as the substance has an electron-transport property higher than a hole-transport property.

[0173] An organic compound including a π-electron deficient heteroaromatic ring is preferable as the organic compound having an electron-transport property. The organic compound including a π-electron deficient heteroaromatic ring is preferably one or more of an organic compound including a heteroaromatic ring having a polyazole skeleton, an organic compound including a heteroaromatic ring having a pyridine skeleton, an organic compound including a heteroaromatic ring having a diazine skeleton, and an organic compound including a heteroaromatic ring having a triazine skeleton.

[0174] Specific examples of the organic compound having an electron-transport property include the following compounds: organic compounds having an azole skeleton, such as 2-(4-biphenylyl)-5-(4-tert-butylphenyl)-1,3,4-oxadiazole (abbreviation: PBD), 3-(4-biphenylyl)-4-phenyl-5-(4-tert-butylphenyl)-1,2,4-triazole (abbreviation: TAZ), 1,3-bis[5-(p-tert-butylphenyl)-1,3,4-oxadiazol-2-yl]benzene (abbreviation: OXD-7), 9-[4-(5-phenyl-1,3,4-oxadiazol-2-yl)phenyl]-9H-carbazole (abbreviation: CO11), 2,2′,2″-(1,3,5-benzenetriyl)tris(1-phenyl-1H-benzimidazole) (abbreviation: TPBI), 2-[3-(dibenzothiophen-4-yl)phenyl]-1-phenyl-1H-benzimidazole (abbreviation: mDBTBIm-II), and 4,4′-bis(5-methylbenzoxazol-2-yl) stilbene (abbreviation: BzOs); organic compounds including a heteroaromatic ring having a pyridine skeleton, such as 3,5-bis[3-(9H-carbazol-9-yl)phenyl]pyridine (abbreviation: 35DCzPPy), 1,3,5-tri[3-(3-pyridyl)phenyl]benzene (abbreviation: TmPyPB), 4.7-diphenyl-1,10-phenanthroline (abbreviation: BPhen), bathocuproine (abbreviation: BCP), 2,9-di(naphthalen-2-yl)-4,7-diphenyl-1,10-phenanthroline (abbreviation: NBPhen), and 2,2′-(1,3-phenylene)bis(9-phenyl-1,10-phenanthroline) (abbreviation: mPPhen2P); organic compounds having a diazine skeleton, such as 2-[3-(dibenzothiophen-4-yl)phenyl]dibenzo[f,h]quinoxaline (abbreviation: 2mDBTPDBq-II), 2-[3′-(dibenzothiophen-4-yl) biphenyl-3-yl]dibenzo[f,h]quinoxaline (abbreviation: 2mDBTBPDBq-II), 2-[3′-(9H-carbazol-9-yl) biphenyl-3-yl]dibenzo[f,h]quinoxaline (abbreviation: 2mCzBPDBq), 2-[4′-(9-phenyl-9H-carbazol-3-yl)-3,1′-biphenyl-1-yl]dibenzo[f,h]quinoxaline (abbreviation: 2mpPCBPDBq), 2-[4-(3,6-diphenyl-9H-carbazol-9-yl)phenyl]dibenzo[f,h]quinoxaline (abbreviation: 2CzPDBq-III), 7-[3-(dibenzothiophen-4-yl)phenyl]dibenzo[f,h]quinoxaline 7mDBTPDBq-II), 6-[3-(dibenzothiophen-4-yl)phenyl]dibenzo[f,h]quinoxaline (abbreviation: (abbreviation: 6mDBTPDBq-II), 9-[3′-(dibenzothiophen-4-yl) biphenyl-3-yl]naphtho[1′,2′: 4,5]furo[2,3-b]pyrazine (abbreviation: 9mDBtBPNfpr), 9-[3′-(dibenzothiophen-4-yl) biphenyl-4-yl]naphtho[1′,2′: 4,5]furo[2,3-b]pyrazine (abbreviation: 9pmDBtBPNfpr), 4,6-bis[3-(phenanthren-9-yl)phenyl]pyrimidine (abbreviation: 4,6mPnP2Pm), 4,6-bis[3-(dibenzothiophen-4-yl)phenyl]pyrimidine (abbreviation: 4,6mDBTP2Pm-II), 4,6-bis[3-(9H-carbazol-9-yl)phenyl]pyrimidine (abbreviation: 4,6mCzP2Pm), 9,9′-[pyrimidine-4,6-diylbis(biphenyl-3,3′-diyl)]bis(9H-carbazole) (abbreviation: 4,6mCzBP2Pm), 8-(biphenyl-4-yl)-4-[3-(dibenzothiophen-

4-yl)phenyl]-[1]benzofuro[3,2-d]pyrimidine (abbreviation: 8BP-4mDBtPBfpm), 3,8-bis[3-(dibenzothiophen-4-yl)phenyl]benzofuro[2,3-b]pyrazine (abbreviation: 4,8-bis[3-(dibenzothiophen-4-yl)phenyl]-[1]benzofuro[3,2-d]pyrimidine 3,8mDBtP2Bfpr), (abbreviation: 4,8mDBtP2Bfpm), 8-[3′-(dibenzothiophen-4-yl) (1,1′-biphenyl-3-yl)]naphtho[1′,2′: 4,5]furo[3,2-d]pyrimidine (abbreviation: 8mDBtBPNfpm), 8-[(2,2′-binaphthalen)-6-yl]-4-[3-(dibenzothiophen-4-yl)phenyl]-[1]benzofuro[3,2-d]pyrimidine (abbreviation: 8 (βN2)-4mDBtPBfpm), 2,2′-(pyridine-2,6-diyl)bis(4-phenylbenzo[h]quinazoline) (abbreviation: 2,6 (P-Bqn) 2Py), 2,2′ (2,2′-bipyridine-6,6′-diyl)bis(4-phenylbenzo[h]quinazoline) (abbreviation: 6,6′ (P-Bqn) 2BPy), 2,2′-(pyridine-2,6-diyl)bis {4-[4-(2-naphthyl)phenyl]-6-phenylpyrimidine} (abbreviation: 2,6 (NP-PPm) 2Py), 6-(biphenyl-3-yl)-4-[3,5-bis(9H-carbazol-9-yl)phenyl]-2-phenylpyrimidine (abbreviation: 6mBP-4Cz2PPm), 2,6-bis(4-naphthalen-1-ylphenyl)-4-[4-(3-pyridyl)phenyl]pyrimidine (abbreviation: 2,4NP-6PyPPm), 4-[3,5-bis(9H-carbazol-9-yl)phenyl]-2-phenyl-6-(biphenyl-4-yl)pyrimidine (abbreviation: 6BP-4Cz2PPm), and 7-[4-(9-phenyl-9H-carbazol-2-yl) quinazolin-2-yl]-7H-dibenzo[c,g]carbazole (abbreviation: PC-cgDBCzQz); and organic compounds having a triazine skeleton, such as 2-(biphenyl-4-yl)-4-phenyl-6-(9,9′-spirobi[9H-fluoren]-2-yl)-1,3,5-triazine (abbreviation: BP-SFTzn), 2-{3-[3-(benzo[b]naphtho[1,2-d]furan-8-yl)phenyl]phenyl}-4,6-diphenyl-1,3,5-triazine (abbreviation: mBnfBPTzn), 2-{3-[3-(benzo[b]naphtho[1,2-d]furan-6-yl)phenyl]phenyl}-4,6-diphenyl-1,3,5-triazine (abbreviation: mBnfBPTzn-02), 2-{4-[3-(N-phenyl-9H-carbazol-3-yl)-9H-carbazol-9-yl]phenyl}-4,6-diphenyl-1,3,5-triazine (abbreviation: PCCzPTzn), 9-[3-(4,6-diphenyl-1,3,5-triazin-2-yl)phenyl]-9′-phenyl-2,3′-bi-9H-carbazole (abbreviation: mPCCzPTzn-02), 2-[3′-(9,9-dimethyl-9H-fluoren-2-yl) biphenyl-3-yl]-4,6-diphenyl-1,3,5-triazine (abbreviation: mFBPTzn), 5-[3-(4,6-diphenyl-1,3,5-triazin-2-yl)phenyl]-7,7-dimethyl-5H,7H-indeno[2,1-b]carbazole (abbreviation: mINc(II)PTzn), 2-{3-[3-(dibenzothiophen-4-yl)phenyl]phenyl}-4,6-diphenyl-1,3,5-triazine (abbreviation: mDBtBPTzn), 2,4,6-tris[3′-(pyridin-3-yl) biphenyl-3-yl]-1,3,5-triazine (abbreviation: TmPPPyTz), 2-[3-(2,6-dimethyl-3-pyridinyl)-5-(9-phenanthrenyl)phenyl]-4,6-diphenyl-1,3,5-triazine (abbreviation: mPn-mDMePyPTzn), 11-[4-(biphenyl-4-yl)-6-phenyl-1,3,5-triazin-2-yl]-11,12-dihydro-12-phenylindolo[2,3-a]carbazole (abbreviation: BP-Icz(II)Tzn), 2-[3′-(triphenylen-2-yl) biphenyl-3-yl]-4,6-diphenyl-1,3,5-triazine (abbreviation: mTpBPTzn), 3-[9-(4,6-diphenyl-1,3,5-triazin-2-yl)-2-dibenzofuranyl]-9-phenyl-9H-carbazole (abbreviation: PCDBfTzn), and 2-(biphenyl-3-yl)-4-phenyl-6-{8-[(1,1′: 4′,1″-terphenyl)-4-yl]-1-dibenzofuranyl}-1,3,5-triazine (abbreviation: mBP-TPDBfTzn).

[0175] Among the above materials, an organic compound having a phenanthroline ring, specifically a 1,10-phenanthroline ring, such as BPhen, BCP, NBPhen, or mPPhen2P, easily interacts with the first organic compound and the metal and thus is particularly preferable. An organic compound having a phenanthroline ring dimer structure, such as mPPhen2P, is also preferable because of its high stability.

[0176] The second organic compound preferably has 25 to 100 carbon atoms. When having 25 to 100 carbon atoms, the second organic compound can have excellent sublimability, and thus, thermal decomposition of the organic compound during vacuum evaporation can be inhibited and the efficiency of use of the material can be high. An organic compound having a glass transition temperature (T.sub.g) higher than or equal to 100° C. can also be used. In that case, the electron-injection layer **115** can be a layer that is not easily crystallized. Accordingly, the electron-injection layer **115** is not easily crystallized even when affected by oxygen or water in the air and a chemical solution or water during processing by a lithography method for forming the organic compound layer **103** and the second electrode **102**. An increase in driving voltage or a reduction in current efficiency of the light-emitting device due to crystallization of the electron-injection layer **115** can be accordingly prevented. Thus, when the second organic compound has T.sub.g higher than or equal to 100° C., the composite material of the metal, the first organic compound, and the second organic compound can be suitably used for the electron-injection layer **115** of the light-emitting device where the organic compound layer **103** and the second electrode **102** are formed through processing by a lithography method.

[0177] Note that T_g can be measured with a differential scanning calorimeter (DSC8500 produced by PerkinElmer Japan Co., Ltd.) in a state where a powder sample is put on an aluminum cell and the temperature is increased at a rate of 40° C./min.

[0178] Examples of an organic compound having a phenanthroline ring and T_g higher than or equal to 100° C. include NBPhen (T_g: 165° C.), mPPhen2P (T_g: 135° C.), 2,2′-(biphenyl-4,4′-diyl)bis(9-phenyl-1,10-phenanthroline) (abbreviation: PPhen2BP) (T_g: 166° C.), 2,2′-biphenyl-3,3′-diylbis(9-phenyl-1,10-phenanthroline) (abbreviation: mPPhen2BP) (T_g: 144° C.), 2,8-bis(phenanthrolin-5-yl)dibenzofuran (abbreviation: 2,8Phen2DBf) (T_g: 210° C.), and 5,5′,5″-(benzene-1,3,5-triyl)tri-1,10-phenanthroline (abbreviation: Phen3P) (T_g: 257° C.). Note that T_g can be measured with a differential scanning calorimeter (DSC8500 produced by PerkinElmer Japan Co., Ltd.) in a state where a powder sample is put on an aluminum cell and the temperature is increased at a rate of 40° C./min.

[0179] As the second organic compound, an organic compound with an acid dissociation constant pK_a greater than or equal to 4 and less than 8 can be used. The second organic compound preferably has such an acid dissociation constant to have a poor hole-transport property, in which case the hole-transport property in the electron-injection layer **115** can be reduced and hole transport from the electron-injection layer **115** to the second electrode **102** can be prevented, enabling high efficiency of the light-emitting device. An excessively large acid dissociation constant pK_a leads to high solubility in water and thus reduces the resistance to water and a chemical solution used in the process by a lithography method. Thus, the acid dissociation constant pK_a of the second organic compound is preferably greater than or equal to 4 and less than 8.

[0180] In the layer including the combination of the metal, the first organic compound, and the second organic compound, interaction between the materials occurs more efficiently than in a layer including only two of the materials (e.g., a layer including the metal and the first organic compound or a layer including the metal and the second organic compound). This can be confirmed when films including some or all of the materials are formed using an odd-numbered metal as the metal and the spin densities of the films are measured by an electron spin resonance (ESR) method.

[0181] For example, in the case where ESR measurement shows that the spin density of a mixed film including the metal, the first organic compound, and the second organic compound is higher than the spin density of a mixed film including the metal and the first organic compound or a mixed film including the metal and the second organic compound, it can be confirmed that the interaction between the materials has occurred more efficiently in the mixed film including the combination of the metal, the first organic compound, and the second organic compound than in the mixed film including only two of the materials. Note that spin density measurement by an electron spin resonance method is preferably performed at room temperature.

[0182] Specifically, in the case where the density of spins attributed to a signal observed at a g-factor of approximately 2.00 in the ESR measurement is, for example, lower than or equal to 2×10^{16} spins/cm^3 in each of a mixed film including the metal and the first organic compound, a mixed film including the metal and the second organic compound, and a mixed film including the first organic compound and the second organic compound, it can be confirmed that interaction between the materials has occurred more efficiently in a mixed film including the metal, the first organic compound, and the second organic compound than in the mixed film including only two of the materials, when the density of spins attributed to a signal observed at a g-factor of approximately 2.00 in the ESR measurement is higher than or equal to 5×10^{16} spins/cm^3, preferably higher than or equal to 1×10^{17} spins/cm^3, for example. Note that spin density measurement by an electron spin resonance method is preferably performed at room temperature.

[0183] The molar ratio of the metal to the sum of the first organic compound and the second organic compound is preferably greater than or equal to 0.1 and less than or equal to 10, further preferably greater than or equal to 0.2 and less than or equal to 5, still further preferably greater than or equal to 0.5 and less than or equal to 2. Alternatively, the volume ratio of the metal to the sum of the first organic compound and the second organic compound is preferably greater than or equal to 0.01 and less than or equal to 0.3, further preferably greater than or equal to 0.02 and less than or equal to 0.2, still further preferably greater than or equal to 0.05 and less than or equal to 0.1. Mixing the metal, the first organic compound, and the second organic compound in such a ratio enables providing the electron-injection layer having a favorable electron-injection property. The volume ratio of the first organic compound to the second organic compound is preferably greater than or equal to 0.1 and less than or equal to 10, further preferably greater than or equal to 0.2 and less than or equal to 5, still further preferably greater than or equal to 0.5 and less than or equal to 2. Mixing the first organic compound and the second organic compound in such a ratio enables providing the electron-injection layer having a high electron-transport property.

[0184] The thickness of the electron-injection layer is preferably greater than or equal to 3 nm and less than or equal to 20 nm, further preferably greater than or equal to 5 nm and less than or equal to 10 nm. In that case, the composite material in which the metal, the first organic compound, and the second organic compound are mixed can favorably function, enabling high emission efficiency of the light-emitting device.

[0185] The structures described in this embodiment can be used in appropriate combination with any of the structures described in the other embodiments.

Embodiment 2

[0186] In this embodiment, light-emitting devices of one embodiment of the present invention will be described in detail.

[0187] FIGS. **1**A to **1**C are each a schematic diagram of the light-emitting device of one embodiment of the present invention. The light-emitting device includes the first electrode **101** over the insulating layer **1000**, and the organic compound layer **103** between the first electrode **101** and the second electrode **102**. The organic compound layer **103** includes at least the light-emitting layer **113** and the electron-injection layer **115**. The light-emitting layer **113** contains a light-emitting substance, and the light-emitting device of one embodiment of the present invention emits light when voltage is applied between the first electrode **101** and the second electrode **102**.

[0188] The organic compound layer **103** preferably includes, besides the light-emitting layer **113** and the electron-injection layer **115**, functional layers such as the hole-injection layer **111**, the hole-transport layer **112**, and the electron-transport layer **114**, as illustrated in FIG. **1**A. The organic compound layer **103** may include functional layers other than the above functional layers, such as a hole-blocking layer, an exciton-blocking layer, and an intermediate layer. Alternatively, any of the above-described layers may be omitted.

[0189] The electron-injection layer **115** includes a metal, the first organic compound, and the second organic compound as described in Embodiment 1. Since the specific structure of the electron-injection layer **115** has been described in detail in Embodiment 1, the repetitive description thereof is omitted.

[0190] The first electrode **101** and the second electrode **102** may each be formed to have a single-layer structure or a stacked-layer structure.

[0191] In the light-emitting device of one embodiment of the present invention, processing by a photolithography method is performed after formation of the film to be the second electrode **102**; thus, the light-emitting device has a feature that, in a cross-sectional view, the end portion of the second electrode **102** and the end portion of the organic compound layer **103** are aligned in a direction substantially perpendicular to the surface of the insulating layer **1000** as illustrated in FIGS. **1**A to **1**C. The end portion of the second electrode and the end portion of the organic compound layer may be positioned inward from an end portion of the first electrode as illustrated in FIGS. **1**A and **1**B, or may be positioned outward from the first electrode as illustrated in FIG. **1**C.

[0192] The first electrode **101** is preferably formed using any of metals, alloys, and conductive compounds with a high work function (specifically, higher than or equal to 4.0 eV), mixtures thereof, and the like. Specific examples include indium oxide-tin oxide (ITO: indium tin oxide), indium oxide-tin oxide containing silicon or silicon oxide (ITSO: indium tin silicon oxide), indium oxide-zinc oxide, and indium oxide containing tungsten oxide and zinc oxide (IWZO). Films of such conductive metal oxides are usually formed by a sputtering method, but may be formed by application of a sol-gel method or the like. For example, a film of indium oxide-zinc oxide is formed by a sputtering method using a target in which 1 wt % to 20 wt % zinc oxide is added to indium oxide. Furthermore, a film of indium oxide containing tungsten oxide and zinc oxide (IWZO) can be formed by a sputtering method using a target in which 0.5 wt % to 5 wt % tungsten oxide and 0.1 wt % to 1 wt % zinc oxide are added to indium oxide. Alternatively, gold (Au), platinum (Pt), nickel (Ni), tungsten (W), chromium (Cr), molybdenum (Mo), iron (Fe), cobalt (Co), copper (Cu), palladium (Pd), titanium (Ti), aluminum (Al), nitride of a metal material (e.g., titanium nitride), or the like can be used for the first electrode **101**. The first electrode **101** may be a stack of layers formed using any of these materials. For example, a film in which Al, Ti, and ITSO are stacked in this order over Ti is preferable because the film has high efficiency owing to high reflectivity and enables a high resolution of several thousand ppi. Graphene can also be used for the first electrode **101**. When a composite material that can be included in the hole-injection layer **111** described later is used for a layer (typically, the hole-injection layer) in contact with the first electrode **101**, an electrode material can be selected regardless of its work function.

[0193] The hole-injection layer **111** is provided in contact with the first electrode **101** and has a function of facilitating injection of holes into the organic compound layer **103**. The hole-injection layer **111** can be formed using a phthalocyanine-based compound such as phthalocyanine (abbreviation: H.sub.2Pc), a phthalocyanine-based complex compound such as copper phthalocyanine (abbreviation: CuPc), an aromatic amine compound such as 4,4′-bis[N-(4-diphenylaminophenyl)-N-phenylamino]biphenyl (abbreviation: DPAB) or 4,4′-bis(N-{4-[N′-(3-methylphenyl)-N′-phenylamino]phenyl}-N-phenylamino) biphenyl (abbreviation: DNTPD), or a high molecular compound such as poly(3,4-ethylenedioxythiophene)/poly(styrenesulfonic acid) (abbreviation: PEDOT/PSS).

[0194] The hole-injection layer **111** may be formed using a substance with an electron-accepting property. Examples of the substance with an acceptor property include organic compounds having an electron-withdrawing group (a halogen group or a cyano group), such as 7,7,8,8-tetracyano-2,3,5,6-tetrafluoroquinodimethane (abbreviation: F4-TCNQ), chloranil, 2,3,6,7,10,11-hexacyano-1,4,5,8,9,12-hexaazatriphenylene (abbreviation: HAT-CN), 1,3,4,5,7,8-hexafluorotetracyano-naphthoquinodimethane (abbreviation: F6-TCNNQ), and 2-(7-dicyanomethylene-1,3,4,5,6,8,9,10-octafluoro-7H-pyren-2-ylidene) malononitrile. A compound in which electron-withdrawing groups are bonded to a condensed aromatic ring having a plurality of heteroatoms, such as HAT-CN, is particularly preferable because it is thermally stable. A [3]radialene derivative having an electron-withdrawing group (in particular, a cyano group, a halogen group such as a fluoro group, or the like) has a significantly high electron-accepting property and thus is preferable. Specific examples include α,α′,α″-1,2,3-cyclopropanetriylidenetris[4-cyano-2,3,5,6-tetrafluorobenzeneacetonitrile], α,α′,α″-1,2,3-cyclopropanetriylidenetris[2,6-dichloro-3,5-difluoro-4-(trifluoromethyl)benzeneacetonitrile], and α,α′,α″-1,2,3-cyclopropanetriylidenetris[2,3,4,5,6-pentafluorobenzeneacetonitrile]. As the substance with an acceptor property, a transition metal oxide such as molybdenum oxide, vanadium oxide, ruthenium oxide, tungsten oxide, or manganese oxide can be used, other than the above-described organic compounds.

[0195] The hole-injection layer **111** is preferably formed using a composite material containing any of the aforementioned materials with an acceptor property and an organic compound with a hole-transport property.

[0196] As the organic compound with a hole-transport property that is used in the composite material, any of a variety of organic compounds such as aromatic amine compounds, heteroaromatic compounds, aromatic hydrocarbons, and high molecular compounds (e.g., oligomers, dendrimers, and polymers) can be used. Note that the organic compound with a hole-transport property that is used in the composite material preferably has a hole mobility higher than or equal to 1×10.sup.−6 cm.sup.2/Vs. The organic compound with a hole-transport property that is used in the composite material preferably has a condensed aromatic hydrocarbon ring or a π-electron rich heteroaromatic ring. As the condensed aromatic hydrocarbon ring, an anthracene ring, a naphthalene ring, or the like is preferable. As the π-electron rich heteroaromatic ring, a condensed aromatic ring having at least one of a pyrrole skeleton, a furan skeleton, and a thiophene skeleton is preferable; specifically, a carbazole ring, a dibenzothiophene ring, or a ring in which an aromatic ring or a heteroaromatic ring is further condensed to the carbazole ring or the dibenzothiophene ring is preferable. Note that in this specification and the like, an organic compound with a hole-transport property is sometimes referred to as a material with a hole-transport property.

[0197] Such an organic compound with a hole-transport property further preferably has any of a carbazole skeleton, a dibenzofuran skeleton, a dibenzothiophene skeleton, and an anthracene skeleton. In particular, an aromatic amine including a substituent having a dibenzofuran ring or a dibenzothiophene ring, an aromatic monoamine having a naphthalene ring, or an aromatic monoamine in which a 9-fluorenyl group is bonded to nitrogen of an amine through an arylene group may be used. Note that the organic compound with a hole-transport property preferably has an N,N-bis(4-biphenyl)amino group to enable fabricating a light-emitting device having a long lifetime.

[0198] Specific examples of the organic compound with a hole-transport property include N-(4-biphenyl)-6,N-diphenylbenzo[b]naphtho[1,2-d]furan-8-amine (abbreviation: BnfABP), N,N-bis(4-biphenyl)-6-phenylbenzo[b]naphtho[1,2-d]furan-8-amine (abbreviation: BBABnf), 4,4′-bis(6-phenylbenzo[b]naphtho[1,2-d]furan-8-yl)-4″-phenyltriphenylamine (abbreviation: BnfBB1BP), N,N-bis(4-biphenyl)benzo[b]naphtho[1,2-d]furan-6-amine (abbreviation: BBABnf(6)), N,N-bis(4-biphenyl)benzo[b]naphtho[1,2-d]furan-8-amine (abbreviation: BBABnf(8)), N,N-bis(4-biphenyl)benzo[b]naphtho[2,3-d]furan-4-amine (abbreviation: BBABnf(II)(4)), N,N-bis[4-(dibenzofuran-4-yl)phenyl]-4-amino-p-terphenyl (abbreviation: DBfBB1TP), N-[4-(dibenzothiophen-4-yl)phenyl]-N-phenyl-4-biphenylamine (abbreviation: ThBA1BP), 4-(2-naphthyl)-4′,4″-diphenyltriphenylamine (abbreviation: BBABNB), 4-[4-(2-naphthyl)phenyl]-4′,4″-diphenyltriphenylamine (abbreviation: BBABNBi), 4,4′-diphenyl-4″-(6; 1′-binaphthyl-2-yl)triphenylamine (abbreviation: BBAαNβNB), 4,4′-diphenyl-4″-(7; 1′-binaphthyl-2-yl)triphenylamine (abbreviation: BBAαNβNB-03), 4,4′-diphenyl-4″-(7-phenyl) naphthyl-2-yltriphenylamine (abbreviation: BBAPβNB-03), 4,4′-diphenyl-4″-(6; 2′-binaphthyl-2-yl)triphenylamine (abbreviation: BBA(βN2)B), 4,4′-diphenyl-4″-(7; 2′-binaphthyl-2-yl)triphenylamine (abbreviation: BBA(βN2)B-03), 4,4′-diphenyl-4″-(4; 2′-binaphthyl-1-yl)triphenylamine (abbreviation: BBAβNαNB), 4,4′-diphenyl-4″-(5; 2′-binaphthyl-1-yl)triphenylamine (abbreviation: BBAβNαNB-02), 4-(4-biphenylyl)-4′-(2-naphthyl)-4″-phenyltriphenylamine (abbreviation: TPBiAβNB), 4-(3-biphenylyl)-4′-[4-(2-naphthyl)phenyl]-4″-phenyltriphenylamine (abbreviation: mTPBiAβNBi), 4-(4-biphenylyl)-4′-[4-(2-naphthyl)phenyl]-4″-phenyltriphenylamine (abbreviation: TPBiAβNBi), 4-phenyl-4′-(1-naphthyl)triphenylamine (abbreviation: αNBA1BP), 4,4′-bis(1-naphthyl)triphenylamine (abbreviation: αNBB1BP), 4,4′-diphenyl-4″-[4′-(carbazol-9-yl) biphenyl-4-yl]triphenylamine (abbreviation: YGTBi1BP), 4′-[4-(3-phenyl-9H-carbazol-9-yl)phenyl]tris(biphenyl-4-yl)amine (abbreviation: YGTBi1BP-02), 4-[4′-(carbazol-9-yl) biphenyl-4-yl]-4′-(2-naphthyl)-4″-phenyltriphenylamine (abbreviation: YGTBiβNB), N-[4-(9-phenyl-9H-carbazol-3-yl)phenyl]-N-[4-(1-naphthyl)phenyl]-9,9′-spirobi[9H-fluoren]-2-amine (abbreviation: PCBNBSF), N,N-bis(biphenyl-4-yl)-9,9′-spirobi[9H-fluoren]-2-amine (abbreviation: BBASF), N,N-bis(biphenyl-4-yl)-9,9′-spirobi[9H-fluoren]-4-amine (abbreviation: BBASF(4)), N-(biphenyl-2-yl)-N-(9,9-

dimethyl-9H-fluoren-2-yl)-9,9′-spirobi[9H-fluoren]-4-amine (abbreviation: oFBiSF), N-(biphenyl-4-yl)-N-(9,9-dimethyl-9H-fluoren-2-yl)dibenzofuran-4-amine (abbreviation: FrBiF), N-[4-(1-naphthyl)phenyl]-N-[3-(6-phenyldibenzofuran-4-yl)phenyl]-1-naphthylamine (abbreviation: mPDBfBNBN), 4-phenyl-4′-(9-phenylfluoren-9-yl)triphenylamine (abbreviation: BPAFLP), 4-phenyl-3′-(9-phenylfluoren-9-yl)triphenylamine (abbreviation: mBPAFLP), 4-phenyl-4′-[4-(9-phenylfluoren-9-yl)phenyl]triphenylamine (abbreviation: BPAFLBi), 4-phenyl-4′-(9-phenyl-9H-carbazol-3-yl)triphenylamine (abbreviation: PCBA1BP), 4,4′-diphenyl-4″-(9-phenyl-9H-carbazol-3-yl)triphenylamine (abbreviation: PCBBi1BP), 4-(1-naphthyl)-4′-(9-phenyl-9H-carbazol-3-yl)triphenylamine (abbreviation: PCBANB), 4,4′-di(1-naphthyl)-4″-(9-phenyl-9H-carbazol-3-yl)triphenylamine (abbreviation: PCBNBB), N-phenyl-N-[4-(9-phenyl-9H-carbazol-3-yl)phenyl]-9,9′-spirobi[9H-fluoren]-2-amine (abbreviation: PCBASF), N-(biphenyl-4-yl)-N-[4-(9-phenyl-9H-carbazol-3-yl)phenyl]-9,9-dimethyl-9H-fluoren-2-amine (abbreviation: PCBBiF), N,N-bis(9,9-dimethyl-9H-fluoren-2-yl)-9,9′-spirobi-9H-fluoren-4-amine, N,N-bis(9,9-dimethyl-9H-fluoren-2-yl)-9,9′-spirobi-9H-fluoren-3-amine, N,N-bis(9,9-dimethyl-9H-fluoren-2-yl)-9,9′-spirobi-9H-fluoren-2-amine, and N,N-bis(9,9-dimethyl-9H-fluoren-2-yl)-9,9′-spirobi-9H-fluoren-1-amine.

[0199] Other examples of the organic compound with a hole-transport property include N,N′-di(p-tolyl)-N,N′-diphenyl-p-phenylenediamine (abbreviation: DTDPPA), 4,4′-bis[N-(4-diphenylaminophenyl)-N-phenylamino]biphenyl (abbreviation: DPAB), 4,4′-bis(N-{4-[N-(3-methylphenyl)-N′-phenylamino]phenyl}-N-phenylamino) biphenyl (abbreviation: DNTPD), and 1,3,5-tris[N-(4-diphenylaminophenyl)-N-phenylamino]benzene (abbreviation: DPA3B).

[0200] The formation of the hole-injection layer **111** can improve the hole-injection property, which allows the light-emitting device to be driven at a low voltage.

[0201] Among substances with an acceptor property, the organic compound with an acceptor property is easy to use because it is easily deposited by evaporation.

[0202] The hole-transport layer **112** is formed using an organic compound with a hole-transport property. The organic compound with a hole-transport property preferably has a hole mobility of 1×10^{-6} cm^2/Vs or higher.

[0203] Examples of the aforementioned organic compound with a hole-transport property include the following compounds: compounds having an aromatic amine skeleton, such as 4,4′-bis[N-(1-naphthyl)-N-phenylamino]biphenyl (abbreviation: NPB), N,N′-diphenyl-N,N′-bis(3-methylphenyl)-4,4′-diaminobiphenyl (abbreviation: TPD), N,N-bis(9,9′-spirobi[9H-fluoren]-2-yl)-N,N′-diphenyl-4,4′-diaminobiphenyl (abbreviation: BSPB), 4-phenyl-4′-(9-phenylfluoren-9-yl)triphenylamine (abbreviation: BPAFLP), 4-phenyl-3′-(9-phenylfluoren-9-yl)triphenylamine (abbreviation: mBPAFLP), 4-phenyl-4′-(9-phenyl-9H-carbazol-3-yl)triphenylamine (abbreviation: PCBA1BP), 4,4′-diphenyl-4″-(9-phenyl-9H-carbazol-3-yl)triphenylamine (abbreviation: PCBBi1BP), 4-(1-naphthyl)-4′-(9-phenyl-9H-carbazol-3-yl)triphenylamine (abbreviation: PCBANB), 4,4′-di(1-naphthyl)-4″-(9-phenyl-9H-carbazol-3-yl)triphenylamine (abbreviation: PCBNBB), 9,9-dimethyl-N-phenyl-N-[4-(9-phenyl-9H-carbazol-3-yl)phenyl]fluoren-2-amine (abbreviation: PCBAF), and N-phenyl-N-[4-(9-phenyl-9H-carbazol-3-yl)phenyl]-9,9′-spirobi[9H-fluoren]-2-amine (abbreviation: PCBASF); compounds having a carbazole skeleton, such as 1,3-bis(N-carbazolyl)benzene (abbreviation: mCP), 4,4′-di(N-carbazolyl) biphenyl (abbreviation: CBP), 3,6-bis(3,5-diphenylphenyl)-9-phenylcarbazole (abbreviation: CzTP), 9,9′-diphenyl-9H,9′H-3,3′-bicarbazole (abbreviation: PCCP), 9,9′-bis(biphenyl-4-yl)-3,3′-bi-9H-carbazole (abbreviation: BisBPCz), 9,9′-bis(biphenyl-3-yl)-3,3′-bi-9H-carbazole (abbreviation: BismBPCz), 9-(biphenyl-3-yl)-9′-(biphenyl-4-yl)-9H,9′H-3,3′-bicarbazole (abbreviation: mBPCCBP), 9-(2-naphthyl)-9′-phenyl-9H,9′H-3,3′-bicarbazole (abbreviation: BNCCP), 9-(3-biphenyl)-9′-(2-naphthyl)-3,3′-bi-9H-carbazole (abbreviation: βNCCmBP), 9-(4-biphenyl)-9′-(2-naphthyl)-3,3′-bi-9H-carbazole (abbreviation: BNCCBP), 9,9′-di-2-naphthyl-3,3′-9H,9′H-bicarbazole (abbreviation: BisBNCz), 9-(2-naphthyl)-9′-[1,1′: 4′,1″-terphenyl]-3-yl-3,3′-9H,9′H-bicarbazole, 9-(2-naphthyl)-9′-[1,1′: 3′,1″-terphenyl]-3-yl-3,3′-9H,9′H-bicarbazole, 9-(2-naphthyl)-9′-[1,1′: 3′,1″-terphenyl]-5′-yl-

3,3′-9H,9′H-bicarbazole, 9-(2-naphthyl)-9′-[1,1′: 4′,1″-terphenyl]-4-yl-3,3′-9H,9′H-bicarbazole, 9-(2-naphthyl)-9′-[1,1′: 3′,1″-terphenyl]-4-yl-3,3′-9H,9′H-bicarbazole, 9-(2-naphthyl)-9′-(triphenylen-2-yl)-3,3′-9H,9′H-bicarbazole, 9-phenyl-9′-(triphenylen-2-yl)-3,3′-9H,9′H-bicarbazole (abbreviation: PCCzTp), 9,9′-bis(triphenylen-2-yl)-3,3′-9H,9′H-bicarbazole, 9-(4-biphenyl)-9′-(triphenylen-2-yl)-3,3′-9H,9′H-bicarbazole, and 9-(triphenylen-2-yl)-9′-[1,1′: 3′,1″-terphenyl]-4-yl-3,3′-9H,9′H-bicarbazole; compounds having a thiophene skeleton, such as 4,4′,4″-(benzene-1,3,5-triyl)tri (dibenzothiophene) (abbreviation: DBT3P-II), 2,8-diphenyl-4-[4-(9-phenyl-9H-fluoren-9-yl)phenyl]dibenzothiophene (abbreviation: DBTFLP-III), and 4-[4-(9-phenyl-9H-fluoren-9-yl)phenyl]-6-phenyldibenzothiophene (abbreviation: DBTFLP-IV); and compounds having a furan skeleton, such as 4,4′,4″-(benzene-1,3,5-triyl)tri (dibenzofuran) (abbreviation: DBF3P-II) and 4-{3-[3-(9-phenyl-9H-fluoren-9-yl)phenyl]phenyl}dibenzofuran (abbreviation: mmDBFFLBi-II). Among the above materials, the compound having an aromatic amine skeleton or the compound having a carbazole skeleton is preferable because the compound is highly reliable and has a high hole-transport property to contribute to a reduction in driving voltage. Note that any of the substances given as examples of the organic compound with a hole-transport property used in the composite material for the hole-injection layer **111** can also be suitably used as the material contained in the hole-transport layer **112**.

[0204] The light-emitting layer **113** is a layer containing a light-emitting substance and preferably contains a light-emitting substance and a host material. The light-emitting layer may additionally contain another material. Alternatively, the light-emitting layer **113** may be a stack of two or more layers with different compositions.

[0205] As the light-emitting substance, fluorescent substances, phosphorescent substances, substances exhibiting thermally activated delayed fluorescence (TADF), or other light-emitting substances may be used.

[0206] Examples of the material that can be used as a fluorescent substance in the light-emitting layer are as follows. Other fluorescent substances can also be used.

[0207] The examples include 5,6-bis[4-(10-phenyl-9-anthryl)phenyl]-2,2′-bipyridine (abbreviation: PAP2BPy), 5,6-bis[4′-(10-phenyl-9-anthryl) biphenyl-4-yl]-2,2′-bipyridine (abbreviation: PAPP2BPy), N,N-diphenyl-N,N′-bis[4-(9-phenyl-9H-fluoren-9-yl)phenyl]pyrene-1,6-diamine (abbreviation: 1,6FLPAPrn), N,N-bis(3-methylphenyl)-N,N-bis[3-(9-phenyl-9H-fluoren-9-yl)phenyl]pyrene-1,6-diamine (abbreviation: 1,6mMemFLPAPrn), N,N′-bis[4-(9H-carbazol-9-yl)phenyl]-N,N′-diphenylstilbene-4,4′-diamine (abbreviation: YGA2S), 4-(9H-carbazol-9-yl)-4′-(10-phenyl-9-anthryl)triphenylamine (abbreviation: YGAPA), 4-(9H-carbazol-9-yl)-4′-(9,10-diphenyl-2-anthryl)triphenylamine (abbreviation: 2YGAPPA), N,9-diphenyl-N-[4-(10-phenyl-9-anthryl)phenyl]-9H-carbazol-3-amine (abbreviation: PCAPA), perylene, 2,5,8,11-tetra-tert-butylperylene (abbreviation: TBP), 4-(10-phenyl-9-anthryl)-4′-(9-phenyl-9H-carbazol-3-yl)triphenylamine (abbreviation: PCBAPA), N,N″-(2-tert-butylanthracene-9,10-diyldi-4,1-phenylene)bis(N,N′,N′-triphenyl-1,4-phenylenediamine) (abbreviation: DPABPA), N,9-diphenyl-N-[4-(9,10-diphenyl-2-anthryl)phenyl]-9H-carbazol-3-amine (abbreviation: 2PCAPPA), N-[4-(9,10-diphenyl-2-anthryl)phenyl]-N,N′,N′-triphenyl-1,4-phenylenediamine (abbreviation: 2DPAPPA), N,N,N′,N′,N″,N″,N‴,N‴-octaphenyldibenzo[g,p]chrysene-2,7,10,15-tetraamine (abbreviation: DBC1), coumarin 30, N-(9,10-diphenyl-2-anthryl)-N,9-diphenyl-9H-carbazol-3-amine (abbreviation: 2PCAPA), N-[9,10-bis(biphenyl-2-yl)-2-anthryl]-N,9-diphenyl-9H-carbazol-3-amine (abbreviation: 2PCABPhA), N-(9,10-diphenyl-2-anthryl)-N,N′,N′-triphenyl-1,4-phenylenediamine (abbreviation: 2DPAPA), N-[9,10-bis(biphenyl-2-yl)-2-anthryl]-N,N′,N′-triphenyl-1,4-phenylenediamine (abbreviation: 2DPABPhA), 9,10-bis(biphenyl-2-yl)-N-[4-(9H-carbazol-9-yl)phenyl]-N-phenylanthracen-2-amine (abbreviation: 2YGABPhA), N,N,9-triphenylanthracen-9-amine (abbreviation: DPhAPhA), coumarin 545T, N,N′-diphenylquinacridone (abbreviation: DPQd), rubrene, 5,12-bis(biphenyl-4-yl)-6,11-diphenyltetracene (abbreviation: BPT), 2-(2-{2-[4-(dimethylamino)phenyl]ethenyl}-6-methyl-4H-pyran-4-ylidene) propanedinitrile

(abbreviation: DCM1), 2-{2-methyl-6-[2-(2,3,6,7-tetrahydro-1H,5H-benzo[ij]quinolizin-9-yl) ethenyl]-4H-pyran-4-ylidene}propanedinitrile (abbreviation: DCM2), N,N,N′,N′-tetrakis(4-methylphenyl)tetracene-5,11-diamine (abbreviation: p-mPhTD), 7,14-diphenyl-N,N,N′,N′-tetrakis(4-methylphenyl) acenaphtho[1,2-a]fluoranthene-3,10-diamine (abbreviation: p-mPhAFD), 2-{2-isopropyl-6-[2-(1,1,7,7-tetramethyl-2,3,6,7-tetrahydro-1H,5H-benzo[ij]quinolizin-9-yl) ethenyl]-4H-pyran-4-ylidene}propanedinitrile (abbreviation: DCJTI), 2-{2-tert-butyl-6-[2-(1,1,7,7-tetramethyl-2,3,6,7-tetrahydro-1H,5H-benzo[ij]quinolizin-9-yl) ethenyl]-4H-pyran-4-ylidene}propanedinitrile (abbreviation: DCJTB), 2-(2,6-bis {2-[4-(dimethylamino)phenyl]ethenyl}-4H-pyran-4-ylidene) propanedinitrile (abbreviation: BisDCM), 2-{2,6-bis[2-(8-methoxy-1,1,7,7-tetramethyl-2,3,6,7-tetrahydro-1H,5H-benzo[ij]quinolizin-9-yl) ethenyl]-4H-pyran-4-ylidene}propanedinitrile (abbreviation: BisDCJTM), N,N-diphenyl-N,N′-(1,6-pyrene-diyl)bis[(6-phenylbenzo[b]naphtho[1,2-d]furan)-8-amine](abbreviation: 1,6BnfAPrn-03), N,N′-diphenyl-N,N′-bis(9-phenyl-9H-carbazol-2-yl) naphtho[2,3-b; 6,7-b′]bisbenzofuran-3,10-diamine (abbreviation: 3,10PCA2Nbf (IV)-02), and 3,10-bis[N-(dibenzofuran-3-yl)-N-phenylamino]naphtho[2,3-b; 6,7-b′]bisbenzofuran (abbreviation: 3,10FrA2Nbf (IV)-02). Condensed aromatic diamine compounds typified by pyrenediamine compounds such as 1,6FLPAPrn, 1,6mMemFLPAPrn, and 1,6BnfAPrn-03 are particularly preferable because of their high hole-trapping properties, high emission efficiency, or high reliability.
[0208] A condensed heteroaromatic compound containing nitrogen and boron, especially a compound having a diaza-boranaphtho-anthracene skeleton, exhibits a narrow emission spectrum, emits blue light with high color purity, and can thus be suitably used. Examples of the compound include 5,9-diphenyl-5,9-diaza-13b-boranaphtho[3,2,1-de]anthracene (abbreviation: DABNA1), 9-(biphenyl-3-yl)-N,N,5,11-tetraphenyl-5H,9H-[1,4]benzazaborino[2,3,4-kl]phenazaborin-3-amine (abbreviation: DABNA2), 2,12-di(tert-butyl)-5,9-di(4-tert-butylphenyl)-N,N-diphenyl-5H,9H-[1,4]benzazaborino[2,3,4-kl]phenazaborin-7-amine (abbreviation: DPhA-tBu4DABNA), 2,12-di(tert-butyl)-N,N,5,9-tetra(4-tert-butylphenyl)-5H,9H-[1,4]benzazaborino[2,3,4-kl]phenazaborin-7-amine (abbreviation: tBuDPhA-tBu4DABNA), 2,12-di(tert-butyl)-5,9-di(4-tert-butylphenyl)-7-methyl-5H,9H-[1,4]benzazaborino[2,3,4-k/]phenazaborine (abbreviation: Me-tBu4DABNA), N7,N7,N13,N13,5,9,11,15-octaphenyl-5H,9H,11H,15H-[1,4]benzazaborino[2,3,4-kl][1,4]benzazaborino[4′,3′,2′:4,5][1,4]benzazaborino[3,2-b]phenazaborine-7,13-diamine (abbreviation: v-DABNA), and 2-(4-tert-butylphenyl)benz[5,6]indolo[3,2,1-jk]benzo[b]carbazole (abbreviation: tBuPBibc).
[0209] Besides the above compounds, 9,10,11-tris[3,6-bis(1,1-dimethylethyl)-9H-carbazolyl-9-yl]-2,5,15,18-tetrakis(1,1-dimethylethyl) indolo[3,2,1-de]indolo[3′,2′,1′:8,1][1,4]benzazaborino[2,3,4-kl]phenazaborine (abbreviation: BBCz-G), 9,11-bis[3,6-bis(1,1-dimethylethyl)-9H-carbazolyl-9-yl]-2,5,15,18-tetrakis(1,1-dimethylethyl)indolo[3,2,1-de]indolo[3′,2′,1′:8,1][1,4]benzazaborino[2,3,4-kl]phenazaborine (abbreviation: BBCz-Y), or the like can be suitably used.
[0210] Examples of the material that can be used when a phosphorescent substance is used as the light-emitting substance in the light-emitting layer are as follows.
[0211] The examples include organometallic iridium complexes having a 4H-triazole skeleton, such as tris {2-[5-(2-methylphenyl)-4-(2,6-dimethylphenyl)-4H-1,2,4-triazol-3-yl-κN.sup.2]phenyl-κC}iridium(III) (abbreviation: [Ir(mpptz-dmp).sub.3]) and tris(5-methyl-3,4-diphenyl-4H-1,2,4-triazolato)iridium(III) (abbreviation: [Ir(Mptz).sub.3]); organometallic iridium complexes having a 1H-triazole skeleton, such as tris[3-methyl-1-(2-methylphenyl)-5-phenyl-1H-1,2,4-triazolato]iridium(III) (abbreviation: [Ir(Mptz1-mp).sub.3]) and tris(1-methyl-5-phenyl-3-propyl-1H-1,2,4-triazolato)iridium(III) (abbreviation: [Ir(Prptz1-Me).sub.3]); organometallic iridium complexes having an imidazole skeleton, such as fac-tris[1-(2,6-diisopropylphenyl)-2-phenyl-1H-imidazole]iridium(III) (abbreviation: [Ir(iPrpim).sub.3]), tris[3-(2,6-dimethylphenyl)-7-methylimidazo[1,2-f]phenanthridinato]iridium(III) (abbreviation: [Ir(dmpimpt-Me).sub.3]), and

tris(2-{1-[2,6-bis(1-methylethyl)phenyl]-1H-imidazol-2-yl-κN.sup.3}-4-cyanophenyl-KC)iridium(III) (abbreviation: CNImIr); organometallic complexes having a benzimizazolidene skeleton, such as tris[(6-tert-butyl-3-phenyl-2H-imidazo[4,5-b]pyrazin-1-yl-κC.sup.2)phenyl-κC]iridium(III) (abbreviation: [Ir(cb).sub.3]); and organometallic iridium complexes in which a phenylpyridine derivative having an electron-withdrawing group is a ligand, such as bis[2-(4′,6′-difluorophenyl)pyridinato-N,C.sup.2]iridium(III) tetrakis(1-pyrazolyl)borate (abbreviation: FIr6), bis[2-(4′,6′-difluorophenyl)pyridinato-N,C.sup.2′]iridium(III) picolinate (abbreviation: FIrpic), bis{2-[3′,5′-bis(trifluoromethyl)phenyl]pyridinato-N,C.sup.2′}iridium(III) picolinate (abbreviation: [Ir(CF.sub.3ppy).sub.2(pic)]), and bis[2-(4′,6′-difluorophenyl)pyridinato-N,C.sup.2′]iridium(III) acetylacetonate (abbreviation: FIracac). These compounds emit blue phosphorescent light and have an emission peak in the wavelength range of 450 nm to 520 nm.

[0212] Other examples include organometallic iridium complexes having a pyrimidine skeleton, such as tris(4-methyl-6-phenylpyrimidinato)iridium(III) (abbreviation: [Ir(mppm).sub.3]), tris(4-t-butyl-6-phenylpyrimidinato)iridium(III) (abbreviation: [Ir(tBuppm).sub.3]), (acetylacetonato)bis(6-methyl-4-phenylpyrimidinato)iridium(III) (abbreviation: [Ir(mppm).sub.2 (acac)]), (acetylacetonato)bis(6-tert-butyl-4-phenylpyrimidinato)iridium(III) (abbreviation: [Ir(tBuppm).sub.2 (acac)]), (acetylacetonato)bis[6-(2-norbornyl)-4-phenylpyrimidinato]iridium(III) (abbreviation: [Ir(nbppm).sub.2 (acac)]), (acetylacetonato)bis[5-methyl-6-(2-methylphenyl)-4-phenylpyrimidinato]iridium(III) (abbreviation: [Ir(mpmppm).sub.2 (acac)]), and (acetylacetonato)bis(4,6-diphenylpyrimidinato)iridium(III) (abbreviation: [Ir(dppm).sub.2 (acac)]); organometallic iridium complexes having a pyrazine skeleton, such as (acetylacetonato)bis(3,5-dimethyl-2-phenylpyrazinato)iridium(III) (abbreviation: [Ir(mppr-Me).sub.2 (acac)]) and (acetylacetonato)bis(5-isopropyl-3-methyl-2-phenylpyrazinato)iridium(III) (abbreviation: [Ir(mppr-iPr).sub.2 (acac)]); organometallic iridium complexes having a pyridine skeleton, such as tris(2-phenylpyridinato-N,C.sup.2′)iridium(III) (abbreviation: [Ir(ppy).sub.3]), bis(2-phenylpyridinato-N,C.sup.2′)iridium(III) acetylacetonate (abbreviation: [Ir(ppy).sub.2 (acac)]), bis(benzo[h]quinolinato)iridium(III) acetylacetonate (abbreviation: [Ir(bzq).sub.2 (acac)]), tris(benzo[h]quinolinato)iridium(III) (abbreviation: [Ir(bzq).sub.3]), tris(2-phenylquinolinato-N,C.sup.2′)iridium(III) (abbreviation: [Ir(pq).sub.3]), bis(2-phenylquinolinato-N,C.sup.2′)iridium(III) acetylacetonate (abbreviation: [Ir(pq).sub.2 (acac)]), [2-d.sub.3-methyl-8-(2-pyridinyl-κN)benzofuro[2,3-b]pyridine-κC]bis[2-(5-d.sub.3-methyl-2-pyridinyl-κN.sup.2)phenyl-κC]iridium(III) (abbreviation: Ir(5mppy-d.sub.3).sub.2 (mbfpypy-d.sub.3)), [2-d.sub.3-methyl-(2-pyridinyl-κN)benzofuro[2,3-b]pyridine-κC]bis[2-(2-pyridinyl-κN)phenyl-κC]iridium(III) (abbreviation: [Ir(ppy).sub.2 (mbfpypy-d.sub.3)]), [2-(4-d.sub.3-methyl-5-phenyl-2-pyridinyl-κN.sup.2)phenyl-κC]bis[2-(5-d.sub.3-methyl-2-pyridinyl-κN.sup.2)phenyl-κC]iridium(III) (abbreviation: [Ir(5mppy-d.sub.3).sub.2 (mdppy-d.sub.3)]), [2-methyl-(2-pyridinyl-κN)benzofuro[2,3-b]pyridine-κC]bis[2-(2-pyridinyl-κN)phenyl-κC]iridium(III) (abbreviation: [Ir(ppy).sub.2 (mbfpypy)]), and [2-(4-methyl-5-phenyl-2-pyridinyl-κN)phenyl-κC]bis[2-(2-pyridinyl-κN)phenyl-κC]iridium(III) (abbreviation: [Ir(ppy).sub.2(mdppy)]); and a rare earth metal complex such as tris(acetylacetonato) (monophenanthroline) terbium (III) (abbreviation: [Tb(acac).sub.3(Phen)]). These compounds mainly emit green phosphorescent light and have an emission peak in the wavelength range of 500 nm to 600 nm. Note that organometallic iridium complexes having a pyrimidine skeleton have distinctively high reliability or emission efficiency and thus are particularly preferable.

[0213] Other examples include organometallic iridium complexes having a pyrimidine skeleton, such as (diisobutyrylmethanato)bis[4,6-bis(3-methylphenyl)pyrimidinato]iridium(III) (abbreviation: [Ir(5mdppm).sub.2 (dibm)]), bis[4,6-bis(3-methylphenyl)pyrimidinato] (dipivaloylmethanato) iridium(III) (abbreviation: [Ir(5mdppm).sub.2(dpm)]), and bis[4,6-di(naphthalen-1-yl)pyrimidinato](dipivaloylmethanato) iridium(III) (abbreviation: [Ir(dlnpm).sub.2 (dpm)]); organometallic iridium complexes having a pyrazine skeleton, such as

(acetylacetonato)bis(2,3,5-triphenylpyrazinato) iridium(III) (abbreviation: [Ir(tppr).sub.2 (acac)]), bis(2,3,5-triphenylpyrazinato) (dipivaloylmethanato) iridium(III) (abbreviation: [Ir(tppr).sub.2 (dpm)]), and (acetylacetonato)bis[2,3-bis(4-fluorophenyl) quinoxalinato]iridium(III) (abbreviation: [Ir(Fdpq).sub.2 (acac)]); organometallic iridium complexes having a pyridine skeleton, such as tris(1-phenylisoquinolinato-N,C.sup.2′) iridium(III) (abbreviation: [Ir(piq).sub.3]), bis(1-phenylisoquinolinato-N,C.sup.2′) iridium(III) acetylacetonate (abbreviation: [Ir(piq).sub.2 (acac)]), (3,7-diethyl-4,6-nonanedionato-κO.sup.4,κO.sup.6)bis[2,4-dimethyl-6-[7-(1-methylethyl)-1-isoquinolinyl-κN]phenyl-κC]iridium(III), and (3,7-diethyl-4,6-nonanedionato-κO.sup.4,κO.sup.6)bis[2,4-dimethyl-6-[5-(1-methylethyl)-2-quinolinyl-κN]phenyl-κC]iridium(III); platinum complexes such as 2,3,7,8,12,13,17,18-octaethyl-21H,23H-porphyrinplatinum (II) (abbreviation: PtOEP); and rare earth metal complexes such as tris(1,3-diphenyl-1,3-propanedionato) (monophenanthroline) europium (III) (abbreviation: [Eu (DBM).sub.3 (Phen)]) and tris[1-(2-thenoyl)-3,3,3-trifluoroacetonato](monophenanthroline) europium (III) (abbreviation: [Eu (TTA).sub.3 (Phen)]). These compounds emit red phosphorescent light and have an emission peak in the wavelength range of 600 nm to 700 nm. Furthermore, the organometallic iridium complexes having a pyrazine skeleton can provide red light emission with favorable chromaticity.
[0214] Besides the above phosphorescent compounds, known phosphorescent compounds may be selected and used.
[0215] Examples of the TADF material include a fullerene, a derivative thereof, an acridine, a derivative thereof, and an eosin derivative. Furthermore, a metal-containing porphyrin, such as a porphyrin containing magnesium (Mg), zinc (Zn), cadmium (Cd), tin (Sn), platinum (Pt), indium (In), or palladium (Pd), can be given. Examples of the metal-containing porphyrin include a protoporphyrin-tin fluoride complex (SnF.sub.2(Proto IX)), a mesoporphyrin-tin fluoride complex (SnF.sub.2(Meso IX)), a hematoporphyrin-tin fluoride complex (SnF.sub.2(Hemato IX)), a coproporphyrin tetramethyl ester-tin fluoride complex (SnF.sub.2(Copro III-4Me)), an octaethylporphyrin-tin fluoride complex (SnF.sub.2(OEP)), an etioporphyrin-tin fluoride complex (SnF.sub.2(Etio I)), and an octaethylporphyrin-platinum chloride complex (PtCl.sub.2OEP), which are represented by the following structural formulae.
##STR00008## ##STR00009## ##STR00010##
[0216] Alternatively, it is possible to use a heterocyclic compound having one or both of a π-electron rich heteroaromatic ring and a π-electron deficient heteroaromatic ring which is represented by any of the following structural formulae, such as 2-(biphenyl-4-yl)-4,6-bis(12-phenylindolo[2,3-a]carbazol-11-yl)-1,3,5-triazine (abbreviation: PIC-TRZ), 9-(4,6-diphenyl-1,3,5-triazin-2-yl)-9′-phenyl-9H,9′H-3,3′-bicarbazole (abbreviation: PCCzTzn), 2-{4-[3-(N-phenyl-9H-carbazol-3-yl)-9H-carbazol-9-yl]phenyl}-4,6-diphenyl-1,3,5-triazine (abbreviation: PCCzPTzn), 2-[4-(10H-phenoxazin-10-yl)phenyl]-4,6-diphenyl-1,3,5-triazine (abbreviation: PXZ-TRZ), 3-[4-(5-phenyl-5,10-dihydrophenazin-10-yl)phenyl]-4,5-diphenyl-1,2,4-triazole (abbreviation: PPZ-3TPT), 3-(9,9-dimethyl-9H-acridin-10-yl)-9H-xanthen-9-one (abbreviation: ACRXTN), bis[4-(9,9-dimethyl-9,10-dihydroacridine)phenyl]sulfone (abbreviation: DMAC-DPS), or 10-phenyl-10H,10′H-spiro[acridin-9,9′-anthracen]-10′-one (abbreviation: ACRSA). Such a heterocyclic compound is preferable because of their high electron-transport and hole-transport properties owing to a π-electron rich heteroaromatic ring and a π-electron deficient heteroaromatic ring. Among skeletons having the π-electron deficient heteroaromatic ring, a pyridine skeleton, a diazine skeleton (a pyrimidine skeleton, a pyrazine skeleton, or a pyridazine skeleton), and a triazine skeleton are preferable because of their high stability and reliability. In a benzothienopyrimidine skeleton, a particular, a benzofuropyrimidine skeleton, benzofuropyrazine skeleton, and a benzothienopyrazine skeleton are preferable because of their high acceptor properties and high reliability. Among skeletons having the π-electron rich heteroaromatic ring, an acridine skeleton, a phenoxazine skeleton, a phenothiazine skeleton, a furan skeleton, a thiophene skeleton, and a pyrrole skeleton have high stability and reliability; thus, at least one of these skeletons is preferably

included. A dibenzofuran skeleton is preferable as a furan skeleton, and a dibenzothiophene skeleton is preferable as a thiophene skeleton. As a pyrrole skeleton, an indole skeleton, a carbazole skeleton, an indolocarbazole skeleton, a bicarbazole skeleton, and a 3-(9-phenyl-9H-carbazol-3-yl)-9H-carbazole skeleton are particularly preferable. Note that a substance in which the π-electron rich heteroaromatic ring is directly bonded to the π-electron deficient heteroaromatic ring is particularly preferable because the electron-donating property of the π-electron rich heteroaromatic ring and the electron-accepting property of the π-electron deficient heteroaromatic ring are both improved, the energy difference between the S.sub.1 level and the T.sub.1 level becomes small, and thus thermally activated delayed fluorescence can be obtained with high efficiency. Note that an aromatic ring to which an electron-withdrawing group such as a cyano group is bonded may be used instead of the π-electron deficient heteroaromatic ring. As a π-electron rich skeleton, an aromatic amine skeleton, a phenazine skeleton, or the like can be used. As a π-electron deficient skeleton, a xanthene skeleton, a thioxanthene dioxide skeleton, an oxadiazole skeleton, a triazole skeleton, an imidazole skeleton, an anthraquinone skeleton, a skeleton containing boron such as phenylborane or boranthrene, an aromatic ring or a heteroaromatic ring having a cyano group or a nitrile group such as benzonitrile or cyanobenzene, a carbonyl skeleton such as benzophenone, a phosphine oxide skeleton, a sulfone skeleton, or the like can be used. As described above, a π-electron deficient skeleton and a π-electron rich skeleton can be used instead of at least one of the π-electron deficient heteroaromatic ring and the π-electron rich heteroaromatic ring.
##STR00011## ##STR00012##
[0217] A TADF material has a small difference between the lower singlet excitation energy level (S.sub.1 level) and the lowest triplet excitation energy level (T.sub.1 level) and has a function of converting triplet excitation energy into singlet excitation energy by reverse intersystem crossing. Thus, a TADF material can upconvert triplet excitation energy into singlet excitation energy (i.e., reverse intersystem crossing) using a small amount of thermal energy and efficiently generate a singlet excited state. In addition, the triplet excitation energy can be converted into light emission.
[0218] An exciplex whose excited state is formed of two kinds of substances has an extremely small difference between the S.sub.1 level and the T.sub.1 level and functions as a TADF material capable of converting triplet excitation energy into singlet excitation energy.
[0219] A phosphorescent spectrum observed at a low temperature (e.g., 77 K to 10 K) is used for an index of the T.sub.1 level. When the level of energy with a wavelength of the line obtained by extrapolating a tangent to the fluorescent spectrum at a tail on the short wavelength side is the S.sub.1 level and the level of energy with a wavelength of the line obtained by extrapolating a tangent to the phosphorescent spectrum at a tail on the short wavelength side is the T.sub.1 level, the difference between the Si level and the T.sub.1 level of the TADF material is preferably smaller than or equal to 0.3 eV, further preferably smaller than or equal to 0.2 eV.
[0220] When a TADF material is used as the light-emitting substance, the S.sub.1 level of the host material is preferably higher than that of the TADF material. In addition, the T.sub.1 level of the host material is preferably higher than that of the TADF material.
[0221] As the host material in the light-emitting layer, various carrier-transport materials such as materials having an electron-transport property and/or materials having a hole-transport property, and the TADF materials can be used.
[0222] The material having a hole-transport property is preferably an organic compound including an amine skeleton or a π-electron rich heteroaromatic ring skeleton, for example. As the π-electron rich heteroaromatic ring, a condensed aromatic ring having at least one of an acridine skeleton, a phenoxazine skeleton, a phenothiazine skeleton, a furan skeleton, a thiophene skeleton, and a pyrrole skeleton is preferable; specifically, a carbazole ring, a dibenzothiophene ring, or a ring in which an aromatic ring or a heteroaromatic ring is further condensed to a carbazole ring or a dibenzothiophene ring is preferable.
[0223] Such an organic compound having a hole-transport property further preferably has any of a

carbazole skeleton, a dibenzofuran skeleton, a dibenzothiophene skeleton, and an anthracene skeleton. In particular, an aromatic amine including a substituent having a dibenzofuran ring or a dibenzothiophene ring, an aromatic monoamine having a naphthalene ring, or an aromatic monoamine in which a 9-fluorenyl group is bonded to nitrogen of an amine through an arylene group may be used. Note that the organic compound having a hole-transport property preferably includes an N,N-bis(4-biphenyl)amino group to enable fabricating a light-emitting device having a long lifetime.

[0224] Examples of such an organic compound include compounds having an aromatic amine skeleton, such as 4,4′-bis[N-(1-naphthyl)-N-phenylamino]biphenyl (abbreviation: NPB), N,N′-diphenyl-N,N′-bis(3-methylphenyl)-4,4′-diaminobiphenyl (abbreviation: TPD), N,N′-bis(9,9′-spirobi[9H-fluoren]-2-yl)-N,N′-diphenyl-4,4′-diaminobiphenyl (abbreviation: BSPB), 4-phenyl-4′-(9-phenylfluoren-9-yl)triphenylamine (abbreviation: BPAFLP), 4-phenyl-3′-(9-phenylfluoren-9-yl)triphenylamine (abbreviation: mBPAFLP), 4-phenyl-4′-(9-phenyl-9H-carbazol-3-yl)triphenylamine (abbreviation: PCBA1BP), 4,4′-diphenyl-4″-(9-phenyl-9H-carbazol-3-yl)triphenylamine (abbreviation: PCBBi1BP), 4-(1-naphthyl)-4′-(9-phenyl-9H-carbazol-3-yl)triphenylamine (abbreviation: PCBANB), 4,4′-di(1-naphthyl)-4″-(9-phenyl-9H-carbazol-3-yl)triphenylamine (abbreviation: PCBNBB), 9,9-dimethyl-N-phenyl-N-[4-(9-phenyl-9H-carbazol-3-yl)phenyl]fluoren-2-amine (abbreviation: PCBAF), and N-phenyl-N-[4-(9-phenyl-9H-carbazol-3-yl)phenyl]-9,9′-spirobi[9H-fluoren]-2-amine (abbreviation: PCBASF); compounds having a carbazole skeleton, such as 1,3-bis(N-carbazolyl)benzene (abbreviation: mCP), 4,4′-di(N-carbazolyl) biphenyl (abbreviation: CBP), 3,6-bis(3,5-diphenylphenyl)-9-phenylcarbazole (abbreviation: CzTP), and 9,9′-diphenyl-9H,9′H-3,3′-bicarbazole (abbreviation: PCCP); a compound having a thiophene skeleton, such as 4,4′,4″-(benzene-1,3,5-triyl)tri (dibenzothiophene) (abbreviation: DBT3P-II), 2,8-diphenyl-4-[4-(9-phenyl-9H-fluoren-9-yl)phenyl]dibenzothiophene (abbreviation: DBTFLP-III), and 4-[4-(9-phenyl-9H-fluoren-9-yl)phenyl]-6-phenyldibenzothiophene (abbreviation: DBTFLP-IV); and compounds having a furan skeleton, such as 4,4′,4″-(benzene-1,3,5-triyl)tri (dibenzofuran) (abbreviation: DBF3P-II) and 4-{3-[3-(9-phenyl-9H-fluoren-9-yl)phenyl]phenyl}dibenzofuran (abbreviation: mmDBFFLBi-II). Among the above materials, the compound having an aromatic amine skeleton and the compound having a carbazole skeleton are preferable because these compounds are highly reliable and have high hole-transport properties to contribute to a reduction in driving voltage. In addition, the organic compounds given as examples of the material having a hole-transport property that can be used for the hole-transport layer can also be used.

[0225] As the material having an electron-transport property, for example, a metal complex such as bis(10-hydroxybenzo[h]quinolinato)beryllium(II) (abbreviation: BeBq.sub.2), bis(2-methyl-8-quinolinolato) (4-phenylphenolato)aluminum(III) (abbreviation: BAlq), bis(8-quinolinolato)zinc(II) (abbreviation: Znq), bis[2-(2-benzoxazolyl)phenolato]zinc(II) (abbreviation: ZnPBO), or bis[2-(2-benzothiazolyl)phenolato]zinc(II) (abbreviation: ZnBTZ); or an organic compound including a π-electron deficient heteroaromatic ring is preferably used. Examples of the organic compound including a π-electron deficient heteroaromatic ring skeleton include an organic compound including heteroaromatic ring having an azole skeleton, an organic compound including a heteroaromatic ring having a pyridine skeleton, an organic compound including a heteroaromatic ring having a diazine skeleton, and an organic compound including a heteroaromatic ring having a triazine skeleton.

[0226] Among the above materials, the organic compound including a heteroaromatic ring having a diazine skeleton (a pyrimidine skeleton, a pyrazine skeleton, or a pyridazine skeleton), the organic compound including a heteroaromatic ring having a pyridine skeleton, and the organic compound including a heteroaromatic ring having a triazine skeleton have high reliability and thus are preferable. In particular, the organic compound including a heteroaromatic ring having a diazine (pyrimidine or pyrazine) skeleton and the organic compound including a heteroaromatic ring having a triazine skeleton have a high electron-transport property to contribute to a reduction in driving voltage. A benzofuropyrimidine skeleton, a benzothienopyrimidine skeleton, a benzofuropyrazine skeleton, and a benzothienopyrazine skeleton are preferable because of their high acceptor properties and high reliability.

[0227] Examples of the organic compound including a n-electron deficient heteroaromatic ring skeleton include organic compounds having an azole skeleton, such as 2-(4-biphenylyl)-5-(4-tert-butylphenyl)-1,3,4-oxadiazole (abbreviation: PBD), 3-(4-biphenylyl)-4-phenyl-5-(4-tert-butylphenyl)-1,2,4-triazole (abbreviation: TAZ), 1,3-bis[5-(p-tert-butylphenyl)-1,3,4-oxadiazol-2-yl]benzene (abbreviation: OXD-7), 9-[4-(5-phenyl-1,3,4-oxadiazol-2-yl)phenyl]-9H-carbazole (abbreviation: CO11), 2,2′,2″-(1,3,5-benzenetriyl)tris(1-phenyl-1H-benzimidazole) (abbreviation: TPBI), 2-[3-(dibenzothiophen-4-yl)phenyl]-1-phenyl-1H-benzimidazole (abbreviation: mDBTBIm-II), and 4,4′-bis(5-methylbenzoxazol-2-yl) stilbene (abbreviation: BzOS); organic compounds including a heteroaromatic ring having a pyridine skeleton, such as 3,5-bis[3-(9H-carbazol-9-yl)phenyl]pyridine (abbreviation: 35DCzPPy), 1,3,5-tri[3-(3-pyridyl)phenyl]benzene (abbreviation: TmPyPB), 4,7-diphenyl-1,10-phenanthroline (abbreviation: BPhen), bathocuproine (abbreviation: BCP), 2,9-di(naphthalen-2-yl)-4,7-diphenyl-1,10-phenanthroline (abbreviation: NBPhen), 2,2′-(1,3-phenylene)bis(9-phenyl-1,10-phenanthroline) (abbreviation: mPPhen2P), 2-[3-(2-triphenylenyl)phenyl]-1,10-phenanthroline (abbreviation: mTpPPhen), 2-phenyl-9-(2-triphenylenyl)-1,10-phenanthroline (abbreviation: Ph-TpPhen), 2-[4-(9-phenanthrenyl)-1-naphthalenyl]-1,10-phenanthroline (abbreviation: PnNPhen), and 2-[4-(2-triphenylenyl)phenyl]-1,10-phenanthroline (abbreviation: pTpPPhen); organic compounds having a diazine skeleton, such as 2-[3-(dibenzothiophen-4-yl)phenyl]dibenzo[f,h]quinoxaline (abbreviation: 2-[3′-(dibenzothiophen-4-yl) biphenyl-3-yl]dibenzo[f,h]quinoxaline 2mDBTPDBq-II), (abbreviation: 2mDBTBPDBq-II), 2-[3′-(9H-carbazol-9-yl) biphenyl-3-yl]dibenzo[f,h]quinoxaline (abbreviation: 2mCzBPDBq), 2-[4′-(9-phenyl-9H-carbazol-3-yl)-3,1′-biphenyl-1-yl]dibenzo[f,h]quinoxaline (abbreviation: 2mpPCBPDBq), 2-[4-(3,6-diphenyl-9H-carbazol-9-yl)phenyl]dibenzo[f,h]quinoxaline (abbreviation: 2CzPDBq-III), 7-[3-(dibenzothiophen-4-yl)phenyl]dibenzo[f,h]quinoxaline (abbreviation: 7mDBTPDBq-II), 6-[3-(dibenzothiophen-4-yl)phenyl]dibenzo[f,h]quinoxaline (abbreviation: 6mDBTPDBq-II), 9-[3′-(dibenzothiophen-4-yl) biphenyl-3-yl]naphtho[1′,2′: 4,5]furo[2,3-b]pyrazine (abbreviation: 9mDBtBPNfpr), 9-[3′-(dibenzothiophen-4-yl) biphenyl-4-yl]naphtho[1′,2′: 4,5]furo[2,3-b]pyrazine (abbreviation: 9pmDBtBPNfpr), 4,6-bis[3-(phenanthren-9-yl)phenyl]pyrimidine (abbreviation: 4,6mPnP2Pm), 4,6-bis[3-(dibenzothiophen-4-yl)phenyl]pyrimidine (abbreviation: 4,6mDBTP2Pm-II), 4,6-bis[3-(9H-carbazol-9-yl)phenyl]pyrimidine (abbreviation: 4,6mCzP2Pm), 9,9′-[pyrimidine-4,6-diylbis(biphenyl-3,3′-diyl)]bis(9H-carbazole) (abbreviation: 4,6mCzBP2Pm), 8-(biphenyl-4-yl)-4-[3-(dibenzothiophen-4-yl)phenyl]-[1]benzofuro[3,2-d]pyrimidine (abbreviation: 8BP-4mDBtPBfpm), 3,8-bis[3-(dibenzothiophen-4-yl)phenyl]benzofuro[2,3-b]pyrazine (abbreviation: 3,8mDBtP2Bfpr), 4,8-bis[3-(dibenzothiophen-4-yl)phenyl]-[1]benzofuro[3,2-d]pyrimidine (abbreviation: 4,8mDBtP2Bfpm), 8-[3′-(dibenzothiophen-4-yl) (1,1′-biphenyl-3-yl)]naphtho[1′,2′: 4,5]furo[3,2-d]pyrimidine (abbreviation: 8mDBtBPNfpm), 8-[(2,2′-binaphthalen)-6-yl]-4-[3-(dibenzothiophen-4-yl)phenyl]-[1]benzofuro[3,2-d]pyrimidine (abbreviation: 8 (βN2)-4mDBtPBfpm), 2,2′-(pyridine-2,6-diyl)bis(4-phenylbenzo[h]quinazoline) (abbreviation: 2,6 (P-Bqn) 2Py), 2,2′-(pyridine-2,6-diyl)bis {4-[4-(2-naphthyl)phenyl]-6-phenylpyrimidine} (abbreviation: 2,6 (NP-PPm) 2Py), 6-(biphenyl-3-yl)-4-[3,5-bis(9H-carbazol-9-yl)phenyl]-2-phenylpyrimidine (abbreviation: 6mBP-4Cz2PPm), 2,6-bis(4-naphthalen-1-ylphenyl)-4-[4-(3-pyridyl)phenyl]pyrimidine (abbreviation: 2,4NP-6PyPPm), 4-[3,5-bis(9H-carbazol-9-yl)phenyl]-2-phenyl-6-(biphenyl-4-yl)pyrimidine (abbreviation: 6BP-4Cz2PPm), and 7-[4-(9-phenyl-9H-carbazol-2-yl) quinazolin-2-yl]-7H-dibenzo[c,g]carbazole (abbreviation: PC-cgDBCzQz); and organic compounds having a heteroaromatic ring having a triazine skeleton, such as 2-(biphenyl-4-yl)-4-phenyl-6-(9,9′-spirobi[9H-fluoren]-2-yl)-1,3,5-triazine (abbreviation: BP-SFTzn), 2-{3-[3-

(benzo[b]naphtho[1,2-d]furan-8-yl)phenyl]phenyl}-4,6-diphenyl-1,3,5-triazine (abbreviation: mBnfBPTzn), 2-{3-[3-(benzo[b]naphtho[1,2-d]furan-6-yl)phenyl]phenyl}-4,6-diphenyl-1,3,5-triazine (abbreviation: mBnfBPTzn-02), 2-{4-[3-(N-phenyl-9H-carbazol-3-yl)-9H-carbazol-9-yl]phenyl}-4,6-diphenyl-1,3,5-triazine (abbreviation: PCCzPTzn), 9-[3-(4,6-diphenyl-1,3,5-triazin-2-yl)phenyl]-9′-phenyl-2,3′-bi-9H-carbazole (abbreviation: mPCCzPTzn-02), 2-[3′-(9,9-dimethyl-9H-fluoren-2-yl) biphenyl-3-yl]-4,6-diphenyl-1,3,5-triazine (abbreviation: mFBPTzn), 5-[3-(4,6-diphenyl-1,3,5-triazin-2-yl)phenyl]-7,7-dimethyl-5H,7H-indeno[2,1-b]carbazole (abbreviation: mINc(II)PTzn), 2-{3-[3-(dibenzothiophen-4-yl)phenyl]phenyl}-4,6-diphenyl-1,3,5-triazine (abbreviation: mDBtBPTzn), 2,4,6-tris(3′-(pyridin-3-yl) biphenyl-3-yl)-1,3,5-triazine (abbreviation: TmPPPyTz), 2-[3-(2,6-dimethyl-3-pyridinyl)-5-(9-phenanthrenyl)phenyl]-4,6-diphenyl-1,3,5-triazine (abbreviation: mPn-mDMePyPTzn), 11-[4-(biphenyl-4-yl)-6-phenyl-1,3,5-triazin-2-yl]-11,12-dihydro-12-phenylindolo[2,3-a]carbazole (abbreviation: BP-Icz (II) Tzn), 2-[3′-(triphenylen-2-yl) biphenyl-3-yl]-4,6-diphenyl-1,3,5-triazine (abbreviation: mTpBPTzn), 3-[9-(4,6-diphenyl-1,3,5-triazin-2-yl)-2-dibenzofuranyl]-9-phenyl-9H-carbazole (abbreviation: PCDBfTzn), and 2-(biphenyl-3-yl)-4-phenyl-6-{8-[(1,1′: 4′,1″-terphenyl)-4-yl]-1-dibenzofuranyl}-1,3,5-triazine (abbreviation: mBP-TPDBfTzn). The organic compound including a heteroaromatic ring having a diazine skeleton, the organic compound including a heteroaromatic ring having a pyridine skeleton, and the organic compound including a heteroaromatic ring having a triazine skeleton are preferable because of their high reliability. In particular, the organic compound including a heteroaromatic ring having a diazine (pyrimidine or pyrazine) skeleton and the organic compound including a heteroaromatic ring having a triazine skeleton have a high electron-transport property to contribute to a reduction in driving voltage.

[0228] As the TADF material that can be used as the host material, the above materials mentioned as the TADF material can also be used. When the TADF material is used as the host material, triplet excitation energy generated in the TADF material is converted into singlet excitation energy by reverse intersystem crossing and transferred to the light-emitting substance, whereby the emission efficiency of the light-emitting device can be increased. Here, the TADF material functions as an energy donor, and the light-emitting substance functions as an energy acceptor.

[0229] This is very effective in the case where the light-emitting substance is a fluorescent substance. In that case, the S.sub.1 level of the TADF material is preferably higher than that of the fluorescent substance in order that high emission efficiency can be achieved. Furthermore, the T.sub.1 level of the TADF material is preferably higher than the S.sub.1 level of the fluorescent substance. Therefore, the T.sub.1 level of the TADF material is preferably higher than that of the fluorescent substance.

[0230] It is also preferable to use a TADF material that emits light whose wavelength overlaps with the wavelength on the lowest-energy-side absorption band of the fluorescent substance. This enables smooth transfer of excitation energy from the TADF material to the fluorescent substance and accordingly enables efficient light emission, which is preferable.

[0231] In order to efficiently generate singlet excitation energy from the triplet excitation energy by reverse intersystem crossing, carrier recombination preferably occurs in the TADF material. It is also preferable that the triplet excitation energy generated in the TADF material not be transferred to the triplet excitation energy of the fluorescent substance. For that reason, the fluorescent substance preferably has a protective group around a luminophore (a skeleton that brings about light emission) of the fluorescent substance. As the protective group, a substituent having no π bond and a saturated hydrocarbon are preferably used. Specific examples include an alkyl group having 3 to 10 carbon atoms, a substituted or unsubstituted cycloalkyl group having 3 to 10 carbon atoms, and a trialkylsilyl group having 3 to 10 carbon atoms. It is further preferable that the fluorescent substance have a plurality of protective groups. The substituents having no π bond are poor in carrier transport performance, whereby the TADF material and the luminophore of the fluorescent substance can be made away from each other with little influence on carrier transportation or carrier recombination. Here, the luminophore refers to an atomic group (skeleton) that brings about light emission in a fluorescent substance. The luminophore is preferably a skeleton having a π bond, further preferably has an aromatic ring, and still further preferably has a condensed aromatic ring or a condensed heteroaromatic ring. Examples of the luminophore include a phenanthrene skeleton, a stilbene skeleton, an acridone skeleton, a phenoxazine skeleton, a phenothiazine skeleton, a naphthalene skeleton, an anthracene skeleton, a fluorene skeleton, a chrysene skeleton, a triphenylene skeleton, a tetracene skeleton, a pyrene skeleton, a perylene skeleton, a coumarin skeleton, a quinacridone skeleton, and a naphthobisbenzofuran skeleton. Specifically, a fluorescent substance having any of a naphthalene skeleton, an anthracene skeleton, a fluorene skeleton, a chrysene skeleton, a triphenylene skeleton, a tetracene skeleton, a pyrene skeleton, a perylene skeleton, a coumarin skeleton, a quinacridone skeleton, and a naphthobisbenzofuran skeleton is preferable because of its high fluorescence quantum yield.

[0232] In the case where a fluorescent substance is used as the light-emitting substance, a material having an anthracene skeleton is suitably used as the host material. The use of a substance having an anthracene skeleton as the host material for the fluorescent substance makes it possible to obtain a light-emitting layer with high emission efficiency and high durability. Among the substance having an anthracene skeleton that is used as the host material, a substance having a diphenylanthracene skeleton, in particular, a substance having a 9,10-diphenylanthracene skeleton, is chemically stable and thus is preferably used as the host material. The host material preferably has a carbazole skeleton because the hole-injection and hole-transport properties are improved; further preferably, the host material has a benzocarbazole skeleton in which a benzene ring is further condensed to a carbazole skeleton because the HOMO level thereof is higher than that of the host material having a carbazole skeleton by approximately 0.1 eV and thus holes enter the host material easily. In particular, the host material preferably has a dibenzocarbazole skeleton because the HOMO level thereof is higher than that of the host material having a carbazole skeleton by approximately 0.1 eV so that holes enter the host material easily, the hole-transport property is improved, and the heat resistance is increased. Accordingly, a substance having both a 9,10-diphenylanthracene skeleton and a carbazole skeleton (or a benzocarbazole or dibenzocarbazole skeleton) is further preferable as the host material. Note that in terms of the hole-injection and hole-transport properties described above, instead of a carbazole skeleton, a benzofluorene skeleton or a dibenzofluorene skeleton may be used. Examples of such a substance include 9-phenyl-3-[4-(10-phenyl-9-anthryl)phenyl]-9H-carbazole (abbreviation: PCzPA), 3-[4-(1-naphthyl)phenyl]-9-phenyl-9H-carbazole (abbreviation: PCPN), 9-[4-(10-phenyl-9-anthracenyl)phenyl]-9H-carbazole (abbreviation: CzPA), 7-[4-(10-phenyl-9-anthryl)phenyl]-7H-dibenzo[c,g]carbazole (abbreviation: cgDBCzPA), 6-[3-(9,10-diphenyl-2-anthryl)phenyl]benzo[b]naphtho[1,2-d]furan (abbreviation: 2mBnfPPA), 9-phenyl-10-[4′-(9-phenyl-9H-fluoren-9-yl) biphenyl-4′-yl]anthracene (abbreviation: FLPPA), 9-(1-naphthyl)-10-[4-(2-naphthyl)phenyl]anthracene (abbreviation: αN-βNPAnth), 9-(1-naphthyl)-10-(2-naphthyl) anthracene (abbreviation: α,β-ADN), 2-(10-phenylanthracen-9-yl)dibenzofuran, 2-(10-phenyl-9-anthracenyl)benzo[b]naphtho[2,3-d]furan (abbreviation: Bnf(II)PhA), 9-(2-naphthyl)-10-[3-(2-naphthyl)phenyl]anthracene (abbreviation: βN-mβNPAnth), and 1-{4-[10-(biphenyl-4-yl)-9-anthracenyl]phenyl}-2-ethyl-1H-benzimidazole (abbreviation: EtBImPBPhA). In particular, CzPA, cgDBCzPA, 2mBnfPPA, and PCzPA exhibit excellent properties and thus are preferably selected.

[0233] Note that the host material may be a mixture of a plurality of kinds of substances; in the case of using a mixed host material, it is preferable to mix a material having an electron-transport property with a material having a hole-transport property. By mixing the material having an electron-transport property with the material having a hole-transport property, the transport property of the light-emitting layer **113** can be easily adjusted and a recombination region can be easily controlled. The weight ratio of the content of the material having a hole-transport property to the content of the material having an electron-transport property may be 1:19 to 19:1.

[0234] Note that a phosphorescent substance can be used as part of the mixed material. When a fluorescent substance is used as the light-emitting substance, a phosphorescent substance can be used as an energy donor for supplying excitation energy to the fluorescent substance.
[0235] These mixed materials may form an exciplex. These mixed materials are preferably selected so as to form an exciplex that exhibits light emission whose wavelength overlaps with the wavelength on the lowest-energy-side absorption band of the light-emitting substance, in which case energy can be transferred smoothly and light emission can be obtained efficiently. The use of such a structure is preferable because the driving voltage can also be reduced.
[0236] Note that at least one of the materials forming an exciplex may be a phosphorescent substance. In this case, triplet excitation energy can be efficiently converted into singlet excitation energy by reverse intersystem crossing.
[0237] In order to form an exciplex efficiently, a material having an electron-transport property is preferably combined with a material having a hole-transport property and a HOMO level higher than or equal to that of the material having an electron-transport property. In addition, the LUMO level of the material having a hole-transport property is preferably higher than or equal to that of the material having an electron-transport property. Note that the LUMO levels and the HOMO levels of the materials can be calculated from the electrochemical characteristics (the reduction potentials and the oxidation potentials) of the materials that are measured by cyclic voltammetry (CV).
[0238] The formation of an exciplex can be confirmed by a phenomenon in which the emission spectrum of the mixed film in which the material having a hole-transport property and the material having an electron-transport property are mixed is shifted to the longer wavelength side than the emission spectrum of each of the materials (or has another peak on the longer wavelength side) observed by comparison of the emission spectra of the material having a hole-transport property, the material having an electron-transport property, and the mixed film of these materials, for example. Alternatively, the formation of an exciplex can be confirmed by a difference in transient response, such as a phenomenon in which the transient photoluminescence (PL) lifetime of the mixed film has longer lifetime components or has a larger proportion of delayed components than that of each of the materials, observed by comparison of transient PL of the material having a hole-transport property, the material having an electron-transport property, and the mixed film of these materials. The transient PL can be rephrased as transient electroluminescence (EL). That is, the formation of an exciplex can also be confirmed by a difference in transient response observed by comparison of the transient EL of the material having a hole-transport property, the material having an electron-transport property, and the mixed film of these materials.
[0239] The electron-transport layer **114** contains a substance with an electron-transport property. The substance with an electron-transport property is preferably a substance having an electron mobility higher than or equal to 1×10.sup.−7 cm.sup.2/Vs, further preferably higher than or equal to 1×10.sup.−6 cm.sup.2/Vs, when the square root of electric field strength [V/cm] is 600. Note that any other substance can also be used as long as the substance has an electron-transport property higher than a hole-transport property. The above organic compound is preferably an organic compound including a T-electron deficient heteroaromatic ring. The organic compound including a T-electron deficient heteroaromatic ring is preferably one or more of an organic compound including a heteroaromatic ring having a polyazole skeleton, an organic compound including a heteroaromatic ring having a pyridine skeleton, an organic compound including a heteroaromatic ring having a diazine skeleton, and an organic compound including a heteroaromatic ring having a triazine skeleton.
[0240] As the organic compound with an electron-transport property that can be used for the electron-transport layer **114**, any of the organic compounds that can be used as the organic compound with an electron-transport property in the light-emitting layer **113** and the second organic compound in the electron-injection layer **115** in Embodiment 1 can be similarly used.

Among the above organic compounds, the organic compound including a heteroaromatic ring having a diazine skeleton, the organic compound including a heteroaromatic ring having a pyridine skeleton, and the organic compound including a heteroaromatic ring having a triazine skeleton have high reliability and thus are preferable. In particular, the organic compound including a heteroaromatic ring having a diazine (pyrimidine or pyrazine) skeleton and the organic compound including a heteroaromatic ring having a triazine skeleton have a high electron-transport property to contribute to a reduction in driving voltage. An organic compound having a phenanthroline ring skeleton such as mTpPPhen, PnNPhen, or mPPhen2P is especially preferable, and an organic compound having a phenanthroline dimer structure such as mPPhen2P is further preferable because of its excellent stability.
[0241] The electron-transport layer preferably includes an organic compound having an electron-transport property with an acid dissociation constant pK.sub.a of less than 4.
[0242] Note that the electron-transport layer **114** may have a stacked-layer structure. In the case where the electron-transport layer **114** has a stacked-layer structure, the layer in contact with the light-emitting layer **113** may function as a hole-blocking layer. In the case where the electron-transport layer in contact with the light-emitting layer functions as a hole-blocking layer, the electron-transport layer is preferably formed using a material having a lower HOMO level than a material contained in the light-emitting layer by greater than or equal to 0.5 eV.
[0243] The electron-injection layer **115** is formed between the electron-transport layer **114** and the second electrode **102**. Since the structure of the electron-injection layer **115** has been described in detail in Embodiment 1, the repetitive description thereof is omitted.
[0244] The second electrode **102** is preferably formed in contact with the electron-injection layer **115**. A metal material can be used for the second electrode **102**, for example. Specifically, it is possible to use a metal such as aluminum (Al), titanium (Ti), chromium (Cr), manganese (Mn), iron (Fe), cobalt (Co), nickel (Ni), copper (Cu), gallium (Ga), zinc (Zn), indium (In), tin (Sn), molybdenum (Mo), tantalum (Ta), tungsten (W), palladium (Pd), gold (Au), platinum (Pt), silver (Ag), yttrium (Y), neodymium (Nd), or magnesium (Mg) or an alloy containing an appropriate combination of any of these metals, for example.
[0245] For the second electrode **102**, an oxide containing one or more selected from indium, tin, zinc, gallium, titanium, aluminum, and silicon can be used. For example, it is preferable to use a conductive oxide containing one or more of indium oxide, indium tin oxide, indium zinc oxide, zinc oxide, zinc oxide containing gallium, titanium oxide, indium zinc oxide containing gallium, indium zinc oxide containing aluminum, indium tin oxide containing silicon, indium zinc oxide containing silicon, and the like. Note that in the case where the light-emitting device **130** is a top-emission light-emitting device, a conductive metal oxide having a light-transmitting property is preferably used for the second electrode **102**.
[0246] When the second electrode **102** is formed using a material that transmits visible light, the light-emitting device can emit light from the second electrode **102** side. When the first electrode **101** is formed using a material that transmits visible light, the light-emitting device can emit light from the first electrode **101** side.
[0247] Films of these conductive materials of the second electrode **102** can be formed by a dry process such as a vacuum evaporation method or a sputtering method, an ink-jet method, a spin coating method, or the like. Alternatively, a wet process using a sol-gel method or a wet process using a paste of a metal material may be employed.
[0248] Note that in the case of a top-emission light-emitting device, forming a cap layer by evaporation of an organic compound over the second electrode can improve light extraction efficiency. The cap layer may have a single-layer structure or a stacked-layer structure. In the case of a stacked-layer structure, the use of organic compounds with different refractive indexes can further increase the light extraction efficiency.
[0249] The organic compound layer **103** can be formed by any of a variety of methods, including a

dry process and a wet process. For example, a vacuum evaporation method, a gravure printing method, an offset printing method, a screen printing method, an ink-jet method, a spin coating method, or the like may be used.
[0250] Different film formation methods may be used to form the electrodes or the layers described above.
[0251] In the case where the film to be the second electrode **102** is formed by a film formation method that causes great damage to a base, such as a sputtering method, a p-type layer **117** may be provided as illustrated in FIG. **1**B in order to protect the electron-injection layer **115**. The p-type layer **117** can be formed using the composite material described above as a material that can be used for the hole-injection layer **111**. A transition metal oxide such as molybdenum oxide, vanadium oxide, ruthenium oxide, tungsten oxide, or manganese oxide is more robust than an organic compound, and thus is preferably used as the substance having an acceptor property in the p-type layer, in which case damage at the time of forming the film to be the second electrode **102** can be prevented.
[0252] Although not illustrated, an electron-relay layer may be provided between the electron-injection layer **115** and the p-type layer **117**. The electron-relay layer contains at least a substance with an electron-transport property and has a function of preventing an interaction between the electron-injection layer **115** and the p-type layer **117** and smoothly transferring electrons. The LUMO level of the substance with an electron-transport property included in the electron-relay layer is preferably between the LUMO level of the acceptor substance in the p-type layer **117** and the LUMO level of a substance contained in a layer of the electron-transport layer **114** that is in contact with the electron-injection layer **115**. Specifically, the LUMO level of the substance with an electron-transport property in the electron-relay layer is preferably higher than or equal to −5.0 eV, further preferably higher than or equal to −5.0 eV and lower than or equal to −3.0 eV. Note that as the substance with an electron-transport property in the electron-relay layer, a phthalocyanine-based material or a metal complex having a metal-oxygen bond and an aromatic ligand is preferably used. As the substance with an electron-transport property used for the electron-relay layer, specifically, it is possible to use diquinoxalino[2,3-a: 2′,3′-c]phenazine (abbreviation: HATNA), 2,3,8,9,14,15-hexafluorodiquinoxalino[2,3-a: 2′,3′-c]phenazine (abbreviation: HATNA-F6), a perylenetetracarboxylic acid derivative such as 3,4,9,10-perylenetetracarboxylic diimide (abbreviation: PTCDI) or 3,4,9,10-perylenetetracarboxyl-bis-benzimidazole (abbreviation: PTCBI), (C.sub.60-Ih) [5,6]fullerene (abbreviation: C.sub.60), (C.sub.70-D.sub.5h) [5,6]fullerene (abbreviation: C.sub.70), or the like. It is also possible to use a compound having a heterophane skeleton, which is a cyclophane skeleton having a hetero ring; for example, a phthalocyanine compound such as phthalocyanine (abbreviation: H.sub.2Pc) can be used as the compound. Alternatively, it is possible to use a metal phthalocyanine containing copper, zinc, cobalt, iron, chromium, nickel, or the like or a derivative thereof, such as copper phthalocyanine (abbreviation: CuPc), zinc phthalocyanine (abbreviation: ZnPc), cobalt phthalocyanine (abbreviation: CoPc), iron phthalocyanine (abbreviation: FePc), tin phthalocyanine (abbreviation: SnPc), tin oxide phthalocyanine (abbreviation: SnOPc), titanium oxide phthalocyanine (abbreviation: TiOPc), or vanadium oxide phthalocyanine (abbreviation: VOPc). It is particularly preferable to use a phthalocyanine-based metal complex such as copper phthalocyanine or zinc phthalocyanine or 2,3,8,9,14,15-hexafluorodiquinoxalino[2,3-a: 2′,3′-c]phenazine.
[0253] The thickness of the electron-relay layer is preferably greater than or equal to 1 nm and less than or equal to 10 nm, further preferably greater than or equal to 2 nm and less than or equal to 5 nm.
[0254] Next, an embodiment of a light-emitting device having a structure where a plurality of light-emitting units are stacked (this type of light-emitting device is also referred to as a stacked or tandem device) is described with reference to FIG. **1**C. This light-emitting device includes a plurality of light-emitting units between an anode and a cathode. One light-emitting unit has

substantially the same structure as the organic compound layer **103** illustrated in FIG. **1**A. In other words, the light-emitting device illustrated in FIG. **1**C includes a plurality of light-emitting units, and the light-emitting device illustrated in FIG. **1**A includes a single light-emitting unit.

[0255] In FIG. **1**C, a first light-emitting unit **511** and a second light-emitting unit **512** are stacked between a first electrode **501** and a second electrode **502**, and an intermediate layer **513** is provided between the first light-emitting unit **511** and the second light-emitting unit **512**. The first electrode **501** and the second electrode **502** correspond, respectively, to the first electrode **101** and the second electrode **102** illustrated in FIG. **1**A, and can be formed using the materials given in the description for FIG. **1**A. Furthermore, the first light-emitting unit **511** and the second light-emitting unit **512** may have the same layer structure or different layer structures. Materials for the layers included in the first light-emitting unit **511** and materials for the layers included in the second light-emitting unit **512** may be the same or different from each other.

[0256] The intermediate layer **513** has a function of injecting electrons into one of the light-emitting units and injecting holes into the other of the light-emitting units when voltage is applied between the first electrode **501** and the second electrode **502**. That is, in FIG. **1**C, the intermediate layer **513** injects electrons into the first light-emitting unit **511** and holes into the second light-emitting unit **512** when voltage is applied such that the potential of the anode becomes higher than the potential of the cathode.

[0257] The intermediate layer **513** includes a charge-generation layer. The charge-generation layer includes at least the p-type layer **117**. The p-type layer **117** is preferably formed using any of the composite materials given above as examples of materials that can be used for the hole-injection layer **111**. The p-type layer **117** may be formed by stacking a film containing the above-described acceptor material of the composite material and a film containing the above-described hole-transport material of the composite material. When a potential is applied to the p-type layer **117**, electrons are injected into the electron-transport layer **114** and holes are injected into the cathode; thus, the light-emitting device operates.

[0258] Note that the intermediate layer **513** preferably includes one or both of an electron-relay layer **118** and an n-type layer **119** in addition to the p-type layer **117**.

[0259] The electron-relay layer **118** has a structure similar to that of the electron-relay layer mentioned in the description for FIG. **1**B; thus, the repetitive description thereof is omitted.

[0260] The n-type layer **119** can be formed using a substance having a high electron-injection property, e.g., an alkali metal, an alkaline earth metal, a rare earth metal, or a compound thereof (an alkali metal compound (including an oxide such as lithium oxide, a halide, and a carbonate such as lithium carbonate or cesium carbonate), an alkaline earth metal compound (including an oxide, a halide, and a carbonate), or a rare earth metal compound (including an oxide, a halide, and a carbonate)).

[0261] In the case where the n-type layer **119** contains a substance having an electron-transport property and a donor substance, the donor substance can be an organic compound such as tetrathianaphthacene (abbreviation: TTN), nickelocene, or decamethylnickelocene, as well as an alkali metal, an alkaline earth metal, a rare earth metal, or a compound thereof (e.g., an alkali metal compound (including an oxide such as lithium oxide, a halide, and a carbonate such as lithium carbonate or cesium carbonate), an alkaline earth metal compound (including an oxide, a halide, and a carbonate), or a rare earth metal compound (including an oxide, a halide, and a carbonate)). As the substance having an electron-transport property, a material similar to the above-described material for the electron-transport layer **114** can be used.

[0262] Instead of the n-type layer **119**, a layer containing the metal, the first organic compound, and the second organic compound described as being used as the electron-injection layer in Embodiment 1 may be formed in the same position as the n-type layer **119**. In the case of such a structure, a tandem light-emitting device with favorable characteristics can be manufactured.

[0263] In the case where the anode-side surface of a light-emitting unit is in contact with the intermediate layer **513**, the charge-generation layer of the intermediate layer **513** can also function as a hole-injection layer of the light-emitting unit; therefore, a hole-injection layer is not necessarily provided in the light-emitting unit. In the case where the cathode-side surface of a light-emitting unit is in contact with the intermediate layer **513**, the intermediate layer **513** can also function as an electron-injection layer of the light-emitting unit; therefore, an electron-injection layer is not necessarily provided in the light-emitting unit.

[0264] The light-emitting device including two light-emitting units is described with reference to FIG. **1**C; however, one embodiment of the present invention can also be applied to a light-emitting device in which three or more light-emitting units are stacked. With a plurality of light-emitting units partitioned by the intermediate layer **513** between a pair of electrodes as in the light-emitting device of this embodiment, it is possible to provide a long-life element that can emit light with high luminance at a low current density. A display apparatus that can be driven at a low voltage and has low power consumption can also be provided.

[0265] When the emission colors of the light-emitting units are different, light emission of a desired color can be obtained from the light-emitting device as a whole. For example, in a light-emitting device including two light-emitting units, the emission colors of the first light-emitting unit may be red and green and the emission color of the second light-emitting unit may be blue, so that the light-emitting device can emit white light as the whole.

[0266] The organic compound layer **103**, the first light-emitting unit **511**, the second light-emitting unit **512**, the layers such as the intermediate layer **513**, and the electrodes that are described above can be formed by a method such as an evaporation method (including a vacuum evaporation method), a droplet discharge method (also referred to as an ink-jet method), a coating method, or a gravure printing method. A low molecular material, a middle molecular material (including an oligomer and a dendrimer), or a high molecular material may be included in the above components.

[0267] FIG. **5**A illustrates two adjacent light-emitting devices (light-emitting devices **130***a* and **130***b*) included in a display apparatus of one embodiment of the present invention.

[0268] The light-emitting device **130***a* includes an organic compound layer **103***a* between a first electrode **101***a* over an insulating layer **175** and a second electrode **102***a* facing the first electrode **101***a*, and the organic compound layer **103***a* includes an electron-injection layer **115***a*. The illustrated organic compound layer **103***a* includes a hole-injection layer **111***a*, a hole-transport layer **112***a*, a light-emitting layer **113***a*, an electron-transport layer **114***a*, and the electron-injection layer **115***a*, but may have a different stacked-layer structure.

[0269] The light-emitting device **130***b* includes an organic compound layer **103***b* between a first electrode **101***b* over the insulating layer **175** and a second electrode **102***b* facing the first electrode **101***b*, and the organic compound layer **103***b* includes an electron-injection layer **115***b*. The illustrated organic compound layer **103***b* includes a hole-injection layer **111***b*, a hole-transport layer **112***b*, a light-emitting layer **113***b*, an electron-transport layer **114***b*, and the electron-injection layer **115***b*, but may have a different stacked-layer structure.

[0270] The structures of the electron-injection layer **115***a* and the second electrode **102***a* in the light-emitting device **130***a* and the structures of the electron-injection layer **115***b* and the second electrode **102***b* in the light-emitting device **130***b* are preferably those described in Embodiment 1.

[0271] The organic compound layers **103***a* and **103***b* are independent of each other and the second electrodes **102***a* and **102***b* are independent of each other because processing by a photolithography method is performed after formation of a film to be the second electrode **102***a* and after formation of a film to be the second electrode **102***b*. In the light-emitting device of one embodiment of the present invention, even when processing by a photolithography method is performed after formation of the film to be the second electrode **102***a* and after formation of the film to be the second electrode **102***b*, the light-emitting device can have favorable characteristics.

[0272] The end portions (outlines) of the second electrode **102***a* and the organic compound layer **103***a* are substantially aligned in the direction perpendicular to the substrate due to the processing by a photolithography method. The end portions (outlines) of the second electrode **102***b* and the organic compound layer **103***b* are substantially aligned in the direction perpendicular to the substrate due to the processing by a photolithography method.

[0273] There is a space d between the organic compound layer **103***a* and the organic compound layer **103***b* due to the processing by a photolithography method. Since the organic compound layers are processed by a photolithography method, the distance between the first electrode **101***a* and the first electrode **101***b* can be small as compared with the case where mask vapor deposition is performed, and can be greater than or equal to 0.5 μm and less than or equal to 5 μm.

[0274] FIG. **5**B illustrates two adjacent tandem light-emitting devices (light-emitting devices **130***c* and **130***d*) manufactured by a photolithography method.

[0275] The light-emitting device **130***c* includes an organic compound layer **103***c* between a first electrode **101***c* over the insulating layer **175** and a second electrode **102***c*. The organic compound layer **103***c* has a structure where a first light-emitting unit **501***c* and a second light-emitting unit **502***c* are stacked with an intermediate layer **116***c* therebetween. Although FIG. **5**B illustrates an example where the two light-emitting units are stacked, three or more light-emitting units may be stacked. FIG. **5**B illustrates a structure where the first light-emitting unit **501***c* includes a hole-injection layer **111***c*, a first hole-transport layer **112***c*_**1**, a first light-emitting layer **113***c*_**1**, and a first electron-transport layer **114***c*_**1**; the intermediate layer **116***c* includes a p-type layer **117***c*, an electron-relay layer **118***c*, and an n-type layer **119***c*; and the second light-emitting unit **502***c* includes a second hole-transport layer **112***c*_**2**, a second light-emitting layer **113***c*_**2**, a second electron-transport layer **114***c*_**2**, and an electron-injection layer **115***c*. The electron-relay layer **118***c* is not necessarily provided.

[0276] The light-emitting device **130***d* includes an organic compound layer **103***d* between a first electrode **101***d* over the insulating layer **175** and a second electrode **102***d*. The organic compound layer **103***d* has a structure where a first light-emitting unit **501***d* and a second light-emitting unit **502***d* are stacked with an intermediate layer **116***d* therebetween. Although FIG. **5**B illustrates an example where the two light-emitting units are stacked, three or more light-emitting units may be stacked. FIG. **5**B illustrates a structure where the first light-emitting unit **501***d* includes a hole-injection layer **111***d*, a first hole-transport layer **112***d*_**1**, a first light-emitting layer **113***d*_**1**, and a first electron-transport layer **114***d*_**1**; the intermediate layer **116***d* includes a p-type layer **117***d*, an electron-relay layer **118***d*, and an n-type layer **119***d*; and the second light-emitting unit **502***d* includes a second hole-transport layer **112***d*_**2**, a second light-emitting layer **113***d*_**2**, a second electron-transport layer **114***d*_**2**, and an electron-injection layer **115***d*. The electron-relay layer **118***d* is not necessarily provided.

[0277] In the light-emitting devices **130***c* and **130***d*, the electron-injection layers **115***c* and **115***d* and the second electrodes **102***c* and **102***d* preferably have the structures described in Embodiment 1. In addition, the structure of the electron-injection layer described in Embodiment 1 is preferably employed for the n-type layer **119***c* in the intermediate layer **116***c* and the n-type layer **119***d* in the intermediate layer **116***d*.

[0278] The organic compound layers **103***c* and **103***d* are independent of each other because processing by a photolithography method is performed after formation of a film to be the second electrode **102***c* and after formation of a film to be the second electrode **102***d*. In the light-emitting device of one embodiment of the present invention, even when processing by a photolithography method is performed after formation of the film to be the second electrode **102***c* and after formation of the film to be the second electrode **102***d*, the light-emitting device can have favorable characteristics.

[0279] The end portions (outlines) of the second electrode **102***c* and the organic compound layer **103***c* are substantially aligned in the direction perpendicular to the substrate due to the processing by a photolithography method. The end portions (outlines) of the second electrode **102***d* and the organic compound layer **103***d* are substantially aligned in the direction perpendicular to the substrate due to the processing by a photolithography method.

[0280] There is the space d between the organic compound layer **103***c* and the organic compound layer **103***d* due to the processing by a photolithography method. Since the organic compound layers are processed by a photolithography method, the distance between the first electrode **101***c* and the first electrode **101***d* can be small as compared with the case where mask vapor deposition is performed, and can be greater than or equal to 0.5 μm and less than or equal to 5 μm.

[0281] Since the second electrodes **102***a* and **102***b* are independent of each other or the second electrodes **102***c* and **102***d* are independent of each other, an auxiliary electrode **105** is preferably formed in order to apply voltage to a plurality of the second electrodes included in the display apparatus. The auxiliary electrode **105** is preferably formed after a short circuit between adjacent organic compound layers or between adjacent first electrodes is prevented by forming an insulating layer **106** between the light-emitting devices **130***a* and **130***b* or between the light-emitting devices **130***c* and **130***d*. The insulating layer **106** is preferably formed using an organic insulating material. For the auxiliary electrode **105**, a material that can be used for the second electrode can be used.

[0282] In the light-emitting device of one embodiment of the present invention, since the organic compound layer is processed by a photolithography method, the organic compound layer can be processed with a sufficient precision to manufacture a high-resolution display apparatus. Furthermore, since a lithography process can be performed on the electron-injection layer far from the light-emitting layer without contamination by an alkali metal, the light-emitting device can have favorable characteristics. As described above, the light-emitting device of one embodiment of the present invention having the above-described structure enables a high-resolution display apparatus and can have favorable characteristics.

[0283] Since the second electrode and the organic compound layer of the light-emitting device of one embodiment of the present invention are processed at a time by a photolithography method after formation of the second electrode, the outlines of the layers included in the organic compound layer are substantially aligned with each other when seen from a direction substantially perpendicular to the surface of the insulating layer where the first electrode is formed. In addition, in a cross-sectional view, the end portion of the second electrode and the end portion of the first layer are aligned with each other in a direction substantially perpendicular to the surface of the insulating layer where the first electrode is formed. Here, “aligned” or “substantially aligned” in this specification means, supposing that the organic compound layer includes a layer A and a layer B, a difference between an outline A of the layer A and an outline B of the layer B is within 5% of the width of the organic compound layer along a line orthogonal to the compared portions of the outlines. Furthermore, “substantially perpendicular” means an angle of 85° to 95°.

[0284] The structure described in this embodiment can be used in combination with any of the other structures as appropriate.

Embodiment 3

[0285] Described in this embodiment is a mode in which the light-emitting device of one embodiment of the present invention is used as a display element of a display apparatus.

[0286] As illustrated in FIGS. **6**A and **6**B, a plurality of light-emitting devices **130** are formed over the insulating layer **175** to constitute a display apparatus.

[0287] A display apparatus includes a pixel portion **177** in which a plurality of pixels **178** are arranged in matrix. The pixel **178** includes a subpixel **110**R, a subpixel **110**G, and a subpixel **110**B.

[0288] In this specification and the like, for example, description common to the subpixels **110**R, **110**G, and **110**B is sometimes made using the collective term “subpixel **110**”. As for other components that are distinguished from each other using letters of the alphabet, matters common to the components are sometimes described using reference numerals without the letters of the alphabet.

[0289] The subpixel **110**R emits red light, the subpixel **110**G emits green light, and the subpixel **110**B emits blue light. Thus, a full-color image can be displayed on the pixel portion **177**. Note that

in this embodiment, three colors of red (R), green (G), and blue (B) are given as examples of colors of light emitted by the subpixels; however, subpixels of a different combination of colors may be employed. The number of subpixels is not limited to three, and may be four or more. Examples of four subpixels include subpixels emitting light of four colors of R, G, B, and white (W), subpixels emitting light of four colors of R, G, B, and yellow (Y), and four subpixels emitting light of R, G, and B and infrared light (IR).
[0290] In this specification and the like, the row direction and the column direction are sometimes referred to as the X direction and the Y direction, respectively. The X direction and the Y direction intersect with each other and are perpendicular to each other, for example.
[0291] FIG. **6**A illustrates an example where subpixels of different colors are arranged in the X direction and subpixels of the same color are arranged in the Y direction. Note that subpixels of different colors may be arranged in the Y direction, and subpixels of the same color may be arranged in the X direction.
[0292] Note that the layout of the subpixels is not limited thereto, and a variety of modes such as stripe arrangement, S-stripe arrangement, matrix arrangement, delta arrangement, Bayer arrangement, and PenTile arrangement can be employed. FIGS. **7**A to **7**G illustrate layout examples of subpixels.
[0293] The pixel **178** illustrated in FIG. **7**A employs S-stripe arrangement. The pixel **178** illustrated in FIG. **7**A includes three subpixels: the subpixel **110**R, the subpixel **110**G, and the subpixel **110**B.
[0294] The pixel **178** illustrated in FIG. **7**B includes the subpixel **110**R having a rough trapezoidal or rough triangle planar shape with rounded corners, the subpixel **110**G having a rough trapezoidal or rough triangle planar shape with rounded corners, and the subpixel **110**B having a rough tetragonal or rough hexagonal planar shape with rounded corners. The subpixel **110**R has a larger light-emitting area than the subpixel **110**G. In this manner, the shapes and sizes of the subpixels can be determined independently. For example, the size of a subpixel including a light-emitting device with higher reliability can be smaller.
[0295] Pixels **124***a* and **124***b* illustrated in FIG. **7**C employ PenTile arrangement. FIG. **7**C illustrates an example where the pixels **124***a* including the subpixels **110**R and **110**G and the pixels **124***b* including the subpixels **110**G and **110**B are alternately arranged.
[0296] The pixels **124***a* and **124***b* illustrated in FIGS. **7**D to **7**F employ delta arrangement. The pixel **124***a* includes two subpixels (the subpixels **110**R and **110**G) in the upper row (first row) and one subpixel (the subpixel **110**B) in the lower row (second row). The pixel **124***b* includes one subpixel (the subpixel **110**B) in the upper row (first row) and two subpixels (the subpixels **110**R and **110**G) in the lower row (second row).
[0297] FIG. **7**D illustrates an example where the top surface of each subpixel has a rough tetragonal shape with rounded corners, FIG. **7**E illustrates an example where the top surface of each subpixel has a circular shape, and FIG. **7**F illustrates an example where the top surface of each subpixel has a rough hexagonal shape with rounded corners.
[0298] In FIG. **7**F, subpixels are placed in respective hexagonal regions that are arranged densely. Focusing on one of the subpixels, the subpixel is placed so as to be surrounded by six subpixels. The subpixels are arranged such that subpixels that emit light of the same color are not adjacent to each other. For example, focusing on the subpixel **110**R, the subpixel **110**R is surrounded by three subpixels **110**G and three subpixels **110**B that are alternately arranged.
[0299] FIG. **7**G illustrates an example where subpixels of different colors are arranged in a zigzag manner. Specifically, the positions of the top sides of two subpixels arranged in the row direction (e.g., the subpixels **110**R and **110**G or the subpixels **110**G and **110**B) are not aligned in the top view.
[0300] In the pixels illustrated in FIGS. **7**A to **7**G, for example, it is preferable that the subpixel **110**R be a subpixel R emitting red light, the subpixel **110**G be a subpixel G emitting green light, and the subpixel **110**B be a subpixel B emitting blue light. Note that the structures of the subpixels

are not limited thereto, and the colors and the order of the subpixels can be determined as appropriate. For example, the subpixel **110**G may be the subpixel R emitting red light, and the subpixel **110**R may be the subpixel G emitting green light.

[0301] In the case of what is called stripe arrangement as illustrated in FIGS. **6**A and **7**G, the second electrodes **102** of the light-emitting devices exhibiting the same emission color can be successively formed. In this case, even when processing by a photolithography method is performed after the components up to the second electrode **102** are formed, voltage can be applied to the light-emitting devices without the auxiliary electrode **105**. In the case where the second electrodes **102** are independent of each other on a light-emitting device basis after processing by a photolithography method, the auxiliary electrode **105** is preferably formed.

[0302] Outside the pixel portion **177**, a connection portion **140** is provided and a region **141** may also be provided. The region **141** is provided between the pixel portion **177** and the connection portion **140**. The organic compound layer **103** is provided in the region **141**. A conductive layer **151**C is provided in the connection portion **140**.

[0303] Although FIG. **6**A illustrates an example where the region **141** and the connection portion **140** are positioned on the right side of the pixel portion **177**, the positions of the region **141** and the connection portion **140** are not particularly limited. The number of the regions **141** and the number of the connection portions **140** can each be one or more.

[0304] FIG. **6**B is an example of a cross-sectional view along the dashed-dotted line A**1**-A**2** in FIG. **6**A. As illustrated in FIG. **6**B, the display apparatus includes an insulating layer **171**, a conductive layer **172** over the insulating layer **171**, an insulating layer **173** over the insulating layer **171** and the conductive layer **172**, an insulating layer **174** over the insulating layer **173**, and the insulating layer **175** over the insulating layer **174**. The insulating layer **171** is provided over a substrate (not illustrated). An opening reaching the conductive layer **172** is provided in the insulating layers **175**, **174**, and **173**, and a plug **176** is provided to fill the opening.

[0305] In the pixel portion **177**, the light-emitting device **130** is provided over the insulating layer **175** and the plug **176**. A protective layer **131** is provided to cover the light-emitting device **130**. A substrate **120** is bonded to the protective layer **131** with a resin layer **122**. An inorganic insulating layer **125** and an insulating layer **127** over the inorganic insulating layer **125** are preferably provided between the adjacent light-emitting devices **130**.

[0306] Examples of materials used for the insulating layer **127** include an acrylic resin, a polyimide resin, an epoxy resin, an imide resin, a polyamide resin, a polyimide-amide resin, a silicone resin, a siloxane resin, a benzocyclobutene-based resin, a phenol resin, and precursors of these resins. The insulating layer **127** may be formed using an organic material such as polyvinyl alcohol (PVA), polyvinyl butyral, polyvinylpyrrolidone, polyethylene glycol, polyglycerin, pullulan, water-soluble cellulose, or an alcohol-soluble polyamide resin.

[0307] Moreover, the insulating layer **127** can be formed using a photosensitive resin. A photoresist may be used as the photosensitive resin. As the photosensitive resin, a positive photosensitive material or a negative photosensitive material can be used.

[0308] The insulating layer **127** may contain a material absorbing visible light. For example, the insulating layer **127** itself may be formed using a material absorbing visible light, or the insulating layer **127** may contain a pigment absorbing visible light. For example, the insulating layer **127** can be formed using a resin that can be used as a color filter transmitting red, blue, or green light and absorbing light of the other colors; or a resin that contains carbon black as a pigment and functions as a black matrix.

[0309] Although FIG. **6**B illustrates cross sections of a plurality of the inorganic insulating layers **125** and a plurality of the insulating layers **127**, the inorganic insulating layers **125** are preferably connected to each other and the insulating layers **127** are preferably connected to each other when the display apparatus is seen from above. In other words, the inorganic insulating layer **125** and the insulating layer **127** preferably include opening portions over first electrodes.

[0310] In FIG. **6**B, a light-emitting device **130**R, a light-emitting device **130**G, and a light-emitting device **130**B are each illustrated as the light-emitting device **130**. The light-emitting devices **130**R, **130**G, and **130**B emit light of different colors. For example, the light-emitting device **130**R can emit red light, the light-emitting device **130**G can emit green light, and the light-emitting device **130**B can emit blue light. Alternatively, the light-emitting device **130**R, the light-emitting device **130**G, or the light-emitting device **130**B may emit visible light of another color or infrared light.
[0311] The display apparatus of one embodiment of the present invention can be, for example, a top-emission display apparatus where light is emitted in the direction opposite to a substrate over which light-emitting devices are formed. Note that the display apparatus of one embodiment of the present invention may be a bottom-emission display apparatus.
[0312] The light-emitting device **130**R has the structure described in Embodiments 1 and 2. The light-emitting device **130**R includes a first electrode **101**R (pixel electrode) including a conductive layer **151**R and a conductive layer **152**R, an organic compound layer **103**R over the first electrode **101**R, and a second electrode **102**R over the organic compound layer **103**R. The electron-injection layer, which is the outermost surface layer of the organic compound layer **103**R, and the second electrode **102**R have the structures described in Embodiments 1 and 2. Such a structure can reduce damage to the light-emitting layer during a photolithography process, enabling the light-emitting device **130**R to have favorable film quality and electrical characteristics.
[0313] The light-emitting device **130**G has the structure described in Embodiments 1 and 2. The light-emitting device **130**G includes a first electrode **101**G (pixel electrode) including a conductive layer **151**G and a conductive layer **152**G, an organic compound layer **103**G over the first electrode **101**G, and a second electrode **102**G over the organic compound layer **103**G. The electron-injection layer, which is the outermost surface layer of the organic compound layer **103**G, and the second electrode **102**G have the structures described in Embodiments 1 and 2. Such a structure can reduce damage to the light-emitting layer during a photolithography process, enabling the light-emitting device **130**G to have favorable film quality and electrical characteristics.
[0314] The light-emitting device **130**B has the structure described in Embodiment 1 and 2. The light-emitting device **130**B includes a first electrode **101**B (pixel electrode) including a conductive layer **151**B and a conductive layer **152**B, an organic compound layer **103**B over the first electrode **101**B, and a second electrode **102**B over the organic compound layer **103**B. The electron-injection layer, which is the outermost surface layer of the organic compound layer **103**B, and the second electrode **102**B have the structures described in Embodiments 1 and 2. Such a structure can reduce damage to the light-emitting layer during a photolithography process, enabling the light-emitting device **130**B to have favorable film quality and electrical characteristics.
[0315] The organic compound layers **103**R, **103**G, and **103**B are island-shaped layers that are independent of each other on a light-emitting device basis or on an emission color basis. It is preferable that the organic compound layers **103**R, **103**G, and **103**B not overlap with one another. Providing the island-shaped organic compound layer **103** in each of the light-emitting devices **130** can inhibit leakage current between the adjacent light-emitting devices **130** even in a high-resolution display apparatus. This can prevent crosstalk, so that a display apparatus with extremely high contrast can be obtained. Specifically, a display apparatus having high current efficiency at low luminance can be obtained.
[0316] The second electrodes **102**R, **102**G, and **102**B are island-shaped layers that are independent of each other on a light-emitting device basis or an emission color basis. It is preferable that the second electrodes **102**R, **102**G, and **102**B not overlap with one another.
[0317] After the insulating layer **127** is formed over the second electrode **102** to cover the side surface of the light-emitting device **130**, the auxiliary electrode **105** is preferably formed, in which case voltage can be easily supplied to the second electrode **102**. A metal material can be used for the auxiliary electrode **105**, for example. Specifically, it is possible to use a metal such as aluminum (Al), titanium (Ti), chromium (Cr), manganese (Mn), iron (Fe), cobalt (Co), nickel (Ni),

copper (Cu), gallium (Ga), zinc (Zn), indium (In), tin (Sn), molybdenum (Mo), tantalum (Ta), tungsten (W), palladium (Pd), gold (Au), platinum (Pt), silver (Ag), yttrium (Y), neodymium (Nd), or magnesium (Mg) or an alloy containing an appropriate combination of any of these metals, for example.

[0318] For the auxiliary electrode **105**, an oxide containing one or more selected from indium, tin, zinc, gallium, titanium, aluminum, and silicon can also be used. For example, it is preferable to use a conductive oxide containing one or more of indium oxide, indium tin oxide, indium zinc oxide, zinc oxide, zinc oxide containing gallium, titanium oxide, indium zinc oxide containing gallium, indium zinc oxide containing aluminum, indium tin oxide containing silicon, indium zinc oxide containing silicon, and the like. In the case where the light-emitting device **130** is a top-emission light-emitting device, a conductive metal oxide having a light-transmitting property is preferably used for the auxiliary electrode **105**.

[0319] The island-shaped organic compound layer **103** is formed by forming an organic compound film, forming the film to be the second electrode **102**, and then processing the organic compound film and the film to be the second electrode **102** by a photolithography method. In the light-emitting device of one embodiment of the present invention, the electron-injection layer and the second electrode have the structure described in Embodiment 1; thus, even when processing is performed by a photolithography method after formation of the second electrode **102**, the light-emitting device can have favorable characteristics in which an increase in driving voltage is suppressed. In addition, processing by a photolithography method after formation of the second electrode **102** enables a light-emitting device with favorable reliability to be obtained at low cost.

[0320] In the display apparatus of one embodiment of the present invention, the first electrode **101** (pixel electrode) of the light-emitting device preferably has a stacked-layer structure. For example, in the example illustrated in FIG. **6**B, the first electrode **101** of the light-emitting device **130** is a stack of the conductive layer **151** on the insulating layer **171** side and the conductive layer **152** on the organic compound layer side. In this specification and the like, for example, description common to the conductive layers **151**R, **151**G, and **151**B is sometimes made using the collective term “conductive layer **151**”. In this specification and the like, description common to the conductive layers **152**R, **152**G, and **152**B is sometimes made using the collective term “conductive layer **152**”.

[0321] A metal material can be used for the conductive layer **151**, for example. Specifically, it is possible to use a metal such as aluminum (Al), titanium (Ti), chromium (Cr), manganese (Mn), iron (Fe), cobalt (Co), nickel (Ni), copper (Cu), gallium (Ga), zinc (Zn), indium (In), tin (Sn), molybdenum (Mo), tantalum (Ta), tungsten (W), palladium (Pd), gold (Au), platinum (Pt), silver (Ag), yttrium (Y), or neodymium (Nd) or an alloy containing an appropriate combination of any of these metals, for example.

[0322] For the conductive layer **152**, an oxide containing one or more selected from indium, tin, zinc, gallium, titanium, aluminum, and silicon can be used. For example, it is preferable to use a conductive oxide containing one or more of indium oxide, indium tin oxide, indium zinc oxide, zinc oxide, zinc oxide containing gallium, titanium oxide, indium zinc oxide containing gallium, indium zinc oxide containing aluminum, indium tin oxide containing silicon, indium zinc oxide containing silicon, and the like. In particular, an indium tin oxide containing silicon can be suitably used for the conductive layer **152** because of having a work function of higher than or equal to 4.0 eV, for example.

[0323] The conductive layer **151** and the conductive layer **152** may each be a stack of a plurality of layers containing different materials. In that case, the conductive layer **151** may include a layer formed using a material that can be used for the conductive layer **152**, such as a conductive oxide. Furthermore, the conductive layer **152** may include a layer formed using a material that can be used for the conductive layer **151**, such as a metal material. In the case where the conductive layer **151** is a stack of two or more layers, for example, a layer in contact with the conductive layer **152** can be formed using a material that can be used for the conductive layer **152**.

[0324] In FIG. **6**B, the conductive layer **151** has a tapered end portion. Specifically, the conductive layer **151** preferably has a tapered end portion with a taper angle less than 90°. In that case, the conductive layer **152** provided along the side surface of the conductive layer **151** also has a tapered shape. When the side surface of the conductive layer **152** has a tapered shape, coverage with the organic compound layer **103** provided along the side surface of the conductive layer **152** can be improved.

[0325] The end portions of the conductive layers **151** and **152** may have no tapered shape, that is, have substantially a vertical shape. The end portion of the organic compound layer **103** is preferably positioned inward from the first electrode **101**. In this case, leakage current through the organic compound layer **103** can be reduced, so that a display apparatus with a low driving voltage and favorable display performance can be obtained.

[0326] Since the light-emitting device **130** has the structure described in Embodiments 1 and 2, the display apparatus of one embodiment of the present invention can be a display apparatus with favorable reliability.

[0327] Next, a method for manufacturing the display apparatus having the structure illustrated in FIG. **6**A is described with reference to FIGS. **8**A to **8**E, FIGS. **9**A and **9**B, FIGS. **10**A to **10**D, FIGS. **11**A to **11**C, FIGS. **12**A to **12**C, and FIGS. **13**A and **13**B.

[Manufacturing Method Example]

[0328] Thin films included in the display apparatus (e.g., insulating films, semiconductor films, and conductive films) can be formed by a sputtering method, a chemical vapor deposition (CVD) method, a vacuum evaporation method, a pulsed laser deposition (PLD) method, an atomic layer deposition (ALD) method, or the like.

[0329] Thin films included in the display apparatus (e.g., insulating films, semiconductor films, and conductive films) can also be formed by a wet process such as spin coating, dipping, spray coating, ink-jetting, dispensing, screen printing, offset printing, doctor blade coating, slit coating, roll coating, curtain coating, or knife coating.

[0330] Thin films included in the display apparatus can be processed by a photolithography method, for example.

[0331] As light used for exposure in the photolithography method, for example, light with the i-line (wavelength: 365 nm), light with the g-line (wavelength: 436 nm), light with the h-line (wavelength: 405 nm), or light in which two or more of the i-line, the g-line, and the h-line are mixed can be used. Alternatively, ultraviolet rays, KrF laser light, ArF laser light, or the like can be used. Exposure may be performed by liquid immersion exposure technique. As the light for exposure, extreme ultraviolet (EUV) light or X-rays may also be used. Furthermore, instead of the light used for the exposure, an electron beam can also be used.

[0332] For etching of thin films, a dry etching method, a wet etching method, a sandblasting method, or the like can be used.

[0333] First, as illustrated in FIG. **8**A, the insulating layer **171** is formed over a substrate (not illustrated). Next, the conductive layer **172** and a conductive layer **179** are formed over the insulating layer **171**, and the insulating layer **173** is formed over the insulating layer **171** so as to cover the conductive layer **172** and the conductive layer **179**. Then, the insulating layer **174** is formed over the insulating layer **173**, and the insulating layer **175** is formed over the insulating layer **174**.

[0334] As the substrate, a substrate that has heat resistance high enough to withstand at least heat treatment performed later can be used. For example, it is possible to use a glass substrate; a quartz substrate; a sapphire substrate; a ceramic substrate; an organic resin substrate; or a semiconductor substrate such as a single crystal semiconductor substrate or a polycrystalline semiconductor substrate of silicon, silicon carbide, or the like, a compound semiconductor substrate of silicon germanium or the like, or an SOI substrate.

[0335] Next, as illustrated in FIG. **8**A, an opening reaching the conductive layer **172** is formed in the insulating layers **175**, **174**, and **173**. Then, the plug **176** is formed to fill the opening.
[0336] Subsequently, as illustrated in FIG. **8**A, a conductive film **151***f* to be the conductive layers **151**R, **151**G, **151**B, and **151**C and a conductive film **152***f* to be the conductive layers **152**R, **152**G, **152**B, and **152**C are formed over the plug **176** and the insulating layer **175**. A metal material can be used for the conductive film **151***f*, for example. For the conductive film **152***f*, an oxide containing one or more selected from indium, tin, zinc, gallium, titanium, aluminum, and silicon can be used.
[0337] Then, as illustrated in FIG. **8**A, a resist mask **191** is formed over the conductive film **152***f*. The resist mask **191** can be formed by application of a photosensitive material (photoresist), light exposure, and development.
[0338] Subsequently, as illustrated in FIG. **8**B, the conductive films **151***f* and **152***f* in a region not overlapping with the resist mask **191** are removed, for example. In this manner, the conductive layers **151** and **152** are formed.
[0339] Next, the resist mask **191** is removed as illustrated in FIG. **8**C. The resist mask **191** can be removed by ashing using oxygen plasma, for example.
[0340] Then, as illustrated in FIG. **8**D, an insulating film **156***f* to be an insulating layer **156**R, an insulating layer **156**G, an insulating layer **156**B, and an insulating layer **156**C is formed over the conductive layers **152**R, **152**G, **152**B, and **152**C and the insulating layer **175**. In this specification and the like, description common to the conductive layers **156**R, **156**G, and **156**B is sometimes made using the collective term “conductive layer **156**”.
[0341] As the insulating film **156***f*, an inorganic insulating film such as an oxide insulating film, a nitride insulating film, an oxynitride insulating film, or a nitride oxide insulating film, e.g., silicon oxynitride, can be used.
[0342] Subsequently, as illustrated in FIG. **8**E, the insulating film **156***f* is processed to form the insulating layers **156**R, **156**G, **156**B, and **156**C.
[0343] Next, as illustrated in FIG. **9**A, an organic compound film **103**Rf is formed over the conductive layers **152**R, **152**G, and **152**B and the insulating layer **175**. As illustrated in FIG. **9**A, the organic compound film **103**Rf is not formed over the conductive layer **152**C.
[0344] Next, as illustrated in FIG. **9**A, a conductive film **102**Rf to be the second electrode is formed over the organic compound film **103**Rf, and then a sacrificial film **158**Rf and a mask film **159**Rf are formed over the conductive film **102**Rf. Providing the sacrificial film **158**Rf and the mask film **159**Rf over the organic compound film **103**Rf with the conductive film **102**Rf therebetween can reduce damage to the organic compound film **103**Rf in the manufacturing process of the display apparatus, resulting in an increase in the reliability of the light-emitting device. Note that in this specification, a sacrificial film is referred to as a mask film in some cases. Similarly, a sacrificial layer is referred to as a mask layer in some cases.
[0345] In the case where the second electrode **102** is the electrode through which light is extracted, a material having a visible-light-transmitting property is preferably used for the conductive film **102**Rf. For example, a conductive material having a visible light reflectivity higher than or equal to 20% and lower than or equal to 80%, preferably higher than or equal to 40% and lower than or equal to 70%, and a resistivity lower than or equal to 1×10.sup.−2 Ω.Math.cm can be used. In the case where a material with low light transmittance, such as metal or alloy, is used, the conductive film **102**Rf may be formed to a thickness that is thin enough to transmit visible light (e.g., a thickness of 1 nm to 10 nm). Specific examples include an oxide semiconductor layer and an organic conductor layer containing an organic substance, in addition to an oxide conductor layer typified by ITO. Examples of the organic conductive layer containing an organic substance include a layer containing a composite material in which an organic compound and an electron donor (donor) are mixed and a layer containing a composite material in which an organic compound and an electron acceptor (acceptor) are mixed. The resistivity of the transparent conductive layer is preferably lower than or equal to 1×10.sup.5 Ω.Math.cm, further preferably lower than or equal to

1×10.sup.4 Ω.Math.cm.

[0346] The conductive film **102**Rf can be formed by a dry process such as a vacuum evaporation method or a sputtering method, an ink-jet method, a spin coating method, or the like. Alternatively, a wet process using a sol-gel method or a wet process using a paste of a metal material may be employed. In particular, the conductive film **102**Rf formed over and in contact with the organic compound film **103**Rf is preferably formed by a formation method that is less likely to damage the organic compound film **103**Rf. For example, the conductive film **102**Rf is preferably formed by an ALD method or a vacuum evaporation method.

[0347] The sacrificial film **158**Rf and the mask film **159**Rf are provided as appropriate. For example, in the case where the organic compound film **103**Rf can be sufficiently protected by the conductive film **102**Rf, the mask film **159**Rf can be formed over the conductive film **102**Rf to omit the step of forming the sacrificial film **158**Rf. Alternatively, for example, in the case where the etching selectivity of the organic compound film **103**Rf to the conductive film **102**Rf and the etching selectivity of the conductive film **102**Rf to the sacrificial film **158**Rf are sufficiently high, the step of forming the mask film **159**Rf may be omitted because the sacrificial film **158**Rf can be used as a mask.

[0348] As the sacrificial film **158**Rf, a film that is highly resistant to the processing conditions for the organic compound film **103**Rf, specifically, a film having high etching selectivity with respect to the organic compound film **103**Rf is used. As the mask film **159**Rf, a film having high etching selectivity with respect to the sacrificial film **158**Rf is used.

[0349] The conductive film **102**Rf, the sacrificial film **158**Rf, and the mask film **159**Rf are preferably formed at a temperature lower than the upper temperature limit of the organic compound film **103**Rf. The typical substrate temperatures in formation of the sacrificial film **158**Rf and the mask film **159**Rf are each higher than or equal to 100° C. and lower than or equal to 200° C., preferably higher than or equal to 100° C. and lower than or equal to 150° C., further preferably higher than or equal to 100° C. and lower than or equal to 120° C. Since the light-emitting device of one embodiment of the present invention contains a first compound, a display apparatus with favorable display quality can be manufactured even through a heating step performed at higher temperatures.

[0350] The sacrificial film **158**Rf and the mask film **159**Rf are preferably films that can be removed by a wet etching method or a dry etching method.

[0351] Note that the sacrificial film **158**Rf preferably has denser film quality than the mask film **159**Rf. For example, the sacrificial film **158**Rf is preferably formed by an ALD method or a vacuum evaporation method rather than a sputtering method.

[0352] As each of the sacrificial film **158**Rf and the mask film **159**Rf, one or more of a metal film, an alloy film, a metal oxide film, a semiconductor film, an organic insulating film, an inorganic insulating film, and the like can be used, for example.

[0353] For each of the sacrificial film **158**Rf and the mask film **159**Rf, it is possible to use a metal material such as gold, silver, platinum, magnesium, nickel, tungsten, chromium, molybdenum, iron, cobalt, copper, palladium, titanium, aluminum, yttrium, zirconium, or tantalum or an alloy material containing any of the metal materials, for example. It is particularly preferable to use a low-melting-point material such as aluminum or silver. It is preferable to use a metal material that can block ultraviolet rays for one or both of the sacrificial film **158**Rf and the mask film **159**Rf, in which case the organic compound film **103**Rf can be inhibited from being irradiated with ultraviolet rays in light exposure for patterning and deterioration of the organic compound film **103**Rf can be inhibited.

[0354] The sacrificial film **158**Rf and the mask film **159**Rf can each be formed using a metal oxide such as an In—Ga—Zn oxide, an indium oxide, an In—Zn oxide, an In—Sn oxide, an indium titanium oxide (In—Ti oxide), an indium tin zinc oxide (In—Sn—Zn oxide), an indium titanium zinc oxide (In—Ti—Zn oxide), an indium gallium tin zinc oxide (In—Ga—Sn—Zn oxide), or an

indium tin oxide containing silicon.
[0355] In the above metal oxide, in place of gallium, an element M (M is one or more of aluminum, silicon, boron, yttrium, copper, vanadium, beryllium, titanium, iron, nickel, germanium, zirconium, molybdenum, lanthanum, cerium, neodymium, hafnium, tantalum, tungsten, and magnesium) may be used.
[0356] The sacrificial film **158**Rf and the mask film **159**Rf are preferably formed using a semiconductor material such as silicon or germanium for excellent compatibility with a semiconductor manufacturing process. Alternatively, a compound containing the above semiconductor material can be used.
[0357] As each of the sacrificial film **158**Rf and the mask film **159**Rf, any of a variety of inorganic insulating films can be used. In particular, an oxide insulating film is preferable because its adhesion to the organic compound film **103**Rf is higher than that of a nitride insulating film.
[0358] Subsequently, a resist mask **190**R is formed as illustrated in FIG. **9**A. The resist mask **190**R can be formed by application of a photosensitive material (photoresist), light exposure, and development.
[0359] The resist mask **190**R is provided at a position overlapping with the conductive layer **152**R. The resist mask **190**R is preferably provided also at a position overlapping with the conductive layer **152**C. This can inhibit the conductive layer **152**C from being damaged during the manufacturing process of the display apparatus.
[0360] Next, as illustrated in FIG. **9**B, part of the mask film **159**Rf is removed using the resist mask **190**R, so that the mask layer **159**R is formed. The mask layer **159**R remains over the conductive layers **152**R and **152**C. After that, the resist mask **190**R is removed. Then, part of the sacrificial film **158**Rf and part of the conductive film **102**Rf are removed using the mask layer **159**R as a mask (also referred to as a hard mask), whereby a sacrificial layer **158**R and the second electrode **102**R are formed.
[0361] The use of a wet etching method can reduce damage to the organic compound film **103**Rf in processing of the conductive film **102**Rf, the sacrificial film **158**Rf, and the mask film **159**Rf, as compared with the case of using a dry etching method. In the case of using a wet etching method, it is preferable to use a developer, an alkaline aqueous solution such as a tetramethylammonium hydroxide (TMAH) aqueous solution, or an acid aqueous solution such as dilute hydrofluoric acid, oxalic acid, phosphoric acid, acetic acid, nitric acid, or a chemical solution containing a mixed solution of any of these acids, for example.
[0362] In the case of using a dry etching method to process the sacrificial film **158**Rf and the conductive film **102**Rf, deterioration of the organic compound film **103**Rf can be inhibited by not using a gas containing oxygen as the etching gas.
[0363] The resist mask **190**R can be removed by a method similar to that for the resist mask **191**.
[0364] Next, as illustrated in FIG. **9**B, the organic compound film **103**Rf is processed, so that the organic compound layer **103**R is formed. For example, part of the conductive film **102**Rf and part of the organic compound film **103**Rf are removed using the mask layer **159**R and the sacrificial layer **158**R as a hard mask, whereby the organic compound layer **103**R is formed.
[0365] Accordingly, as illustrated in FIG. **9**B, the stacked-layer structure of the organic compound layer **103**R, the second electrode **102**R, the sacrificial layer **158**R, and the mask layer **159**R remains over the conductive layer **152**R. The conductive layers **152**G and **152**B are exposed.
[0366] The organic compound film **103**Rf is preferably processed by anisotropic etching. Anisotropic dry etching is particularly preferable. Alternatively, wet etching may be used.
[0367] In the case of using a dry etching method, deterioration of the organic compound film **103**Rf can be inhibited by not using a gas containing oxygen as the etching gas.
[0368] A gas containing oxygen may be used as the etching gas. When the etching gas contains oxygen, the etching rate can be increased. Thus, the etching can be performed under a low-power condition while an adequately high etching rate is maintained. Accordingly, damage to the organic

compound film **103**Rf can be reduced. Furthermore, a defect such as attachment of a reaction product generated during the etching can be inhibited.

[0369] In the case of using a dry etching method, it is preferable to use a gas containing at least one of H.sub.2, CF.sub.4, C.sub.4F.sub.8, SF.sub.6, CHF.sub.3, C.sub.12, H.sub.2O, BCl.sub.3, and a Group 18 element such as He or Ar as the etching gas, for example. Alternatively, a gas containing oxygen and at least one of the above is preferably used as the etching gas. Alternatively, an oxygen gas may be used as the etching gas.

[0370] Next, as illustrated in FIG. **10**A, an organic compound film **103**Gf to be the organic compound layer **103**G and a conductive film **102**Gf to be the second electrode **102**G are formed.

[0371] The organic compound film **103**Gf can be formed by a method similar to that for forming the organic compound film **103**Rf. The organic compound film **103**Gf can have a structure similar to that of the organic compound film **103**Rf. The conductive film **102**Gf can be formed by a method similar to that for forming the conductive film **102**Rf. The conductive film **102**Gf can have a structure similar to that of the conductive film **102**Rf.

[0372] Subsequently, as illustrated in FIG. **10**A, a sacrificial film **158**Gf and a mask film **159**Gf are formed in this order. After that, a resist mask **190**G is formed. The materials and the formation methods of the sacrificial film **158**Gf and the mask film **159**Gf are similar to those of the sacrificial film **158**Rf and the mask film **159**Rf. The material and the formation method of the resist mask **190**G are similar to those of the resist mask **190**R.

[0373] The resist mask **190**G is provided at a position overlapping with the conductive layer **152**G.

[0374] Subsequently, as illustrated in FIG. **10**B, part of the mask film **159**Gf is removed using the resist mask **190**G, so that a mask layer **159**G is formed. The mask layer **159**G remains over the conductive layer **152**G. After that, the resist mask **190**G is removed. Then, part of the sacrificial film **158**Gf and part of the conductive film **102**Gf are removed using the mask layer **159**G as a mask, so that the sacrificial layer **158**G and the second electrode **102**G are formed. Next, the organic compound film **103**Gf is processed to form the organic compound layer **103**G.

[0375] Next, as illustrated in FIG. **10**C, an organic compound film **103**Bf to be the organic compound layer **103**B and a conductive film **102**Bf to be the second electrode **102**B are formed.

[0376] The organic compound film **103**Bf can be formed by a method similar to that for forming the organic compound film **103**Rf. The organic compound film **103**Bf can have a structure similar to that of the organic compound film **103**Rf. The conductive film **102**Bf can be formed by a method similar to that for forming the conductive film **102**Rf. The conductive film **102**Bf can have a structure similar to that of the conductive film **102**Rf.

[0377] Subsequently, as illustrated in FIG. **10**C, a sacrificial film **158**Bf and a mask film **159**Bf are formed in this order. After that, a resist mask **190**B is formed. The materials and the formation methods of the sacrificial film **158**Bf and the mask film **159**Bf are similar to those of the sacrificial film **158**Rf and the mask film **159**Rf. The material and the formation method of the resist mask **190**B are similar to those of the resist mask **190**R.

[0378] The resist mask **190**B is provided at a position overlapping with the conductive layer **152**B.

[0379] Subsequently, as illustrated in FIG. **10**D, part of the mask film **159**Bf is removed using the resist mask **190**B, so that a mask layer **159**B is formed. The mask layer **159**B remains over the conductive layer **152**B. After that, the resist mask **190**B is removed. Then, part of the sacrificial film **158**Bf and part of the conductive film **102**Bf are removed using the mask layer **159**B as a mask, so that a sacrificial layer **158**B and the second electrode **102**B are formed. Next, the organic compound film **103**Bf is processed to form the organic compound layer **103**B. For example, part of the organic compound film **103**Bf is removed using the mask layer **159**B and the sacrificial layer **158**B as a hard mask, whereby the organic compound layer **103**B is formed.

[0380] Accordingly, as illustrated in FIG. **10**D, the stacked-layer structure of the organic compound layer **103**B, the second electrode **102**B, the sacrificial layer **158**B, and the mask layer **159**B remains over the conductive layer **152**B. The mask layers **159**R and **159**G are exposed.

[0381] Note that the side surfaces of the stacked-layer structure of the organic compound layer **103**R and the second electrode **102**R, the stacked-layer structure of the organic compound layer **103**G and the second electrode **102**G, and the stacked-layer structure of the organic compound layer **103**B and the second electrode **102**B are preferably perpendicular or substantially perpendicular to the formation surfaces. For example, the angle between the formation surfaces and these side surfaces is preferably greater than or equal to 60° and less than or equal to 90°.
[0382] The distance between two adjacent stacked-layer structures among the stacked-layer structure of the organic compound layer **103**R and the second electrode **102**R, the stacked-layer structure of the organic compound layer **103**G and the second electrode **102**G, and the stacked-layer structure of the organic compound layer **103**B and the second electrode **102**B, which are formed by a photolithography method as described above, can be reduced to less than or equal to 8 μm, less than or equal to 5 μm, less than or equal to 3 μm, less than or equal to 2 μm, or less than or equal to 1 μm. Here, the distance can be specified, for example, by a distance between facing end portions of two adjacent layers among the organic compound layers **103**R, **103**G, and **103**B. Reducing the distance between the island-shaped organic compound layers makes it possible to provide a display apparatus having high resolution and a high aperture ratio. In addition, the distance between the first electrodes of adjacent light-emitting devices can also be reduced to, for example, less than or equal to 10 μm, less than or equal to 8 μm, less than or equal to 5 μm, less than or equal to 3 μm, or less than or equal to 2 μm. Note that the distance between the first electrodes of adjacent light-emitting devices is preferably greater than or equal to 2 μm and less than or equal to 5 μm.
[0383] Next, as illustrated in FIG. **11**A, the mask layers **159**R, **159**G, and **159**B and the sacrificial layers **158**R, **158**G, and **158**B are preferably removed.
[0384] Note that in the case where the light-emitting devices are arranged in what is called a stripe pattern as illustrated in FIG. **7**G, the second electrode **102** can be formed as a continuous layer over the light-emitting devices that exhibit the same color. In that case, since the auxiliary electrode **105** described later is not necessarily formed, the mask layer **159** is not necessarily removed in the case of a bottom-emission structure; thus, the process may proceed to the step in FIG. **13**B after the step in FIG. **10**D. In the case of the top-emission structure, the sacrificial layer **158** and the mask layer **159** are not necessarily removed when having a light-transmitting property, and the process can proceed to the step in FIG. **13**B after the step in FIG. **10**D. When not having a light-transmitting property, the sacrificial layer **158** and the mask layer **159** are preferably removed, and after removal of the sacrificial layer **158** and/or the mask layer **159** (after the step in FIG. **11**A), the process can proceed to the step in FIG. **13**B.
[0385] The step of removing the mask layers can be performed by a method similar to that for the step of processing the mask films. Specifically, by using a wet etching method, damage to the organic compound layer **103** at the time of removing the mask layers can be reduced as compared with the case of using a dry etching method.
[0386] The mask layers may be removed by being dissolved in a polar solvent such as water or an alcohol. Examples of an alcohol include ethyl alcohol, methyl alcohol, isopropyl alcohol (IPA), and glycerin.
[0387] After the mask layers are removed, drying treatment may be performed in order to remove water adsorbed on surfaces. For example, heat treatment in an inert gas atmosphere or a reduced-pressure atmosphere can be performed. The heat treatment can be performed at a substrate temperature higher than or equal to 50° C. and lower than or equal to 200° C., preferably higher than or equal to 60° C. and lower than or equal to 150° C., further preferably higher than or equal to 70° C. and lower than or equal to 120° C. The heat treatment is preferably performed in a reduced-pressure atmosphere, in which case drying at a lower temperature is possible.
[0388] Next, an inorganic insulating film **125*f*** is formed as illustrated in FIG. **11**B.
[0389] Then, as illustrated in FIG. **11**C, an insulating film **127*f*** to be the insulating layer **127** is

formed over the inorganic insulating film **125***f*.

[0390] The substrate temperatures in formation of the inorganic insulating film **125***f* and the insulating film **127***f* are each preferably higher than or equal to 60° C., higher than or equal to 80° C., higher than or equal to 100° C., or higher than or equal to 120° C. and lower than or equal to 200° C., lower than or equal to 180° C., lower than or equal to 160° C., lower than or equal to 150° C., or lower than or equal to 140° C.

[0391] As the inorganic insulating film **125***f*, an insulating film having a thickness greater than or equal to 3 nm, greater than or equal to 5 nm, or greater than or equal to 10 nm and less than or equal to 200 nm, less than or equal to 150 nm, less than or equal to 100 nm, or less than or equal to 50 nm is preferably formed in the above-described range of the substrate temperature.

[0392] The inorganic insulating film **125***f* is preferably formed by an ALD method, for example. An ALD method is preferably used, in which case deposition damage can be reduced and a film with good coverage can be formed. As the inorganic insulating film **125***f*, an aluminum oxide film is preferably formed by an ALD method, for example.

[0393] The insulating film **127***f* is preferably formed by the aforementioned wet process. The insulating film **127***f* is preferably formed by spin coating using a photosensitive material, for example, and specifically preferably formed using a photosensitive resin composition containing an acrylic resin.

[0394] Then, part of the insulating film **127***f* is exposed to visible light or ultraviolet rays. The insulating layer **127** is formed in regions that are interposed between any two of the conductive layers **152**R, **152**G, and **152**B and around the conductive layer **152**C.

[0395] The width of the insulating layer **127** formed later can be controlled with the exposed region of the insulating film **127***f*. In this embodiment, processing is performed such that the insulating layer **127** includes a portion overlapping with the top surface of the conductive layer **151**.

[0396] Light used for the exposure preferably includes the i-line (wavelength: 365 nm). Furthermore, light used for the exposure may include at least one of the g-line (wavelength: 436 nm) and the h-line (wavelength: 405 nm).

[0397] Next, as illustrated in FIG. **12**A, development is performed to remove the exposed region of the insulating film **127***f*, so that an insulating layer **127***a* is formed.

[0398] Next, as illustrated in FIG. **12**B, etching treatment is performed using the insulating layer **127***a* as a mask to remove part of the inorganic insulating film **125***f*. Thus, the inorganic insulating layer **125** is formed under the insulating layer **127***a*. Note that the etching treatment using the insulating layer **127***a* as a mask may be hereinafter referred to as first etching treatment.

[0399] The first etching treatment can be performed by dry etching or wet etching.

[0400] In the case of performing dry etching, a chlorine-based gas is preferably used. As the chlorine-based gas, one of Cl_2, BCl_3, $SiCl_4$, CCl_4, and the like or a mixture of two or more of them can be used.

[0401] As a dry etching apparatus, a dry etching apparatus including a high-density plasma source can be used. As the dry etching apparatus including a high-density plasma source, an inductively coupled plasma (ICP) etching apparatus can be used, for example. Alternatively, a capacitively coupled plasma (CCP) etching apparatus including parallel-plate electrodes can be used.

[0402] The first etching treatment is preferably performed by wet etching. The use of a wet etching method can reduce damage to the structure to be processed as compared with the case of using a dry etching method. Wet etching can be performed using an alkaline solution, for example. For instance, TMAH, which is an alkaline solution, can be used for the wet etching of an aluminum oxide film. Alternatively, an acid solution containing fluoride can also be used.

[0403] Next, light exposure is preferably performed on the entire substrate so that the insulating layer **127***a* is irradiated with visible light or ultraviolet rays. The energy density for the light exposure is preferably greater than 0 mJ/cm^2 and less than or equal to 800 mJ/cm^2, further preferably greater than 0 mJ/cm^2 and less than or equal to 500 mJ/cm^2.

Performing such light exposure after the development can sometimes increase the degree of transparency of the insulating layer **127***a*. In addition, it is sometimes possible to lower the substrate temperature required for subsequent heat treatment for changing the shape of the insulating layer **127***a* into a tapered shape.

[0404] Then, heat treatment (also referred to as post-baking) is performed. The heat treatment can change the insulating layer **127***a* into the insulating layer **127** having a tapered side surface (FIG. **12**C). The heat treatment is performed at a temperature lower than the upper temperature limit of the EL layer. The heat treatment can be performed at a substrate temperature higher than or equal to 50° C. and lower than or equal to 200° C., preferably higher than or equal to 60° C. and lower than or equal to 150° C., further preferably higher than or equal to 70° C. and lower than or equal to 130° C. The heating atmosphere may be an air atmosphere or an inert gas atmosphere. Moreover, the heating atmosphere may be an atmospheric-pressure atmosphere or a reduced-pressure atmosphere. Accordingly, adhesion between the insulating layer **127** and the inorganic insulating layer **125** can be improved, and corrosion resistance of the insulating layer **127** can be increased.

[0405] Next, as illustrated in FIG. **13**A, the auxiliary electrode **105** is formed over the second electrode **102**R, the second electrode **102**B, the second electrode **102**G, the conductive layer **152**C, and the insulating layer **127**. The auxiliary electrode **105** can be formed by a sputtering method, a vacuum evaporation method, or the like.

[0406] Next, as illustrated in FIG. **13**B, the protective layer **131** is formed over the auxiliary electrode **105**. The protective layer **131** can be formed by a vacuum evaporation method, a sputtering method, a CVD method, an ALD method, or the like.

[0407] Then, the substrate **120** is bonded to the protective layer **131** with the resin layer **122**, so that the display apparatus can be manufactured. In the method for manufacturing the display apparatus of one embodiment of the present invention, the insulating layer **156** is formed to include a region overlapping with the side surface of the conductive layer **151** and the conductive layer **152** is formed to cover the conductive layer **151** and the insulating layer **156** as described above. This can increase the yield of the display apparatus and inhibit generation of defects.

[0408] As described above, in the method for manufacturing the display apparatus of one embodiment of the present invention, the island-shaped organic compound layers **103**R, **103**G, and **103**B are formed not by using a fine metal mask but by processing a film formed on the entire surface; thus, the island-shaped layers can be formed to have a uniform thickness. In addition, a high-resolution display apparatus or a display apparatus with a high aperture ratio can be obtained. Furthermore, even when the resolution or the aperture ratio is high and the distance between the subpixels is extremely short, the organic compound layers **103**R, **103**G, and **103**B can be inhibited from being in contact with each other in the adjacent subpixels. As a result, generation of a leakage current between the subpixels can be inhibited. This can prevent crosstalk, so that a display apparatus with extremely high contrast can be obtained. Moreover, even a display apparatus that includes tandem light-emitting devices formed by a photolithography method can have favorable characteristics.

Embodiment 4

[0409] In this embodiment, display apparatuses of one embodiment of the present invention will be described.

[0410] The display apparatus in this embodiment can be a high-resolution display apparatus. Thus, the display apparatus in this embodiment can be used for display portions of information terminals (wearable devices) such as watch-type and bracelet-type information terminals and display portions of wearable devices capable of being worn on a head, such as a VR device like a head mounted display (HMD) and a glasses-type AR device.

[0411] The display apparatus in this embodiment can be a high-definition display apparatus or a large-sized display apparatus. Accordingly, the display apparatus in this embodiment can be used for display portions of a digital camera, a digital video camera, a digital photo frame, a mobile phone, a portable game console, a portable information terminal, and an audio reproducing device, in addition to display portions of electronic appliances with a relatively large screen, such as a television device, desktop and laptop personal computers, a monitor of a computer and the like, digital signage, and a large game machine such as a pachinko machine. [Display module]

[0412] FIG. **14**A is a perspective view of a display module **280**. The display module **280** includes a display apparatus **100**A and an FPC **290**. Note that the display apparatus included in the display module **280** is not limited to the display apparatus **100**A and may be any of display apparatuses **100**B to **100**G described later.

[0413] The display module **280** includes a substrate **291** and a substrate **292**. The display module **280** includes a display portion **281**. The display portion **281** is a region of the display module **280** where an image is displayed, and is a region where light emitted from pixels provided in a pixel portion **284** described later can be seen.

[0414] FIG. **14**B is a perspective view schematically illustrating a structure on the substrate **291** side. Over the substrate **291**, a circuit portion **282**, a pixel circuit portion **283** over the circuit portion **282**, and the pixel portion **284** over the pixel circuit portion **283** are stacked. In addition, a terminal portion **285** for connection to the FPC **290** is included in a portion over the substrate **291** that does not overlap with the pixel portion **284**. The terminal portion **285** and the circuit portion **282** are electrically connected to each other through a wiring portion **286** formed of a plurality of wirings.

[0415] The pixel portion **284** includes a plurality of pixels **284***a* arranged periodically. An enlarged view of one pixel **284***a* is illustrated on the right side in FIG. **14**B. The pixels **284***a* can employ any of the structures described in the above embodiments. FIG. **14**B illustrates an example where the pixel **284***a* has a structure similar to that of the pixel **178** illustrated in FIG. **6**A.

[0416] The pixel circuit portion **283** includes a plurality of pixel circuits **283***a* arranged periodically.

[0417] One pixel circuit **283***a* is a circuit that controls driving of a plurality of elements included in one pixel **284***a*.

[0418] The circuit portion **282** includes a circuit for driving the pixel circuits **283***a* in the pixel circuit portion **283**. For example, the circuit portion **282** preferably includes one or both of a gate line driver circuit and a source line driver circuit. The circuit portion **282** may also include at least one of an arithmetic circuit, a memory circuit, a power supply circuit, and the like.

[0419] The FPC **290** functions as a wiring for supplying a video signal, a power supply potential, or the like to the circuit portion **282** from the outside. An IC may be mounted on the FPC **290**.

[0420] The display module **280** can have a structure where one or both of the pixel circuit portion **283** and the circuit portion **282** are stacked below the pixel portion **284**; hence, the aperture ratio (effective display area ratio) of the display portion **281** can be significantly high.

[0421] Such a display module **280** has extremely high resolution, and thus can be suitably used for a VR device such as an HMD or a glasses-type AR device. For example, even in the case of a structure where the display portion of the display module **280** is seen through a lens, pixels of the extremely-high-resolution display portion **281** included in the display module **280** are prevented from being recognized when the display portion is enlarged by the lens, so that display providing a high sense of immersion can be performed. Without being limited thereto, the display module **280** can be suitably used for electronic devices including a relatively small display portion.

[Display Apparatus **100**A]

[0422] The display apparatus **100**A illustrated in FIG. **15**A includes a substrate **301**, the light-emitting devices **130**R, **130**G, and **130**B, a capacitor **240**, and a transistor **310**.

[0423] The substrate **301** corresponds to the substrate **291** in FIGS. **14**A and **14**B. The transistor **310** includes a channel formation region in the substrate **301**. As the substrate **301**, a semiconductor substrate such as a single crystal silicon substrate can be used, for example. The transistor **310** includes part of the substrate **301**, a conductive layer **311**, a low-resistance region **312**, an

insulating layer **313**, and an insulating layer **314**. The conductive layer **311** functions as a gate electrode. The insulating layer **313** is positioned between the substrate **301** and the conductive layer **311** and functions as a gate insulating layer. The low-resistance region **312** is a region where the substrate **301** is doped with an impurity, and functions as a source or a drain. The insulating layer **314** is provided to cover the side surface of the conductive layer **311**.
[0424] An element isolation layer **315** is provided between two adjacent transistors **310** to be embedded in the substrate **301**.
[0425] An insulating layer **261** is provided to cover the transistor **310**, and the capacitor **240** is provided over the insulating layer **261**.
[0426] The capacitor **240** includes a conductive layer **241**, a conductive layer **245**, and an insulating layer **243** between the conductive layers **241** and **245**. The conductive layer **241** functions as one electrode of the capacitor **240**, the conductive layer **245** functions as the other electrode of the capacitor **240**, and the insulating layer **243** functions as a dielectric of the capacitor **240**.
[0427] The conductive layer **241** is provided over the insulating layer **261** and is embedded in an insulating layer **254**. The conductive layer **241** is electrically connected to one of the source and the drain of the transistor **310** through a plug **271** embedded in the insulating layer **261**. The insulating layer **243** is provided to cover the conductive layer **241**. The conductive layer **245** is provided in a region overlapping with the conductive layer **241** with the insulating layer **243** therebetween.
[0428] An insulating layer **255** is provided to cover the capacitor **240**. The insulating layer **174** is provided over the insulating layer **255**. The insulating layer **175** is provided over the insulating layer **174**. The light-emitting devices **130**R, **130**G, and **130**B are provided over the insulating layer **175**. An insulator is provided in regions between adjacent light-emitting devices.
[0429] The insulating layer **156**R is provided to include a region overlapping with the side surface of the conductive layer **151**R. The insulating layer **156**G is provided to include a region overlapping with the side surface of the conductive layer **151**G. The insulating layer **156**B is provided to include a region overlapping with the side surface of the conductive layer **151**B. The conductive layer **152**R is provided to cover the conductive layer **151**R and the insulating layer **156**R. The conductive layer **152**G is provided to cover the conductive layer **151**G and the insulating layer **156**G. The conductive layer **152**B is provided to cover the conductive layer **151**B and the insulating layer **156**B. Note that the insulating layer **156** is not necessarily provided.
[0430] Each of the conductive layers **151**R, **151**G, and **151**B is electrically connected to one of the source and the drain of the corresponding transistor **310** through a plug **256** embedded in the insulating layers **243**, **255**, **174**, and **175**, the conductive layer **241** embedded in the insulating layer **254**, and the plug **271** embedded in the insulating layer **261**. Any of a variety of conductive materials can be used for the plugs.
[0431] The protective layer **131** is provided over the light-emitting devices **130**R, **130**G, and **130**B with the auxiliary electrode **105** therebetween. The substrate **120** is bonded to the protective layer **131** with the resin layer **122**. Embodiment 3 can be referred to for the details of the light-emitting device **130** and the components thereover up to the substrate **120**. The substrate **120** corresponds to the substrate **292** in FIG. **14**A.
[0432] FIG. **15**B illustrates a variation example of the display apparatus **100**A illustrated in FIG. **15**A. The display apparatus illustrated in FIG. **15**B includes coloring layers **132**R, **132**G, and **132**B, and each of the light-emitting devices **130** includes a region overlapping with one of the coloring layers **132**R, **132**G, and **132**B. In the display apparatus illustrated in FIG. **15**B, the light-emitting device **130** can emit white light, for example. The coloring layer **132**R, the coloring layer **132**G, and the coloring layer **132**B can transmit red light, green light, and blue light, respectively, for example.
[Display Apparatus **100**B]
[0433] FIG. **16** is a perspective view of the display apparatus **100**B.

[0434] In the display apparatus **100**B, a substrate **352** and a substrate **351** are bonded to each other. In FIG. **16**, the substrate **352** is denoted by a dashed line.
[0435] The display apparatus **100**B includes the pixel portion **177**, the connection portion **140**, a circuit **356**, a wiring **355**, and the like. FIG. **16** illustrates an example where an IC **354** and an FPC **353** are mounted on the display apparatus **100**B. Thus, the structure illustrated in FIG. **16** can be regarded as a display module including the display apparatus **100**B, the integrated circuit (IC), and the FPC. Here, a display apparatus in which a substrate is equipped with a connector such as an FPC or mounted with an IC is referred to as a display module.
[0436] The connection portion **140** is provided outside the pixel portion **177**. The number of connection portions **140** may be one or more. In the connection portion **140**, a common electrode of a light-emitting device is electrically connected to a conductive layer, so that a potential can be supplied to the common electrode.
[0437] As the circuit **356**, a scan line driver circuit can be used, for example.
[0438] The wiring **355** has a function of supplying a signal and power to the pixel portion **177** and the circuit **356**. The signal and power are input to the wiring **355** from the outside through the FPC **353** or from the IC **354**.
[0439] FIG. **16** illustrates an example where the IC **354** is provided over the substrate **351** by a chip on glass (COG) method, a chip on film (COF) method, or the like. An IC including a scan line driver circuit, a signal line driver circuit, or the like can be used as the IC **354**, for example. Note that the display apparatus **100**B and the display module are not necessarily provided with an IC. Alternatively, the IC may be mounted on the FPC by a COF method, for example.
[0440] FIG. **17** illustrates an example of cross sections of part of a region including the FPC **353**, part of the circuit **356**, part of the pixel portion **177**, and a region including part of the connection portion **140** of the display apparatus **100**C.
[Display Apparatus **100**C]
[0441] The display apparatus **100**C illustrated in FIG. **17** includes a transistor **201**, a transistor **205**, the light-emitting device **130**R that emits red light, the light-emitting device **130**G that emits green light, the light-emitting device **130**B that emits blue light, and the like between the substrate **351** and the substrate **352**.
[0442] Embodiment 1 or 2 can be referred to for the details of the light-emitting devices **130**R, **130**G, and **130**B.
[0443] The light-emitting device **130**R includes a conductive layer **224**R, the conductive layer **151**R over the conductive layer **224**R, and the conductive layer **152**R over the conductive layer **151**R. The light-emitting device **130**G includes a conductive layer **224**G, the conductive layer **151**G over the conductive layer **224**G, and the conductive layer **152**G over the conductive layer **151**G. The light-emitting device **130**B includes a conductive layer **224**B, the conductive layer **151**B over the conductive layer **224**B, and the conductive layer **152**B over the conductive layer **151**B.
[0444] The conductive layer **224**R is connected to a conductive layer **222***b* included in the transistor **205** through an opening provided in an insulating layer **214**. An end portion of the conductive layer **151**R is positioned outward from an end portion of the conductive layer **224**R. The insulating layer **156**R is provided to include a region that is in contact with the side surface of the conductive layer **151**R, and the conductive layer **152**R is provided to cover the conductive layer **151**R and the insulating layer **156**R.
[0445] The conductive layers **224**G, **151**G, and **152**G, and the insulating layer **156**G in the light-emitting device **130**G are not described in detail because they are respectively similar to the conductive layers **224**R, **151**R, and **152**R, and the insulating layer **156**R in the light-emitting device **130**R; the same applies to the conductive layers **224**B, **151**B, and **152**B, and the insulating layer **156**B in the light-emitting device **130**B.
[0446] The conductive layers **224**R, **224**G, and **224**B each have a depressed portion covering the opening provided in the insulating layer **214**. A layer **128** is embedded in the depressed portion.

[0447] The layer **128** has a function of filling the depressed portions of the conductive layers **224**R, **224**G, and **224**B to obtain planarity. Over the conductive layers **224**R, **224**G, and **224**B and the layer **128**, the conductive layers **151**R, **151**G, and **151**B that are respectively electrically connected to the conductive layers **224**R, **224**G, and **224**B are provided. Thus, the regions overlapping with the depressed portions of the conductive layers **224**R, **224**G, and **224**B can also be used as light-emitting regions, whereby the aperture ratio of the pixel can be increased.
[0448] The layer **128** may be an insulating layer or a conductive layer. Any of a variety of inorganic insulating materials, organic insulating materials, and conductive materials can be used for the layer **128** as appropriate. Specifically, the layer **128** is preferably formed using an insulating material and is particularly preferably formed using an organic insulating material. The layer **128** can be formed using an organic insulating material that can be used for the insulating layer **127**, for example.
[0449] The protective layer **131** is provided over the light-emitting devices **130**R, **130**G, and **130**B with the auxiliary electrode **105** therebetween. The protective layer **131** and the substrate **352** are bonded to each other with an adhesive layer **142**. The substrate **352** is provided with a light-blocking layer **157**. A solid sealing structure, a hollow sealing structure, or the like can be employed to seal the light-emitting device **130**. In FIG. **17**, a solid sealing structure is employed, in which a space between the substrate **352** and the substrate **351** is filled with the adhesive layer **142**. Alternatively, the space may be filled with an inert gas (e.g., nitrogen or argon), i.e., a hollow sealing structure may be employed. In that case, the adhesive layer **142** may be provided not to overlap with the light-emitting device. Alternatively, the space may be filled with a resin other than the frame-like adhesive layer **142**.
[0450] FIG. **17** illustrates an example where the connection portion **140** includes a conductive layer **224**C obtained by processing the same conductive film as the conductive layers **224**R, **224**G, and **224**B; the conductive layer **151**C obtained by processing the same conductive film as the conductive layers **151**R, **151**G, and **151**B; and the conductive layer **152**C obtained by processing the same conductive film as the conductive layers **152**R, **152**G, and **152**B. In the example illustrated in FIG. **17**, the insulating layer **156**C is provided to include a region overlapping with the side surface of the conductive layer **151**C.
[0451] The display apparatus **100**C has a top-emission structure. Light from the light-emitting device is emitted toward the substrate **352**. For the substrate **352**, a material with a high visible-light-transmitting property is preferably used. The pixel electrode contains a material that reflects visible light, and the counter electrode (the second electrode **102**) and the auxiliary electrode **105** each contain a material that transmits visible light.
[0452] An insulating layer **211**, an insulating layer **213**, an insulating layer **215**, and the insulating layer **214** are provided in this order over the substrate **351**. Part of the insulating layer **211** functions as a gate insulating layer of each transistor. Part of the insulating layer **213** functions as a gate insulating layer of each transistor. The insulating layer **215** is provided to cover the transistors. The insulating layer **214** is provided to cover the transistors and has a function of a planarization layer. Note that the number of gate insulating layers and the number of insulating layers covering the transistors are not limited and may each be one or more.
[0453] An inorganic insulating film is preferably used as each of the insulating layers **211**, **213**, and **215**.
[0454] An organic insulating layer is suitable as the insulating layer **214** functioning as a planarization layer.
[0455] Each of the transistors **201** and **205** includes a conductive layer **221** functioning as a gate, the insulating layer **211** functioning as the gate insulating layer, a conductive layer **222***a* and the conductive layer **222***b* functioning as a source and a drain, a semiconductor layer **231**, the insulating layer **213** functioning as the gate insulating layer, and a conductive layer **223** functioning as a gate.

[0456] A connection portion **204** is provided in a region of the substrate **351** not overlapping with the substrate **352**. In the connection portion **204**, one of the source electrode and the drain electrode of the transistor **201** is electrically connected to the FPC **353** through a conductive layer **166** and a connection layer **242**. As an example, the conductive layer **166** has a stacked-layer structure of a conductive film obtained by processing the same conductive film as the conductive layers **224**R, **224**G, and **224**B; a conductive film obtained by processing the same conductive film as the conductive layers **151**R, **151**G, and **151**B; and a conductive film obtained by processing the same conductive film as the conductive layers **152**R, **152**G, and **152**B. On the top surface of the connection portion **204**, the conductive layer **166** is exposed. Thus, the connection portion **204** and the FPC **353** can be electrically connected to each other through the connection layer **242**.
[0457] The light-blocking layer **157** is preferably provided on the surface of the substrate **352** on the substrate **351** side. The light-blocking layer **157** can be provided over a region between adjacent light-emitting devices, in the connection portion **140**, in the circuit **356**, and the like. A variety of optical members can be arranged on the outer surface of the substrate **352**.
[0458] A material that can be used for the substrate **120** can be used for each of the substrates **351** and **352**.
[0459] A material that can be used for the resin layer **122** can be used for the adhesive layer **142**.
[0460] As the connection layer **242**, an anisotropic conductive film (ACF), an anisotropic conductive paste (ACP), or the like can be used.
[Display Apparatus **100**D]
[0461] The display apparatus **100**D illustrated in FIG. **18** differs from the display apparatus **100**C illustrated in FIG. **17** mainly in having a bottom-emission structure.
[0462] Light from the light-emitting device is emitted toward the substrate **351**. For the substrate **351**, a material with a high visible-light-transmitting property is preferably used. By contrast, there is no limitation on the light-transmitting property of a material used for the substrate **352**.
[0463] A light-blocking layer **317** is preferably formed between the substrate **351** and the transistor **201** and between the substrate **351** and the transistor **205**. FIG. **18** illustrates an example where the light-blocking layer **317** is provided over the substrate **351**, an insulating layer **153** is provided over the light-blocking layer **317**, and the transistors **201** and **205** and the like are provided over the insulating layer **153**.
[0464] The light-emitting device **130**R includes a conductive layer **112**R, a conductive layer **126**R over the conductive layer **112**R, and a conductive layer **129**R over the conductive layer **126**R.
[0465] The light-emitting device **130**B includes a conductive layer **112**B, a conductive layer **126**B over the conductive layer **112**B, and a conductive layer **129**B over the conductive layer **126**B.
[0466] A material with a high visible-light-transmitting property is used for each of the conductive layers **112**R, **112**B, **126**R, **126**B, **129**R, and **129**B. A material that reflects visible light is preferably used for the second electrode **102**.
[0467] Although not illustrated in FIG. **18**, the light-emitting device **130**G is also provided.
[0468] Although FIG. **18** and the like illustrate an example where the top surface of the layer **128** includes a flat portion, the shape of the layer **128** is not particularly limited.
[Display Apparatus **100**E]
[0469] The display apparatus **100**E illustrated in FIG. **19**A is an example of a bottom-emission display apparatus different from the display apparatus **100**D illustrated in FIG. **18**. The display apparatus **100**E is different from the display apparatus **100**D in including an organic resin layer **180**. Note that the reference numerals of the components that are the same as those in FIG. **18** are sometimes omitted and the description for FIG. **18** can be referred to for the details of such components.
[0470] FIG. **19**B is a top-view layout of the pixel **178** (the pixels **178***a* and **178***b*) including the subpixel **110** (the subpixels **110**R, **110**G, and **110**B), and FIG. **19**C is a top view of the organic resin layer **180** in a region where the subpixels **110**R and **110**G included in the pixel **178** are formed. The

width between the light-blocking layers **317** corresponds to a width **110**Rw in a light-emitting region of the subpixel **110**R.

[0471] As illustrated in FIG. **19**A, the organic resin layer **180** is provided over the insulating layer **214**. As illustrated in FIG. **19**C and the region surrounded by the dashed-dotted line in FIG. **19**A, the organic resin layer **180** includes a depressed portion **181** (depressed portions **181***a* and **181***b*) having a curved surface at least in a region where the subpixel is formed. Note that the depressed portion **181** may be provided outside the light-emitting region, like a depressed portion **181***c*. With the depressed portion **181***c*, light emission caused in a region overlapping with the light-blocking layer **317** or light that has progressed to the region overlapping with the light-blocking layer **317** can be refracted and extracted from the light-emitting region, whereby emission efficiency can be improved.

[0472] A plurality of depressed portions **181** may be formed in a matrix. The depressed portions **181***a* and **181***b* may be provided in contact with each other or may have a flat surface therebetween.

[0473] Although the top surface shape and the cross-sectional shape of the depressed portion are hexagonal (FIG. **19**C) and semicircular (FIG. **19**A), respectively, other shapes may be employed as needed. Examples of the top surface shape of the depressed portion include polygons such as a triangle, a tetragon (including a rectangle and a square), and a pentagon; these polygons with rounded corners; an ellipse; and a circle.

[0474] An insulating layer containing an organic material can be used as the organic resin layer **180**. Examples of materials used for the organic resin layer **180** include an acrylic resin, a polyimide resin, an epoxy resin, an imide resin, a polyamide resin, a polyimide-amide resin, a silicone resin, a siloxane resin, a benzocyclobutene-based resin, a phenol resin, and precursors of these resins. The organic resin layer **180** may be formed using an organic material such as polyvinyl alcohol (PVA), polyvinyl butyral, polyvinylpyrrolidone, polyethylene glycol, polyglycerin, pullulan, water-soluble cellulose, or an alcohol-soluble polyamide resin.

[0475] A photosensitive resin can also be used for the organic resin layer **180**. A photoresist may be used as the photosensitive resin. As the photosensitive resin, a positive photosensitive material or a negative photosensitive material can be used.

[0476] The organic resin layer **180** may contain a material absorbing visible light. For example, the organic resin layer **180** itself may be made of a material absorbing visible light, or the organic resin layer **180** may contain a pigment absorbing visible light. For example, the organic resin layer **180** can be formed using a resin that can be used for a color filter transmitting red, blue, or green light and absorbing light of the other colors; or a resin that contains carbon black as a pigment and functions as a black matrix.

[0477] The first electrode **101** is over the organic resin layer **180**, the organic compound layer **103** is over the first electrode **101**, and the second electrode **102** is over the organic compound layer **103**. End portions of the first electrode **101**, the organic compound layer **103**, and the second electrode **102** may be covered with the insulating layer **127**.

[0478] The first electrode **101** formed over the organic resin layer **180** also has a depressed portion along the depressed portion of the organic resin layer **180**. The organic compound layer **103** formed over the first electrode **101** also has a depressed portion along the depressed portion of the first electrode **101**. The second electrode **102** formed over the organic compound layer **103** also has a depressed portion along the depressed portion of the organic compound layer **103**. The auxiliary electrode **105** formed over the second electrode **102** also has a depressed portion along the depressed portion of the second electrode **102**. That is, the depressed portions of the organic resin layer **180**, the first electrode **101**, the organic compound layer **103**, the second electrode **102**, and the auxiliary electrode **105** overlap with each other.

[0479] The second electrode **102** is over the organic compound layer **103** and the insulating layer **127**, and the auxiliary electrode **105** is over the second electrode **102**. The protective layer **131** is provided over the auxiliary electrode **105** and bonded to the substrate **352** with the adhesive layer

142.

[0480] Although the light-emitting devices **130**G and **130**B are not illustrated in FIGS. **19**A to **19**C, the light-emitting devices **130**G and **130**B are also provided.

[0481] The light-emitting device of one embodiment of the present invention including the above-described organic resin layer **180** has a structure described in Embodiment 1 or 2. Accordingly, an organic semiconductor device with a low driving voltage and favorable characteristics can be provided.

[Display Apparatus **100**F]

[0482] The display apparatus **100**F illustrated in FIG. **20** is a variation example of the display apparatus **100**C illustrated in FIG. **17** and differs from the display apparatus **100**C mainly in including the coloring layers **132**R, **132**G, and **132**B.

[0483] In the display apparatus **100**F, the light-emitting device **130** includes a region overlapping with one of the coloring layers **132**R, **132**G, and **132**B. The coloring layers **132**R, **132**G, and **132**B can be provided on a surface of the substrate **352** on the substrate **351** side. End portions of the coloring layers **132**R, **132**G, and **132**B can overlap with the light-blocking layer **157**.

[0484] In the display apparatus **100**F, the coloring layer **132**R, the coloring layer **132**G, and the coloring layer **132**B can transmit red light, green light, and blue light, respectively, for example. Note that in the display apparatus **100**F, the coloring layers **132**R, **132**G, and **132**B may be provided between the protective layer **131** and the adhesive layer **142**.

[Display Apparatus **100**G]

[0485] The display apparatus **100**G illustrated in FIG. **21**A is a variation example of the display apparatus **100**F illustrated in FIG. **20** and includes microlenses **182** over the coloring layers **132**R, **132**G, and **132**B. Note that the reference numerals of the components that are the same as those in FIG. **20** are sometimes omitted and the description for FIG. **20** can be referred to for the details of such components.

[0486] FIG. **21**B is a top-view layout of the pixel **178** (the pixels **178***a* and **178***b*) including the subpixel **110** (the subpixels **110**R, **110**G, and **110**B), and FIG. **21**C is a top view of the microlens **182** in a region where the subpixels **110**R and **110**G included in the pixel **178** are formed. Note that the width of a region where the auxiliary electrode **105** and the organic compound layer **103** are in contact with each corresponds to a width **110**Gw in a light-emitting region of the subpixel **110**G.

[0487] In the display apparatus **100**G illustrated in FIG. **21**A, a planarization film **143** is provided over the protective layer **131**, and the coloring layers **132**R, **132**G, and **132**B are provided over the planarization film **143**. A planarization film **144** is provided to cover the coloring layers **132**R, **132**G, and **132**B. The microlenses **182** are provided over the planarization film **144**.

[0488] Note that as illustrated in FIG. **21**C, the microlens **182** is preferably provided for each of the subpixels in a region where the subpixel is formed.

[0489] Although the top surface shape of the microlens **182** is illustrated as a hexagon in FIG. **21**C, other shapes may be employed as needed. Examples of the top surface shape of the depressed portion include polygons such as a triangle, a tetragon (including a rectangle and a square), and a pentagon; these polygons with rounded corners; an ellipse; and a circle.

[0490] The microlens **182** can be formed using a material similar to that for the organic resin layer **180**.

[0491] This embodiment can be combined as appropriate with the other embodiments or the examples. In this specification, in the case where a plurality of structure examples are shown in one embodiment, the structure examples can be combined as appropriate.

Embodiment 5

[0492] In this embodiment, electronic devices of embodiments of the present invention will be described.

[0493] Electronic devices in this embodiment each include the display apparatus of one embodiment of the present invention in a display portion. The display apparatus of one

embodiment of the present invention has high display performance and can be easily increased in resolution and definition. Thus, the light-emitting apparatus of one embodiment of the present invention can be used for a display portion of a variety of electronic devices.
[0494] Examples of the electronic devices include a digital camera, a digital video camera, a digital photo frame, a mobile phone, a portable game console, a portable information terminal, and an audio reproducing device, in addition to electronic devices with a relatively large screen, such as a television device, desktop and laptop personal computers, a monitor of a computer and the like, digital signage, and a large game machine such as a pachinko machine.
[0495] In particular, the display apparatus of one embodiment of the present invention can have a high resolution, and thus can be favorably used for an electronic device with a relatively small display portion. Examples of such an electronic appliance include watch-type and bracelet-type information terminals (wearable devices) and wearable devices capable of being worn on a head, such as a VR device like a head-mounted display, a glasses-type AR device, and an MR device.
[0496] The electronic device in this embodiment may include a sensor (a sensor having a function of measuring force, displacement, position, speed, acceleration, angular velocity, rotational frequency, distance, light, liquid, magnetism, temperature, a chemical substance, sound, time, hardness, electric field, current, voltage, electric power, radiation, flow rate, humidity, gradient, oscillation, odor, or infrared rays).
[0497] Examples of wearable devices capable of being worn a head are described with reference to FIGS. **22**A to **22**D.
[0498] An electronic device **700**A illustrated in FIG. **22**A and an electronic device **700**B illustrated in FIG. **22**B each include a pair of display panels **751**, a pair of housings **721**, a communication portion (not illustrated), a pair of wearing portions **723**, a control portion (not illustrated), an image capturing portion (not illustrated), a pair of optical members **753**, a frame **757**, and a pair of nose pads **758**.
[0499] The display apparatus of one embodiment of the present invention can be used for the display panels **751**. Thus, a highly reliable electronic device is obtained.
[0500] The electronic devices **700**A and **700**B can each project images displayed on the display panels **751** onto display regions **756** of the optical members **753**. Since the optical members **753** have a light-transmitting property, the user can see images displayed on the display regions, which are superimposed on transmission images seen through the optical members **753**.
[0501] In the electronic devices **700**A and **700**B, a camera capable of capturing images of the front side may be provided as the image capturing portion. Furthermore, when the electronic devices **700**A and **700**B are provided with an acceleration sensor such as a gyroscope sensor, the orientation of the user's head can be sensed and an image corresponding to the orientation can be displayed on the display regions **756**.
[0502] The communication portion includes a wireless communication device, and a video signal, for example, can be supplied by the wireless communication device. Instead of or in addition to the wireless communication device, a connector that can be connected to a cable for supplying a video signal and a power supply potential may be provided.
[0503] The electronic devices **700**A and **700**B are provided with a battery, so that they can be charged wirelessly and/or by wire.
[0504] A touch sensor module may be provided in the housing **721**.
[0505] Various touch sensors can be applied to the touch sensor module. For example, any of touch sensors of the following types can be used: a capacitive type, a resistive type, an infrared type, an electromagnetic induction type, a surface acoustic wave type, and an optical type. In particular, a capacitive sensor or an optical sensor is preferably used for the touch sensor module.
[0506] An electronic device **800**A illustrated in FIG. **22**C and an electronic device **800**B illustrated in FIG. **22**D each include a pair of display portions **820**, a housing **821**, a communication portion **822**, a pair of wearing portions **823**, a control portion **824**, a pair of image capturing portions **825**,

and a pair of lenses **832**.
[0507] The display apparatus of one embodiment of the present invention can be used in the display portions **820**. Thus, a highly reliable electronic device is obtained.
[0508] The display portions **820** are positioned inside the housing **821** so as to be seen through the lenses **832**. When the pair of display portions **820** display different images, three-dimensional display using parallax can be performed.
[0509] The electronic devices **800**A and **800**B preferably include a mechanism for adjusting the lateral positions of the lenses **832** and the display portions **820** so that the lenses **832** and the display portions **820** are positioned optimally in accordance with the positions of the user's eyes.
[0510] The electronic device **800**A or the electronic device **800**B can be mounted on the user's head with the wearing portions **823**.
[0511] The image capturing portion **825** has a function of obtaining information on the external environment. Data obtained by the image capturing portion **825** can be output to the display portion **820**. An image sensor can be used for the image capturing portion **825**. Moreover, a plurality of cameras may be provided so as to support a plurality of fields of view, such as a telescope field of view and a wide field of view.
[0512] The electronic device **800**A may include a vibration mechanism that functions as bone-conduction earphones.
[0513] The electronic devices **800**A and **800**B may each include an input terminal. To the input terminal, a cable for supplying a video signal from a video output device or the like, power for charging a battery provided in the electronic device, and the like can be connected.
[0514] The electronic device of one embodiment of the present invention may have a function of performing wireless communication with earphones **750**.
[0515] The electronic device may include an earphone portion. The electronic device **700**B illustrated in FIG. **22**B includes earphone portions **727**. Part of a wiring that connects the earphone portion **727** and the control portion may be positioned inside the housing **721** or the wearing portion **723**.
[0516] Similarly, the electronic device **800**B illustrated in FIG. **22**D includes earphone portions **827**. For example, the earphone portion **827** can be connected to the control portion **824** by wire.
[0517] As described above, both the glasses-type device (e.g., the electronic devices **700**A and **700**B) and the goggles-type device (e.g., the electronic devices **800**A and **800**B) are preferable as the electronic device of one embodiment of the present invention.
[0518] An electronic device **6500** illustrated in FIG. **23**A is a portable information terminal that can be used as a smartphone.
[0519] The electronic device **6500** includes a housing **6501**, a display portion **6502**, a power button **6503**, buttons **6504**, a speaker **6505**, a microphone **6506**, a camera **6507**, a light source **6508**, and the like. The display portion **6502** has a touch panel function.
[0520] The display apparatus of one embodiment of the present invention can be used in the display portion **6502**. Thus, a highly reliable electronic device is obtained.
[0521] FIG. **23**B is a schematic cross-sectional view including an edge portion of the housing **6501** on the microphone **6506** side.
[0522] A protection member **6510** having a light-transmitting property is provided on the display surface side of the housing **6501**. A display panel **6511**, an optical member **6512**, a touch sensor panel **6513**, a printed circuit board **6517**, a battery **6518**, and the like are provided in a space surrounded by the housing **6501** and the protection member **6510**.
[0523] The display panel **6511**, the optical member **6512**, and the touch sensor panel **6513** are fixed to the protection member **6510** with an adhesive layer (not illustrated).
[0524] Part of the display panel **6511** is folded back in a region outside the display portion **6502**, and an FPC **6515** is connected to the part that is folded back. An IC **6516** is mounted on the FPC **6515**. The FPC **6515** is connected to a terminal provided on the printed circuit board **6517**.

[0525] The display apparatus of one embodiment of the present invention can be used in the display panel **6511**. Thus, an extremely lightweight electronic device can be achieved. Since the display panel **6511** is extremely thin, the battery **6518** with high capacity can be mounted without an increase in the thickness of the electronic device. Moreover, part of the display panel **6511** is folded back so that a connection portion with the FPC **6515** is provided on the back side of the pixel portion, whereby an electronic device with a narrow bezel can be achieved.
[0526] FIG. **23**C illustrates an example of a television device. In a television device **7100**, a display portion **7000** is incorporated in a housing **7171**. Here, the housing **7171** is supported by a stand **7173**.
[0527] The display apparatus of one embodiment of the present invention can be used in the display portion **7000**. Thus, a highly reliable electronic device is obtained.
[0528] Operation of the television device **7100** illustrated in FIG. **23**C can be performed with an operation switch provided in the housing **7171** and a separate remote control **7151**.
[0529] FIG. **23**D illustrates an example of a laptop personal computer. A laptop personal computer **7200** includes a housing **7211**, a keyboard **7212**, a pointing device **7213**, an external connection port **7214**, and the like. The display portion **7000** is incorporated in the housing **7211**.
[0530] The display apparatus of one embodiment of the present invention can be used in the display portion **7000**. Thus, a highly reliable electronic device is obtained.
[0531] FIGS. **23**E and **23**F illustrate examples of digital signage.
[0532] Digital signage **7300** illustrated in FIG. **23**E includes a housing **7301**, the display portion **7000**, a speaker **7303**, and the like. The digital signage **7300** can also include an LED lamp, an operation key (including a power switch or an operation switch), a connection terminal, a variety of sensors, a microphone, and the like.
[0533] FIG. **23**F illustrates digital signage **7400** attached to a cylindrical pillar **7401**. The digital signage **7400** includes the display portion **7000** provided along a curved surface of the pillar **7401**.
[0534] In FIGS. **23**E and **23**F, the display apparatus of one embodiment of the present invention can be used in the display portion **7000**. Thus, a highly reliable electronic device is obtained.
[0535] A larger area of the display portion **7000** can increase the amount of information that can be provided at a time. The larger display portion **7000** attracts more attention, so that the effectiveness of the advertisement can be increased, for example.
[0536] As illustrated in FIGS. **23**E and **23**F, it is preferable that the digital signage **7300** or the digital signage **7400** can work with an information terminal **7311** or an information terminal **7411**, such as a smartphone that a user has, through wireless communication.
[0537] Electronic devices illustrated in FIGS. **24**A to **24**G include a housing **9000**, a display portion **9001**, a speaker **9003**, an operation key **9005** (including a power switch or an operation switch), a connection terminal **9006**, a sensor **9007** (a sensor having a function of measuring force, displacement, position, speed, acceleration, angular velocity, rotational frequency, distance, light, liquid, magnetism, temperature, chemical substance, sound, time, hardness, electric field, current, voltage, electric power, radiation, flow rate, humidity, gradient, oscillation, odor, or infrared rays), a microphone **9008**, and the like.
[0538] The electronic devices illustrated in FIGS. **24**A to **24**G have a variety of functions. For example, the electronic devices can have a function of displaying a variety of information (e.g., a still image, a moving image, and a text image) on the display portion, a touch panel function, a function of displaying a calendar, date, time, and the like, a function of controlling processing with use of a variety of software (programs), a wireless communication function, and a function of reading out and processing a program or data stored in a recording medium.
[0539] The electronic devices illustrated in FIGS. **24**A to **24**G are described in detail below.
[0540] FIG. **24**A is a perspective view of a portable information terminal **9171**. The portable information terminal **9171** can be used as a smartphone, for example. The portable information terminal **9171** may include the speaker **9003**, the connection terminal **9006**, the sensor **9007**, or the

like. The portable information terminal **9171** can display text and image information on its plurality of surfaces. FIG. **24**A illustrates an example where three icons **9050** are displayed. Furthermore, information **9051** indicated by dashed rectangles can be displayed on another surface of the display portion **9001**. Examples of the information **9051** include notification of reception of an e-mail, an SNS message, an incoming call, or the like, the title and sender of an e-mail, an SNS message, or the like, the date, the time, remaining battery, and the radio field intensity. Alternatively, the icon **9050** or the like may be displayed at the position where the information **9051** is displayed.

[0541] FIG. **24**B is a perspective view of a portable information terminal **9172**. The portable information terminal **9172** has a function of displaying information on three or more surfaces of the display portion **9001**. Here, information **9052**, information **9053**, and information **9054** are displayed on different surfaces. For example, the user of the portable information terminal **9172** can check the information **9053** displayed such that it can be seen from above the portable information terminal **9172**, with the portable information terminal **9172** put in a breast pocket of his/her clothes.

[0542] FIG. **24**C is a perspective view of a tablet terminal **9173**. The tablet terminal **9173** is capable of executing a variety of applications such as mobile phone calls, e-mailing, viewing and editing texts, music reproduction, Internet communication, and a computer game, for example. The tablet terminal **9173** includes the display portion **9001**, a camera **9002**, the microphone **9008**, and the speaker **9003** on the front surface of the housing **9000**; the operation keys **9005** as buttons for operation on the left side surface of the housing **9000**; and the connection terminal **9006** on the bottom surface of the housing **9000**.

[0543] FIG. **24**D is a perspective view of a watch-type portable information terminal **9200**. The portable information terminal **9200** can be used as a Smartwatch (registered trademark), for example. The display surface of the display portion **9001** is curved, and an image can be displayed on the curved display surface. Furthermore, for example, mutual communication between the portable information terminal **9200** and a headset capable of wireless communication can be performed, and thus hands-free calling is possible. With the connection terminal **9006**, the portable information terminal **9200** can perform mutual data transmission with another information terminal and charging. Note that the charging operation may be performed by wireless power feeding.

[0544] FIGS. **24**E to **24**G are perspective views of a foldable portable information terminal **9201**. FIG. **24**E is a perspective view illustrating the portable information terminal **9201** that is opened. FIG. **24**G is a perspective view illustrating the portable information terminal **9201** that is folded. FIG. **24**F is a perspective view illustrating the portable information terminal **9201** that is shifted from one of the states in FIGS. **24**E and **24**G to the other. The portable information terminal **9201** is highly portable when folded. When the portable information terminal **9201** is opened, a seamless large display region is highly browsable. The display portion **9001** of the portable information terminal **9201** is supported by three housings **9000** joined together by hinges **9055**. The display portion **9001** can be folded with a radius of curvature greater than or equal to 0.1 mm and less than or equal to 150 mm, for example.

[0545] This embodiment can be combined as appropriate with the other embodiments or the examples. In this specification, in the case where a plurality of structure examples are shown in one embodiment, the structure examples can be combined as appropriate.

Example 1

[0546] In this example, samples imitating the electron-injection layer of the light-emitting device of one embodiment of the present invention were evaluated by an electron spin resonance method, and the results are described. Structural Formulae of organic compounds used in this example are shown below.

##STR00013##

[0547] Next, fabrication methods of the samples used in this example are described.

(Fabrication Method of Sample 1)

[0548] First, a quartz substrate was fixed to a holder in a vacuum evaporation apparatus such that the surface to be subjected to evaporation faced downward. A sample 1 was fabricated in the following manner: the pressure in the vacuum evaporation apparatus was reduced to 1×10.sup.−4 Pa, and then 2,2′-(1,3-phenylene)bis(9-phenyl-1,10-phenanthroline) (abbreviation: mPPhen2P), 4,7-di-1-pyrrolidinyl-1,10-phenanthroline (abbreviation: Pyrrd-Phen), and In were deposited by co-evaporation to a thickness of 50 nm such that the volume ratio between mPPhen2P, Pyrrd-Phen, and In was 0.5:0.5:0.02. Note that the size of the quartz substrate was 3.0 mm×20 mm.

(Fabrication Method of Sample 2)

[0549] A sample 2 was fabricated by replacing In used in the sample 1 with Ag. The other conditions were similar to those of the sample 1.

(Fabrication Method of Comparative Sample 3)

[0550] A comparative sample 3 was fabricated by omitting mPPhen2P from the sample 1; that is, the comparative sample 3 was fabricated by depositing Pyrrd-Phen and In by co-evaporation to a thickness of 50 nm such that the volume ratio of Pyrrd-Phen to In was 1:0.02. The other conditions were similar to those of the sample 1.

(Fabrication Method of Comparative Sample 4)

[0551] A comparative sample 4 was fabricated by omitting Pyrrd-Phen from the sample 1; that is, the comparative sample 4 was fabricated by depositing mPPhen2P and In by co-evaporation to a thickness of 50 nm such that the volume ratio of mPPhen2P to In was 1:0.02. The other conditions were similar to those of the sample 1.

(Fabrication Method of Comparative Sample 5)

[0552] A comparative sample 5 was fabricated by omitting mPPhen2P from the sample 2; that is, the comparative sample 5 was fabricated by depositing Pyrrd-Phen and Ag by co-evaporation to a thickness of 50 nm such that the volume ratio of Pyrrd-Phen to Ag was 1:0.02. The other conditions were similar to those of the sample 2.

(Fabrication Method of Comparative Sample 6)

[0553] A comparative sample 6 was fabricated by omitting In from the sample 1; that is, the comparative sample 6 was fabricated by depositing Pyrrd-Phen and mPPhen2P by co-evaporation to a thickness of 50 nm such that the weight ratio of Pyrrd-Phen to mPPhen2P was 0.5:0.5. The other conditions were similar to those of the sample 1.

(Fabrication Methods of Comparative Samples 7 to 9)

[0554] A comparative sample 7 was fabricated by depositing Pyrrd-Phen by evaporation. A comparative sample 8 was fabricated by depositing mPPhen2P by evaporation. A comparative sample 9 was fabricated by depositing In by evaporation. The other conditions were similar to those of the sample 1.

[0555] The fabricated samples were evaluated by an electron spin resonance (ESR) method. Note that the measurement of the electron spin resonance spectrum by an ESR method was performed with an electron spin resonance spectrometer E500 (produced by Bruker Corporation). The measurement was performed at room temperature under the conditions where the resonance frequency was 9.56 GHz, the output power was 1 mW, the modulated magnetic field was 50 mT, the modulation width was 0.5 mT, the time constant was 0.04 s, and the sweep time was 1 min.

[0556] FIGS. **25** to **33** show ESR measurement results of the fabricated samples. FIGS. **25**, **26**, **27**, **28**, **29**, **30**, **31**, **32**, and **33** respectively show the ESR spectra of the sample 1, the sample 2, the comparative sample 3, the comparative sample 4, the comparative sample 5, the comparative sample 6, the comparative sample 7, the comparative sample 8, and the comparative sample 9. The table below shows the structure of each sample and the density of spins attributed to a signal at a g-factor of approximately 2.00.

TABLE-US-00005 TABLE 5 Spin density Structure (spins/cm.sup.3) Sample 1 mPPhen2P:Pyrrd-Phen:In 8.7 × 10.sup.16 (0.5:0.5:0.02) Sample 2 mPPhen2P:Pyrrd-Phen:Ag 1.3 × 10.sup.17 (0.5:0.5:0.02) Comparative sample 3 Pyrrd-Phen:In (1:0.02) 0 Comparative sample 4 mPPhen2P:In

(1:0.02) 0 Comparative sample 5 Pyrrd-Phen:Ag (1:0.02) 0 Comparative sample 6 mPPhen2P:Pyrrd-Phen (0.5:0.5) 0 Comparative sample 7 Pyrrd-Phen 0 Comparative sample 8 mPPhen2P 0 Comparative sample 9 In 0 *Lower detection limit 1.4 × 10.sup.16 spins/cm.sup.3

[0557] FIGS. **25** to **33** and the above table show that the samples 1 and 2 exhibited a signal at a g-factor of approximately 2.00 and the spin density higher than or equal to 5×10.sup.16 spins/cm.sup.3, whereas the comparative samples 3 to 9 exhibited no signal at a g-factor of approximately 2.00 and the spin density lower than 1.4×10.sup.16 spins/cm.sup.3, which is the lower detection limit of the electron spin resonance spectrometer. It was thus found that the spin density of the samples 1 and 2 in which three kinds of materials were combined was higher than that of the samples in which two kinds of materials were combined and the samples formed using a single material.

[0558] The above showed that in the layer including the combination of the metal, the first organic compound, and the second organic compound, interaction between the materials occurred and a SOMO level was formed.

[0559] Similarly, a thin film was formed over a quartz substrate by co-evaporation of N-(biphenyl-4-yl)-N-[4-(9-phenyl-9H-carbazol-3-yl)phenyl]-9,9-dimethyl-9H-fluoren-2-amine (abbreviation: PCBBiF) and a fluorine-containing electron-acceptor material with a molecular weight of 672 (OCHD-003) to a thickness of 100 nm such that the weight ratio of PCBBiF to OCHD-003 was 1:0.1, and an electron spin resonance spectrum of the thin film was measured at room temperature. The measurement of the electron spin resonance spectrum by an ESR method was performed with an electron spin resonance spectrometer JES FA300 (produced by JEOL Ltd.). The measurement was performed at room temperature under the conditions where the resonance frequency was 9.18 GHz, the output power was 1 mW, the modulated magnetic field was 50 mT, the modulation width was 0.5 mT, the time constant was 0.03 s, and the sweep time was 1 min. The results are shown in FIG. **34**. FIG. **34** shows that a signal was observed at a g-factor of approximately 2.00 and the spin density was 5×10.sup.19 spins/cm.sup.3. This shows that OCHD-003 has an electron-acceptor property with respect to PCBBiF and the layer including PCBBiF and OCHD-003 has a function of a hole-injection layer.

Example 2

[0560] In this example, a light-emitting device **1**, which is a light-emitting device of one embodiment of the present invention, and a reference light-emitting device **2**, which is a light-emitting device for reference, were fabricated, and the evaluation results thereof are described. The light-emitting device **1** was fabricated by a method where an organic compound layer and a second electrode are formed and then exposure to the air and heating are performed, which is modeled on a fabrication method where an organic compound film to be an organic compound layer and a conductive film to be a second electrode are formed and then these films are processed by a photolithography method to form the organic compound layer and the second electrode. Meanwhile, the reference light-emitting device **2** was fabricated by a fabrication method where exposure to the air and heating are not performed after formation of the organic compound layer and the second electrode. The structural formulae of the organic compounds used in the light-emitting device **1** and the reference light-emitting device **2** are shown below.

##STR00014## ##STR00015## ##STR00016##

(Fabrication Method of Light-Emitting Device **1**)

[0561] First, as a reflective electrode, silver (Ag) was deposited over a glass substrate to a thickness of 100 nm by a sputtering method, and then, as a transparent electrode, indium tin oxide containing silicon oxide (ITSO) was deposited to a thickness of 85 nm by a sputtering method, so that a first electrode was formed. The electrode area was set to 4 mm.sup.2 (2 mm×2 mm). Note that the transparent electrode serves as an anode, and the transparent electrode and the reflective electrode can be collectively regarded as the first electrode.

[0562] Next, in pretreatment for forming the light-emitting device over the substrate, a surface of

the substrate was washed with water and baking was performed at 200° C. for 1 hour.
[0563] After that, the substrate was transferred into a vacuum evaporation apparatus where the internal pressure was reduced to approximately 1×10.sup.−4 Pa, and was subjected to heat treatment at 170° C. for 30 minutes in a heating chamber of the vacuum evaporation apparatus, and then, the substrate was cooled down for approximately 30 minutes.
[0564] Then, the substrate was fixed to a substrate holder provided in the vacuum evaporation apparatus such that the surface on which the first electrode was formed faced downward. Over the first electrode, N-(biphenyl-4-yl)-N-[4-(9-phenyl-9H-carbazol-3-yl)phenyl]-9,9-dimethyl-9H-fluoren-2-amine (abbreviation: PCBBiF) and an electron-acceptor material with a molecular weight of 672 and four or more fluorine atoms (OCHD-003) were deposited by co-evaporation to a thickness of 10 nm such that the weight ratio of PCBBiF to OCHD-003 was 1:0.03, so that a hole-injection layer was formed.
[0565] Over the hole-injection layer, PCBBiF was deposited by evaporation to a thickness of 55 nm, so that a hole-transport layer was formed.
[0566] Then, over the hole-transport layer, 8-(1,1′: 4′,1″-terphenyl-3-yl)-4-[3-(dibenzothiophen-4-yl)phenyl]-[1]benzofuro[3,2-d]pyrimidine (abbreviation: 8mpTP-4mDBtPBfpm), 9-(2-naphthyl)-9′-phenyl-9H,9′H-3,3′-bicarbazole (abbreviation: βNCCP), and [2-d.sub.3-methyl-8-(2-pyridinyl-κN)benzofuro[2,3-b]pyridine-κC]bis[2-(5-d.sub.3-methyl-2-pyridinyl-κN.sup.2)phenyl-κC]iridium(III) (abbreviation: Ir(5mppy-d.sub.3).sub.2 (mbfpypy-d.sub.3)) were deposited by co-evaporation to a thickness of 40 nm such that the weight ratio between 8mpTP-4mDBtPBfpm, βNCCP, and Ir(5mppy-d.sub.3).sub.2 (mbfpypy-d.sub.3) was 0.5:0.5:0.1, so that a light-emitting layer was formed.
[0567] Next, over the light-emitting layer, 2-{3-[3-(N-phenyl-9H-carbazol-3-yl)-9H-carbazol-9-yl]phenyl}dibenzo[f,h]quinoxaline (abbreviation: 2mPCCzPDBq) was deposited by evaporation to a thickness of 15 nm, so that an electron-transport layer was formed.
[0568] Then, over the electron-transport layer, 4,7-di(2,3,3a,4,5,6,7,7a-octahydro-1H-isoindol-2-yl)-1,10-phenanthroline (abbreviation: Hid2Phen) (Structural Formula (105)), 2,2′-(2,2′-bipyridine-6,6′-diyl)bis(4-phenylbenzo[h]quinazoline) (abbreviation: 6,6′ (P-Bqn) 2BPy), and indium (In) were deposited by co-evaporation to a thickness of 5 nm such that the volume ratio between Hid2Phen, 6,6′ (P-Bqn) 2BPy, and In was 0.5:0.5:0.02, so that an electron-injection layer was formed.
[0569] Then, over the electron-injection layer, copper phthalocyanine (abbreviation: CuPc) was deposited by evaporation to a thickness of 2 nm, so that an electron-relay layer was formed.
[0570] Next, over the electron-relay layer, PCBBiF and molybdenum oxide(VI) (MoO.sub.3) were deposited by co-evaporation to a thickness of 15 nm such that the volume ratio of PCBBIF to MoO.sub.3 was 1:0.5, so that a p-type layer was formed.
[0571] Then, over the p-type layer, indium tin oxide (ITO) was deposited to a thickness of 40 nm by a sputtering method as a transparent electrode, so that a second electrode was formed.
[0572] After formation of the second electrode, exposure to the air was performed for 1 hour. Then, the substrate was transferred into a vacuum evaporation apparatus where the internal pressure was reduced to approximately 1×10.sup.−4 Pa, and heating was performed at 100° C. for 1 hour in a heating chamber of the vacuum evaporation apparatus.
[0573] Next, over the second electrode, indium tin oxide (ITO) was deposited by a sputtering method to a thickness of 70 nm, so that a cap layer was formed. Note that the second electrode can also be regarded as including the cap layer. That is, the 40-nm-thick ITO formed before exposure to the air and the 70-nm-thick ITO formed after exposure to the air can be collectively regarded as the second electrode, and the second electrode also has a function of the cap layer. The above is the fabricating method of the light-emitting device **1**.
(Fabrication Method of Reference Light-Emitting Device **2**)
[0574] The reference light-emitting device **2** is different from the light-emitting device **1** in that the

second electrode as the transparent electrode was formed by depositing indium tin oxide (ITO) by a sputtering method to a thickness of 110 nm after formation of the p-type layer in the fabrication method of the light-emitting device **1**, and that the steps of exposure to the air and heating were not performed. The other conditions were similar to those of the light-emitting device **1**.

[0575] The table below lists the structures of the light-emitting device **1** and the reference light-emitting device **2**.

TABLE-US-00006 TABLE 6 Reference Light-emitting light-emitting Thickness device 1 device 2 Second electrode — ITO (70 nm) — — — Exposure to the — air and heating Second electrode — ITO (40 nm) ITO (110 nm) P-type layer 3 15 nm PCBBiF:MoO.sub.3 (1:0.5) Electron-relay layer 2 2 nm CuPc Electron-injection 1 5 nm Hid2Phen:6,6′(P-Bqn)2BPy:In layer (0.5:0.5:0.02) Electron-transport layer 15 nm 2mPCCzPDBq Light-emitting layer 40 nm 8mpTP-4mDBtPBfpm:βNCCP:Ir(5mppy- d.sub.3).sub.2(mbfpypy-d.sub.3) (0.5:0.5:0.1) Hole-transport layer 55 nm PCBBiF Hole-injection layer 10 nm PCBBiF:OCHD-003 (1:0.03) First electrode 2 85 nm ITSO 1 100 nm Ag

[0576] The fabricated light-emitting device **1** and reference light-emitting device **2** were each sealed using a glass substrate in a glove box containing a nitrogen atmosphere so as not to be exposed to the air. Specifically, a UV curable sealing material was applied to surround the devices, only the sealing material was irradiated with UV while the light-emitting devices were not irradiated with the UV, and heat treatment was performed at 80° C. under an atmospheric pressure for 1 hour. Then, the initial characteristics of the light-emitting devices were measured.

[0577] FIG. **35** shows the luminance-current density characteristics of the light-emitting device **1** and the reference light-emitting device **2**, FIG. **36** shows the luminance-voltage characteristics thereof, FIG. **37** shows the current efficiency-current density characteristics thereof, FIG. **38** shows the current density-voltage characteristics thereof, and FIG. **39** shows the electroluminescence spectra thereof. Note that in legends in FIGS. **35** to **39**, the light-emitting device **1** is denoted by Device **1**, and the reference light-emitting device **2** is denoted by Reference.

[0578] The table below shows the main characteristics of the light-emitting device **1** and the reference light-emitting device **2** at a luminance of approximately 1000 cd/m.sup.2. The luminance, CIE chromaticity, and electroluminescence spectra were measured at room temperature with a spectroradiometer (SR-UL1R, produced by TOPCON TECHNOHOUSE CORPORATION).

TABLE-US-00007 TABLE 7 Current Current Voltage Current density Luminance efficiency (V) (mA) (mA/cm.sup.2) Chromaticity x Chromaticity y (cd/m.sup.2) (cd/A) Light-emitting device 1 3.2 0.037 0.91 0.30 0.67 1009 110 Reference 3.4 0.040 0.99 0.30 0.67 947 96 light-emitting device 2

[0579] According to FIGS. **35** to **39** and the above table, the light-emitting device **1** showed the results equivalent to or more favorable than those of the reference light-emitting device **2** not subjected to exposure to the air and heating. Since processing by a photolithography method always includes an air exposure step, the light-emitting device of one embodiment of the present invention is found to be highly resistant to processing by a photolithography method and be capable of maintaining favorable characteristics even when fabricated through processing by a photolithography method.

[0580] This application is based on Japanese Patent Application Serial No. 2024-029810 filed with Japan Patent Office on Feb. 29, 2024, the entire contents of which are hereby incorporated by reference.

Claims

1. A light-emitting device comprising: a first electrode; a second electrode; and an organic compound layer, wherein the light-emitting device is over a first insulating layer, wherein the first electrode is in contact with the first insulating layer, wherein the organic compound layer is

between the first electrode and the second electrode, wherein the second electrode and the organic compound layer are separated from at least one of other light-emitting devices adjacent to the light-emitting device, wherein when seen from a direction perpendicular to the first insulating layer where the first electrode is formed, an outline of the second electrode and an outline of the organic compound layer are aligned or substantially aligned with each other, wherein the organic compound layer comprises a light-emitting layer and an electron-injection layer, wherein the electron-injection layer comprises a mixed layer comprising a metal, a first organic compound, and a second organic compound, wherein the first organic compound comprises a phenanthroline ring with an electron-donating group, and wherein the second organic compound comprises a π-electron deficient heteroaromatic ring.
2. A light-emitting device comprising: a first electrode; a second electrode; and an organic compound layer, wherein the light-emitting device is over a first insulating layer, wherein the first electrode is in contact with the first insulating layer, wherein the organic compound layer is between the first electrode and the second electrode, wherein the second electrode and the organic compound layer are separated from at least one of other light-emitting devices adjacent to the light-emitting device, wherein when seen from a direction perpendicular to the first insulating layer where the first electrode is formed, an outline of the second electrode and an outline of the organic compound layer are aligned or substantially aligned with each other, wherein the organic compound layer comprises a light-emitting layer and an electron-injection layer, wherein the electron-injection layer comprises a mixed layer comprising a metal, a first organic compound, and a second organic compound, wherein the first organic compound comprises a phenanthroline ring, wherein a minimum value of an electrostatic potential of the first organic compound is smaller than or equal to −0.085 E.sub.h when a threshold value of electron density distribution in atomic unit is 0.0004 e/a.sub.0.sup.3, and wherein the second organic compound comprises a n-electron deficient heteroaromatic ring.
3. The light-emitting device according to claim 2, wherein an electron-donating group is bonded to the phenanthroline ring of the first organic compound.
4. The light-emitting device according to claim 1, wherein the phenanthroline ring is a 1,10-phenanthroline ring and the electron-donating group is bonded to at least one of a 4-position and a 7-position of the 1,10-phenanthroline ring.
5. The light-emitting device according to claim 1, wherein the electron-donating group is at least one of an alkyl group, an alkoxy group, an aryloxy group, an alkylamino group, an arylamino group, and a heterocyclic amino group.
6. The light-emitting device according to claim 1, wherein an acid dissociation constant pK.sub.a of the first organic compound is greater than or equal to 8.
7. The light-emitting device according to claim 1, wherein a spin density of the electron-injection layer measured by an electron spin resonance method is higher than or equal to 5×10.sup.16 spins/cm.sup.3.
8. The light-emitting device according to claim 7, wherein a spin density of a mixed film comprising the metal and the first organic compound measured by an electron spin resonance method is lower than or equal to 2×10.sup.16 spins/cm.sup.3, and wherein a spin density of a mixed film comprising the metal and the second organic compound measured by an electron spin resonance method is lower than or equal to 2×10.sup.16 spins/cm.sup.3.
9. The light-emitting device according to claim 1, wherein the second organic compound comprises a phenanthroline ring.
10. The light-emitting device according to claim 1, wherein a glass transition temperature of the second organic compound is higher than or equal to 100° C.
11. The light-emitting device according to claim 1, wherein the metal belongs to Group 1, Group 3, Group 11, or Group 13 in a periodic table.
12. The light-emitting device according to claim 1, wherein the organic compound layer comprises

a p-type layer between the electron-injection layer and the second electrode, wherein the p-type layer comprises a third organic compound and a fourth organic compound or a metal oxide, wherein the third organic compound has a hole-transport property, and wherein the fourth organic compound comprises at least one of a halogen group and a cyano group.

13. A display apparatus comprising: the light-emitting device according to claim 1; and at least one of a transistor and a substrate.

14. The light-emitting device according to claim 3, wherein the phenanthroline ring is a 1,10-phenanthroline ring and the electron-donating group is bonded to at least one of a 4-position and a 7-position of the 1,10-phenanthroline ring.

15. The light-emitting device according to claim 3, wherein the electron-donating group is at least one of an alkyl group, an alkoxy group, an aryloxy group, an alkylamino group, an arylamino group, and a heterocyclic amino group.

16. The light-emitting device according to claim 2, wherein an acid dissociation constant pK.sub.a of the first organic compound is greater than or equal to 8.

17. The light-emitting device according to claim 2, wherein a spin density of the electron-injection layer measured by an electron spin resonance method is higher than or equal to 5×10.sup.16 spins/cm.sup.3.

18. The light-emitting device according to claim 2, wherein the second organic compound comprises a phenanthroline ring.

19. The light-emitting device according to claim 2, wherein a glass transition temperature of the second organic compound is higher than or equal to 100° C.

20. The light-emitting device according to claim 2, wherein the metal belongs to Group 1, Group 3, Group 11, or Group 13 in a periodic table.
